

Cardiovascular System

August 2025!

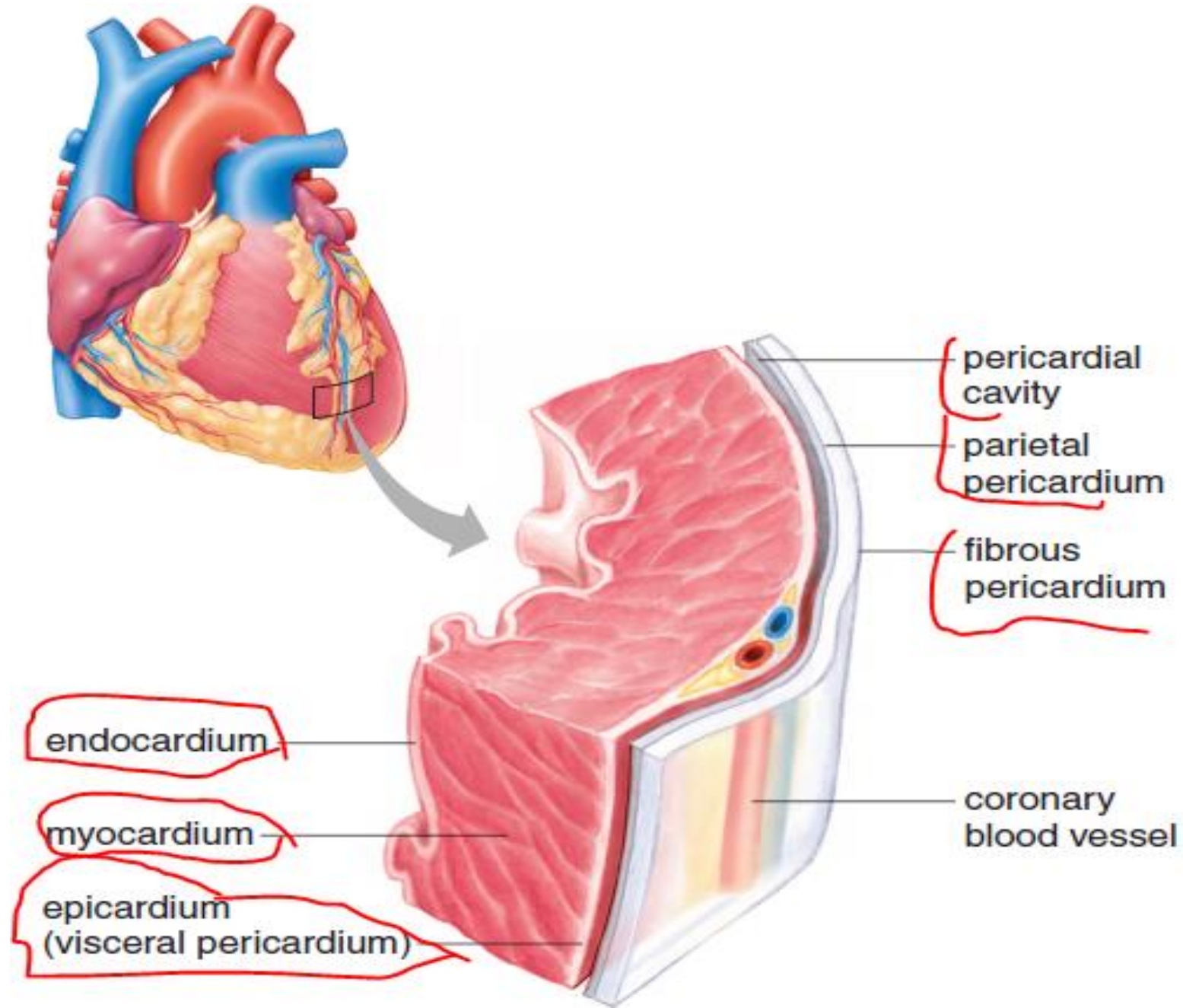
The wall of heart chambers

❖ The wall of the heart has three layers: a superficial *epicardium*, a middle *myocardium*, and a deep *endocardium*:

1. **Endocardium**, a thin internal layer (endothelium & subendothelial connective tissue) or lining membrane.
2. **Myocardium**, a thick, helical middle layer composed of cardiac muscle.
3. **Epicardium**, a thin external layer (mesothelium) formed by the visceral layer of serous pericardium.

❑ The walls of the heart consist mostly of myocardium, especially in the ventricles.

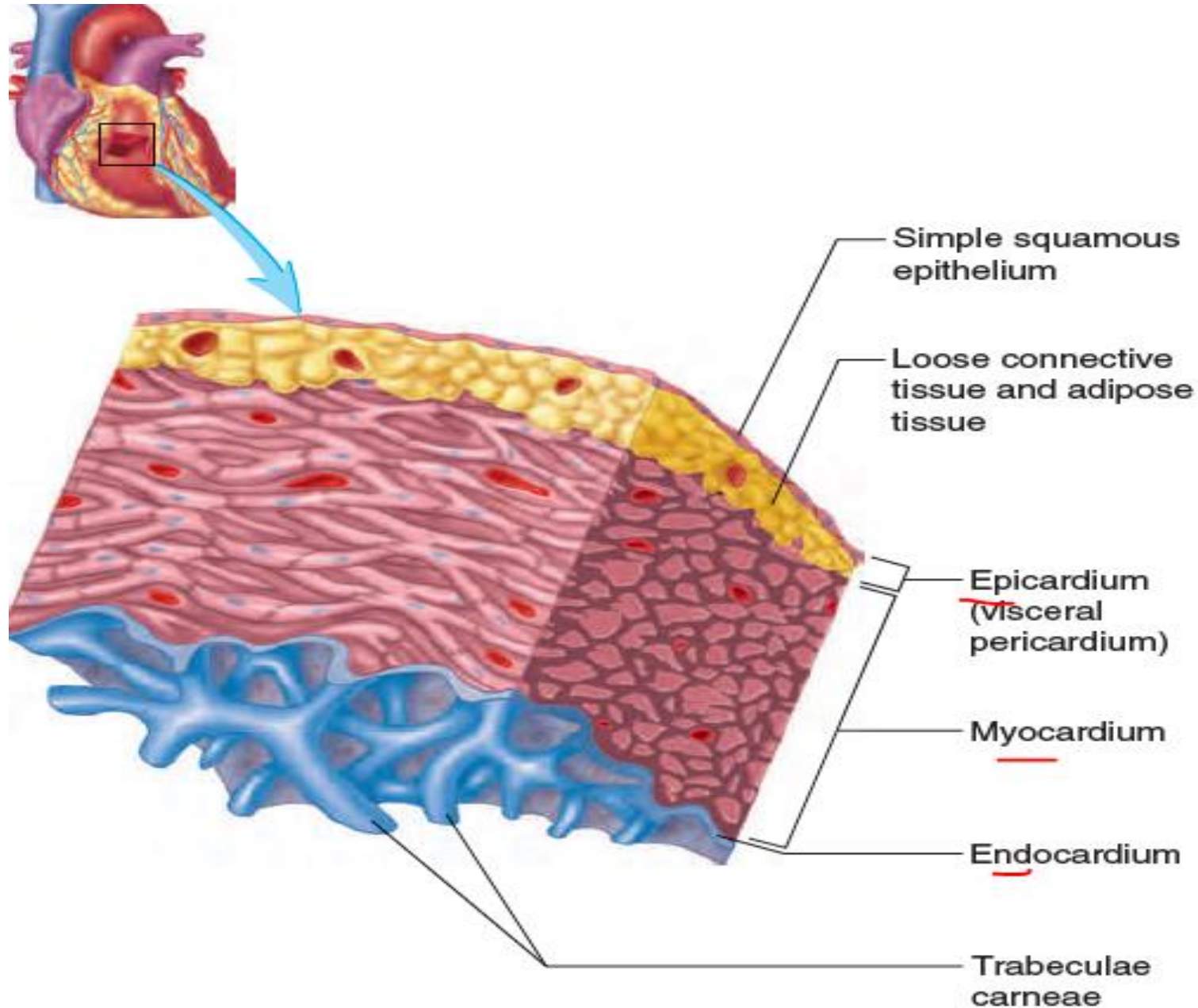
The wall of heart chamber,,



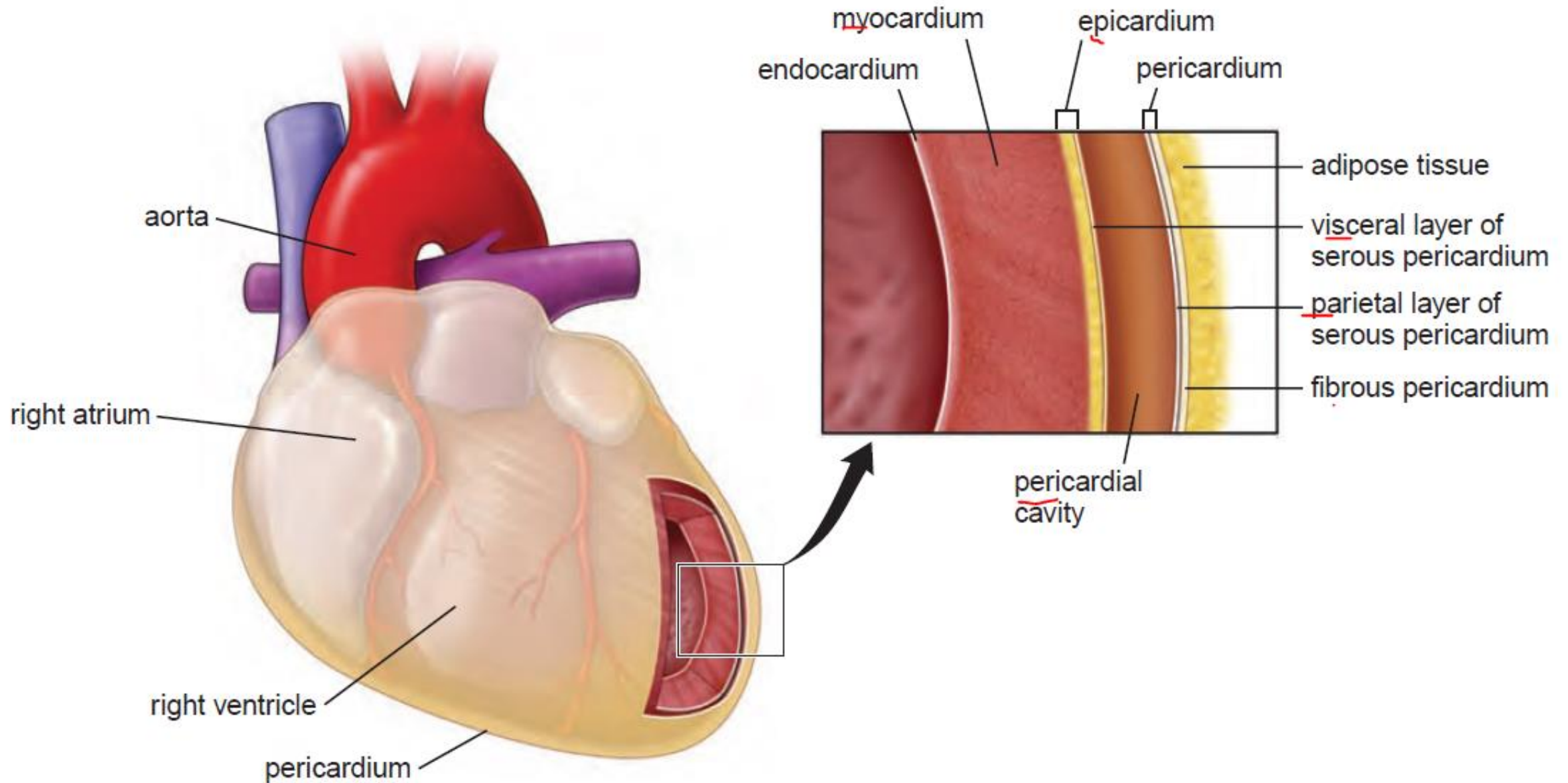
The wall of heart chamber,,

- ❖ The **epicardium**, also called the *visceral pericardium*, is a thin, serous membrane forming the smooth outer surface of the heart.
 - It consists of **simple squamous epithelium** overlying a layer of loose connective tissue and adipose tissue.
- ❖ The **thick, middle layer of the heart**, the **myocardium**, is composed of cardiac muscle cells and is responsible for contraction of the heart chambers.
- ❖ The **smooth inner surface** of the heart chambers is the **endocardium**, which consists of **simple squamous epithelium** over a layer of connective tissue.
 - The smooth endothelial lining minimizes the surface friction as blood passes through the heart.
 - The endocardium is continuous with the endothelial lining of the large blood vessels attached to the heart.

The wall of heart chamber,,



The wall of heart chamber,,



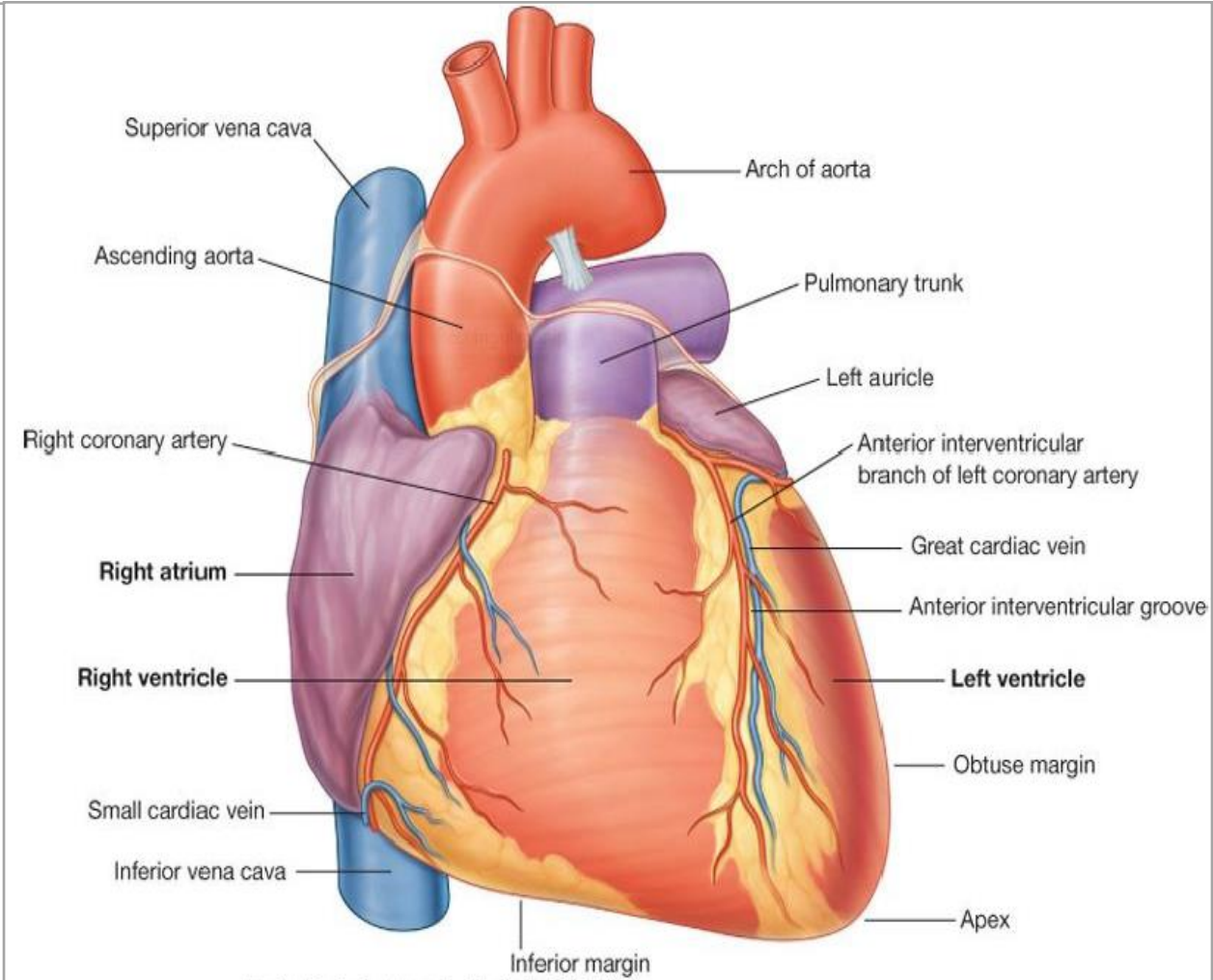
Surfaces of the heart

The heart has 4 surfaces.

1. Anterior surface: consists mostly of **Rt ventricle** with some of **Rt atrium** & some of **Lt ventricle**.

2. Diaphragmatic surface: consists of **Lt ventricle** & a small portion of **Rt ventricle**.

➤ heart in the anatomic position rests on **diaphragmatic surface**.



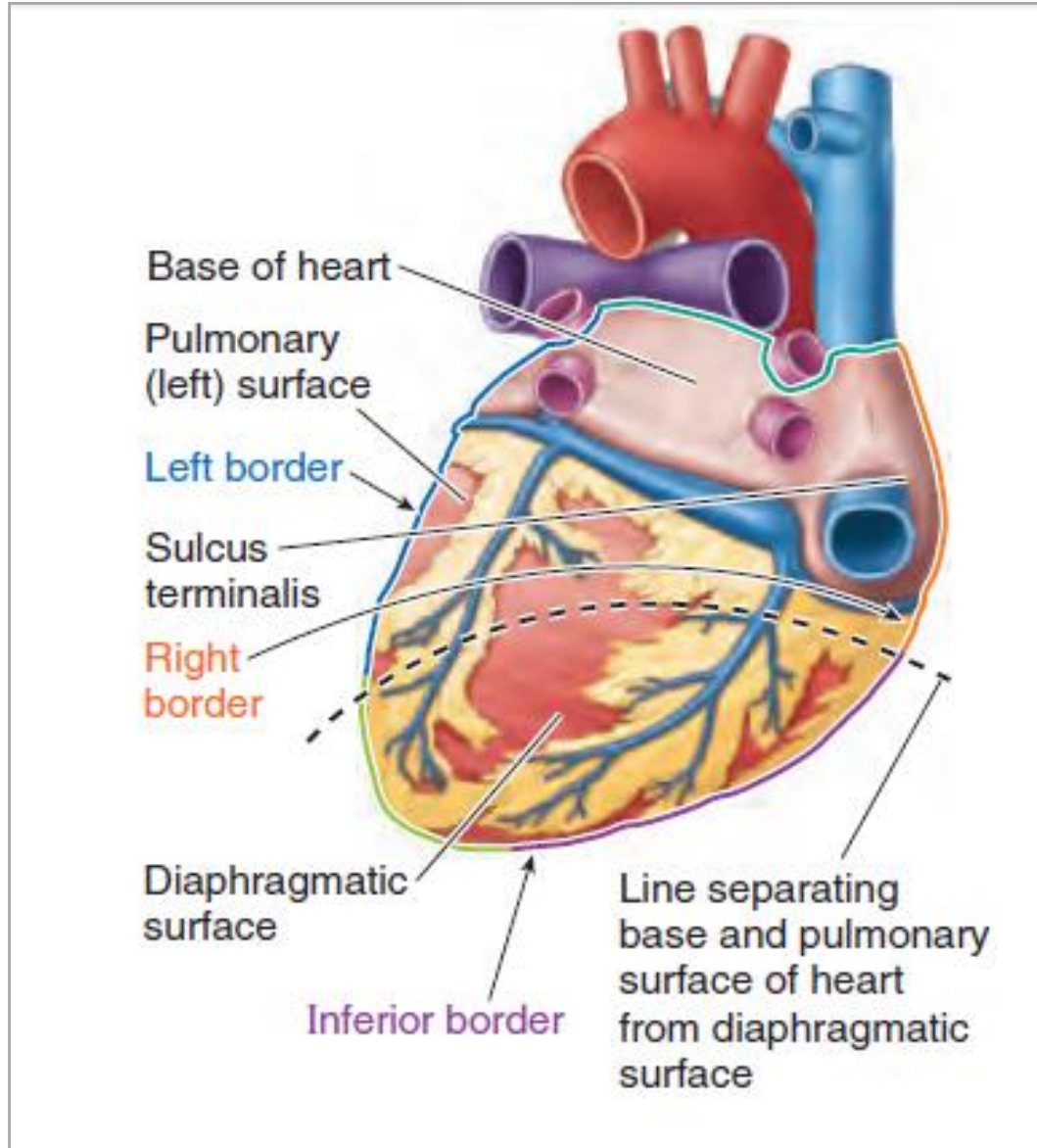
Surfaces of the heart,,,

3. Left pulmonary surface:

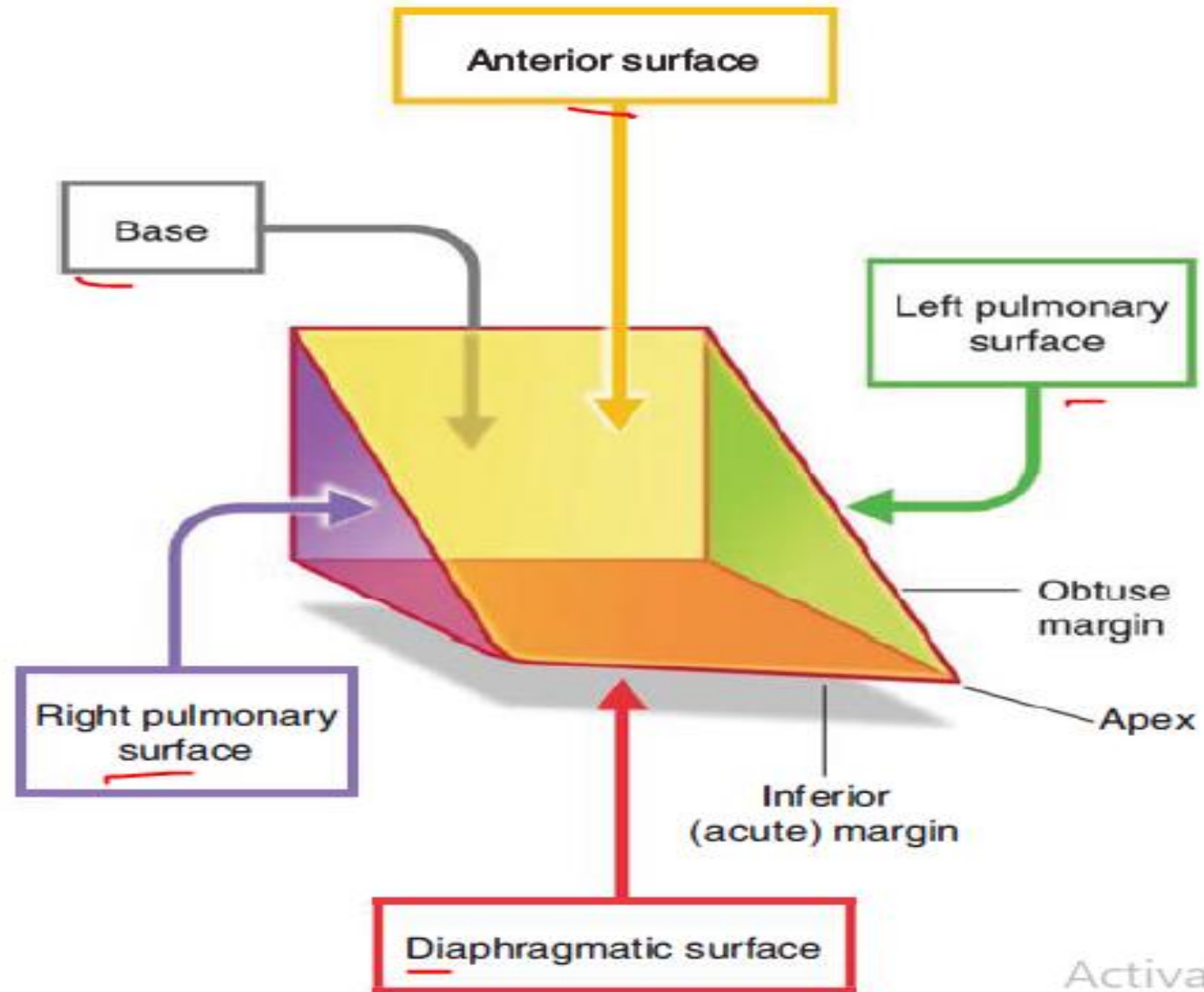
- consists of Lt ventricle & a portion of Lt atrium & faces Lt lung.

4. Right pulmonary surface: consists of Rt atrium & faces Rt lung.

- **Apex** – formed by ***Lt ventricle***: is directed ***downward, forward & to the left***.



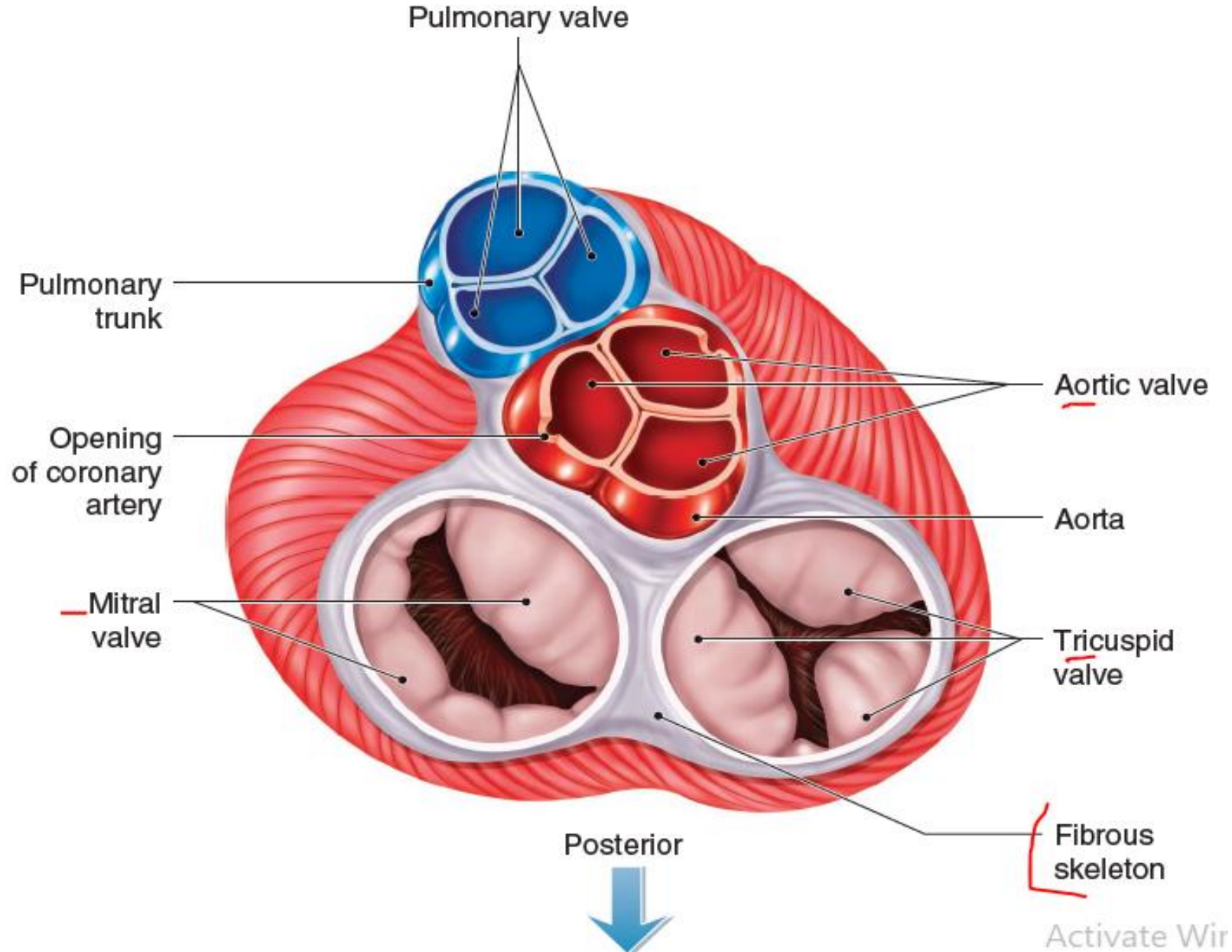
Surfaces of the heart,,



The fibrous skeleton of the heart

- The cardiac skeleton is a collection of dense, fibrous connective tissue in the **form of four rings** with interconnecting areas in a plane between the atria and the ventricles.
- The four rings of the cardiac skeleton surround the two atrioventricular orifices, the aortic orifice and opening of the pulmonary trunks.
- The cardiac muscle fibers are anchored to the **fibrous skeleton of the heart.**
- Fibrous skeleton, fuse with one another, and merge with the interventricular septum.

The fibrous skeleton of the heart,,,



The fibrous skeleton of the heart,,,

- In addition to forming a structural foundation for the heart valves, the fibrous skeleton prevents overstretching of the valves as blood passes through them.

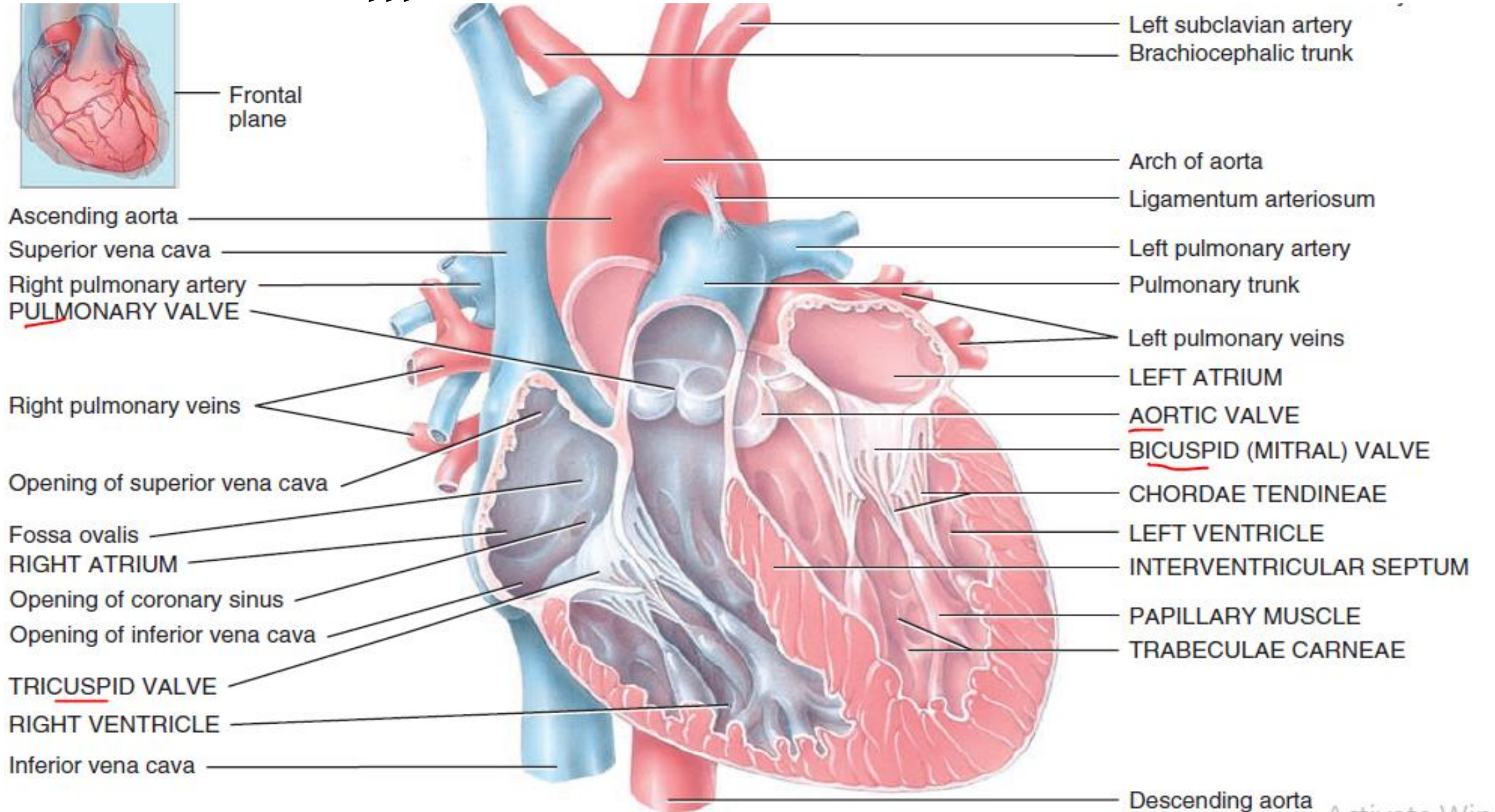
☐The fibrous skeleton of the heart functions in:

- ✓ Keeps the orifices of the AV and semilunar valves patent & prevents over distention.
- ✓ Provides attachments for the leaflets and cusps of the valves.
- ✓ Provides attachment for the myocardium.
- ✓ Forms an electrical “insulator,” by separating the myenterically conducted impulses of the atria and ventricles.

Heart Valves

- Like all pumps, the heart contains valves that allow the blood to flow in only one direction through the heart.
 - There is a valve between **each atrium and its ventricle** and another **at the exit from each ventricle into its great artery**, but the heart **has no valves where the great veins empty into the atria**.
- ❑ The **atrioventricular (AV) valves** regulate the openings between the atria and ventricles.
- ❑ The **semilunar valves** (pulmonary and aortic valves) regulate the flow of blood from the ventricles into the great arteries.

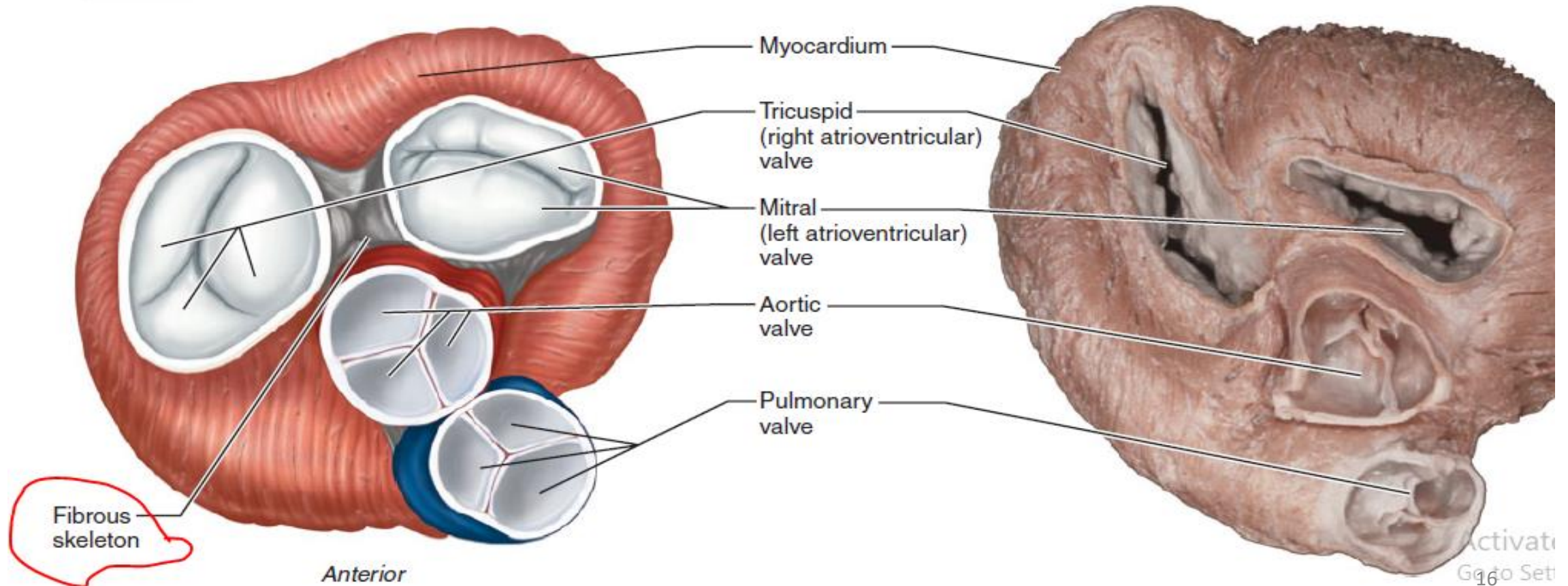
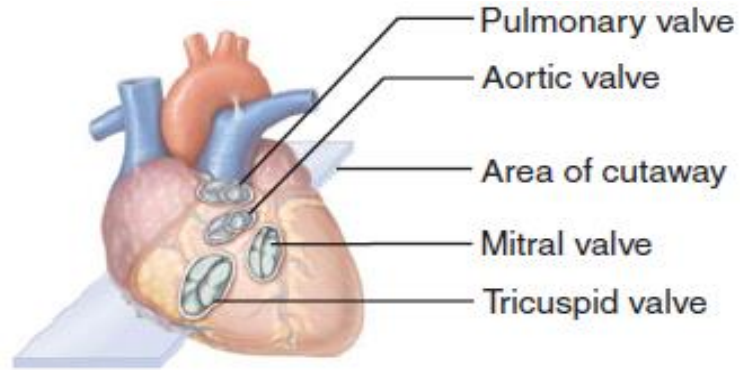
Heart Valves,,,



Heart Valves,,,

- **The AV valve** between the right atrium and the right ventricle has **three cusps** and is called the **tricuspid valve**
- **The AV valve** between the left atrium and the left ventricle has **two cusps** and is called the **bicuspid valve** or *mitral* (resembling a bishop's miter, a two-pointed hat) *valve*.
- **Each semilunar valve** is composed of 3 **pocketlike cusps** of dense irregular connective tissue.
- ❑ **The mitral valve is** located posterior to the sternum at the level of the 4th costal cartilage.
- ❑ **The right AV (tricuspid)** , located posterior to the body of the sternum at the level of the 4th and 5th intercostal spaces

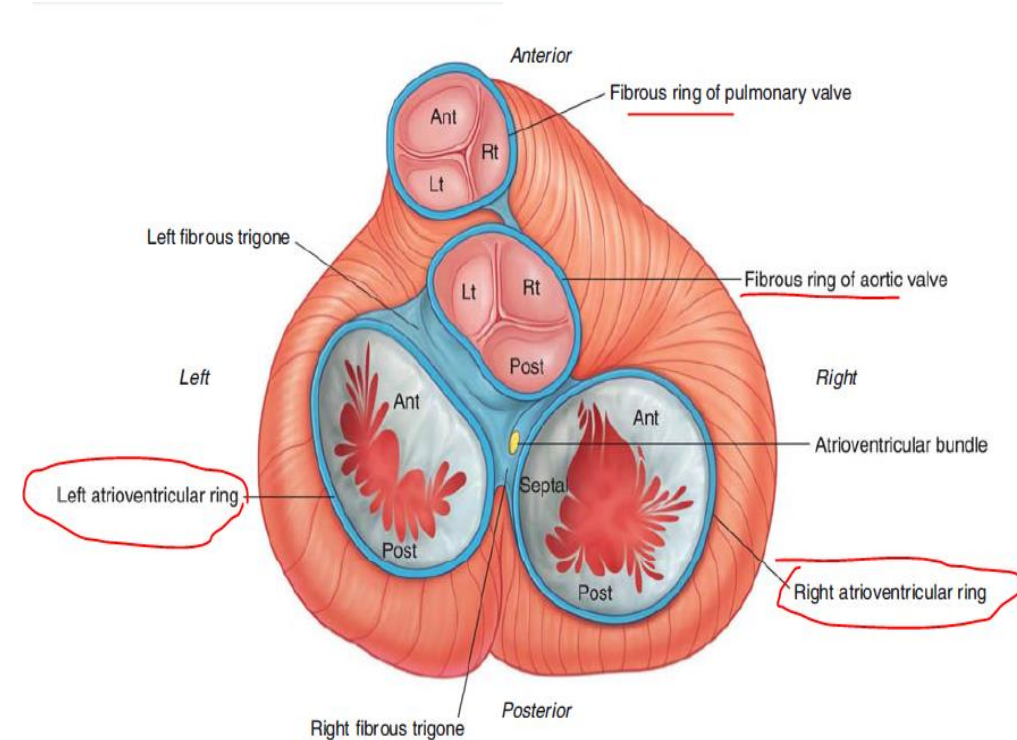
Heart Valves,,,



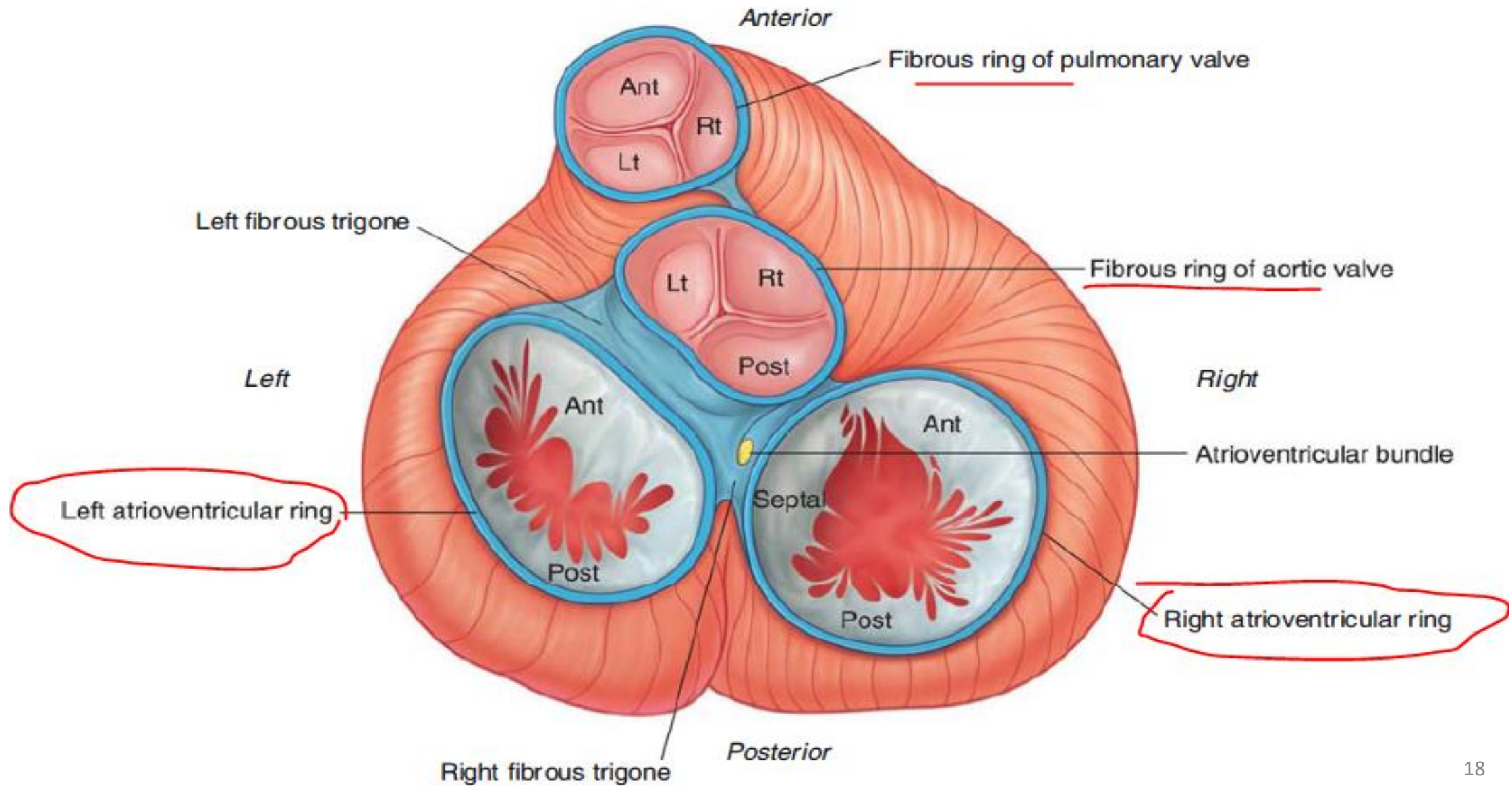
Heart Valves,,,

❑ *The cusps of the valves are named based on their location.*

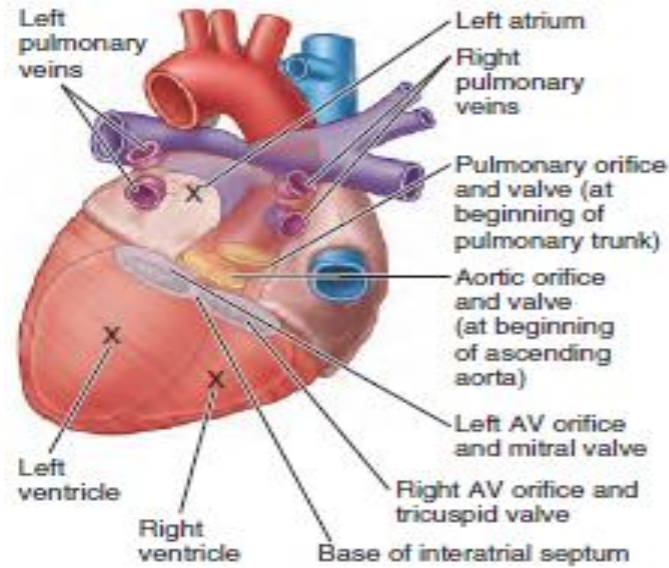
- The names of of the 3 cusps **AV of tricuspid valve** **are**, the **anterior**, **septal**, and **posterior** cusps.
- The names of the 2 cusps of *mitral* are the **anterior** and **posterior** cusps.
- The names of the 3 cusps of *aortic valve* are **right**, **left**, and **posterior** aortic cusps .
- The names of of the three cusps **of pulmonary** **valve** are **right**, **left**, and **anterior** aortic cusps .



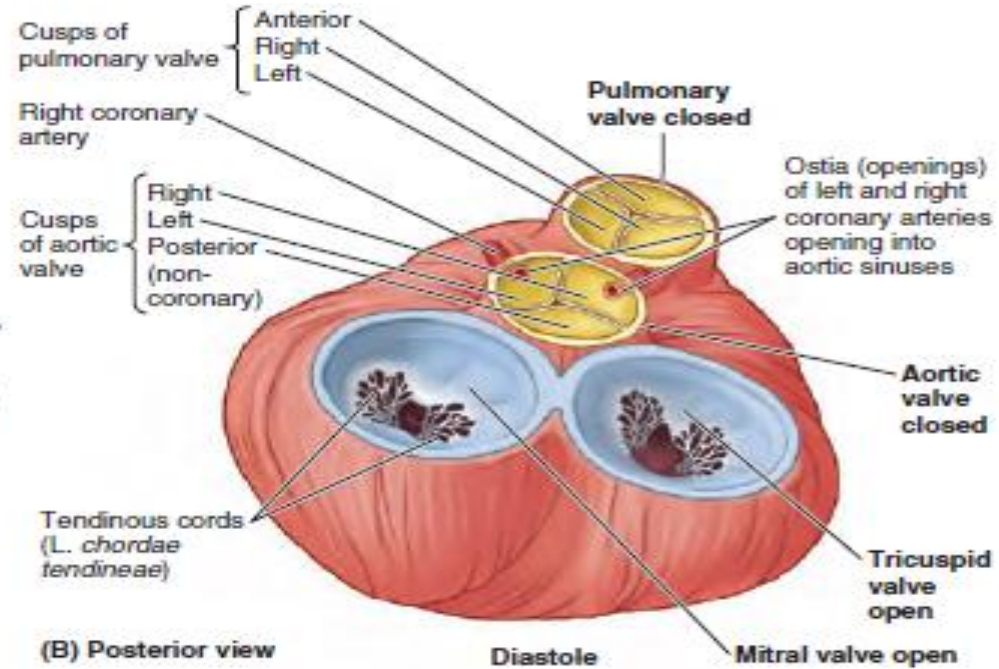
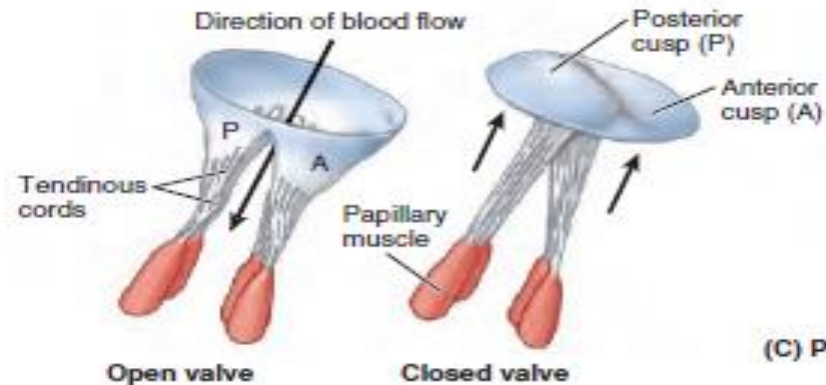
Heart Valves,,,



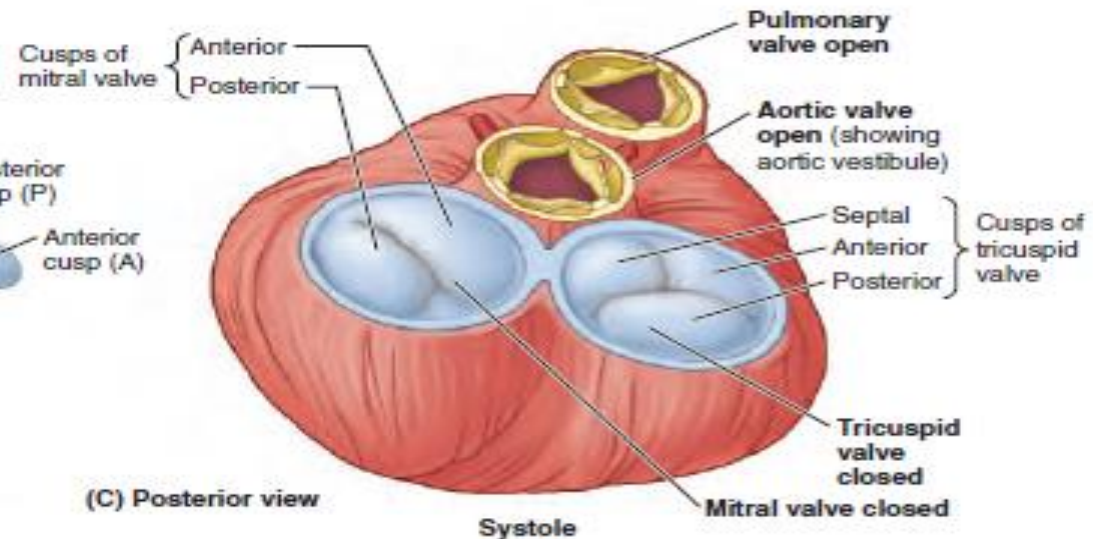
Heart Valves,,,



(A) Posteroinferior view



(B) Posterior view

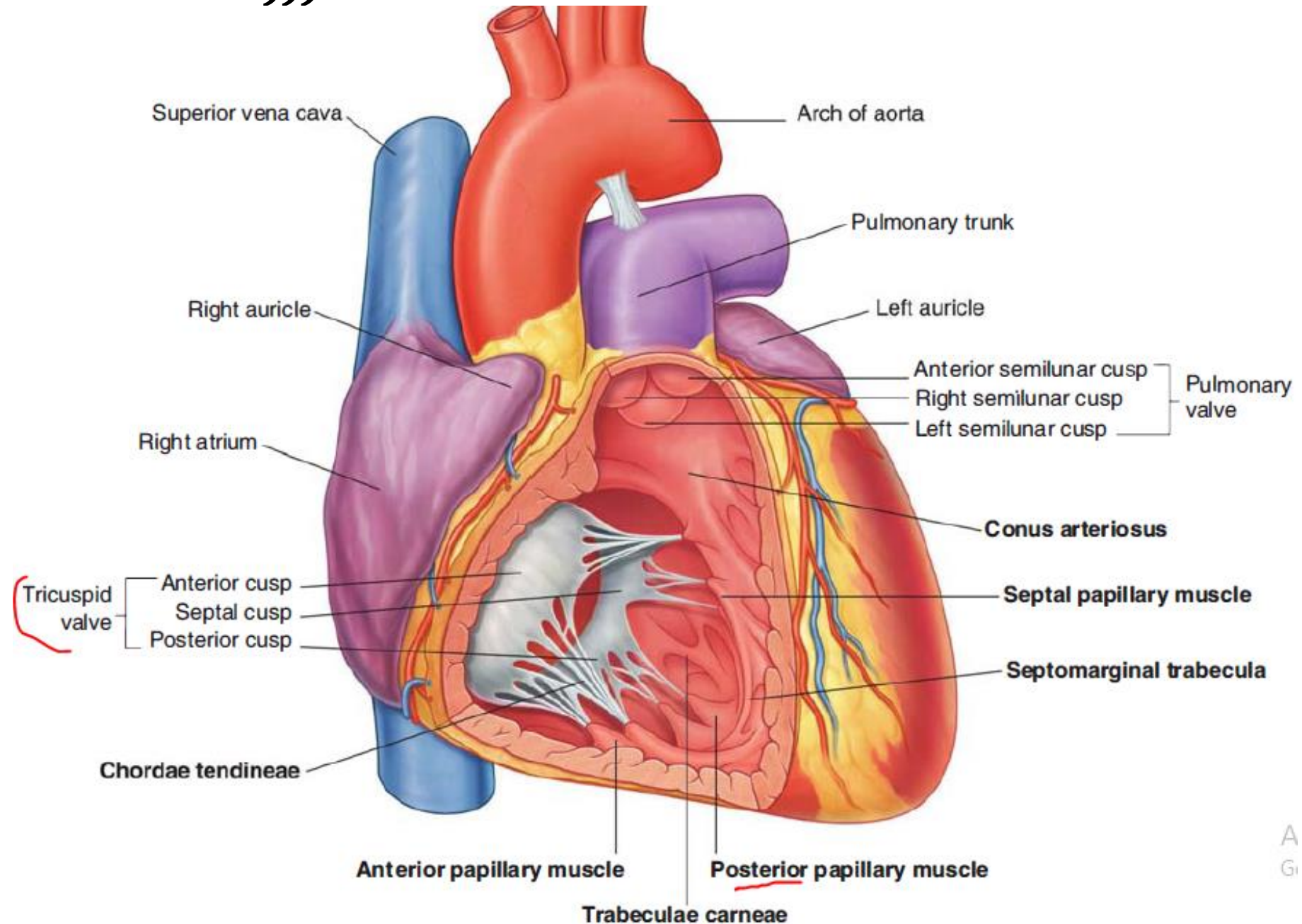


(C) Posterior view

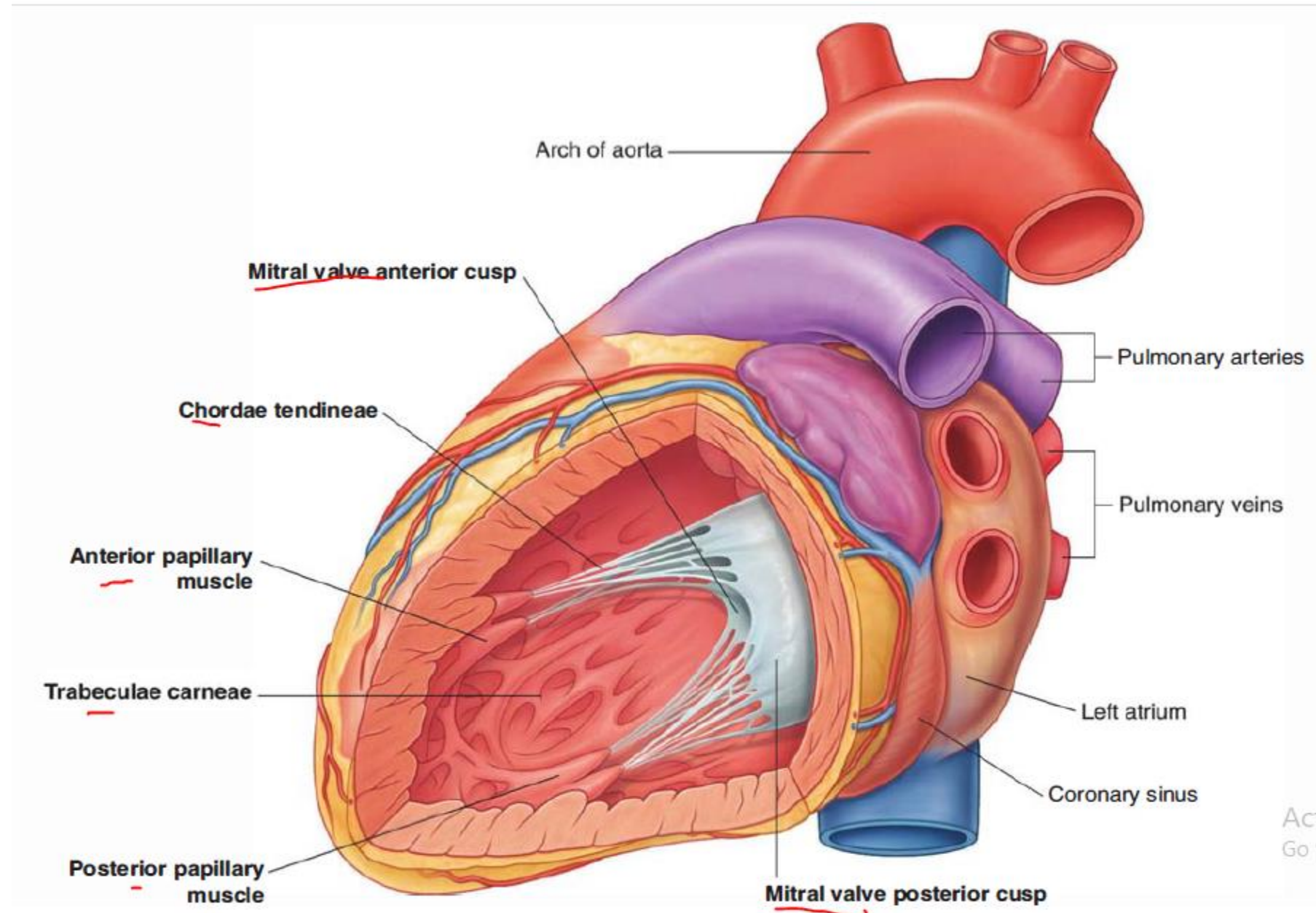
Heart Valves,,,

- The 3 flaps of **right atrioventricular valve**, are connected by **chordae tendineae to the papillary muscles**, which control the opening and closing of the valves.
- **The two cusps** of the mitral valve are attached by chordae tendineae to two papillary muscles that project from the wall of the ventricle
- **Unlike the atrioventricular valves**, there are no papillary muscles or chordae tendineae associated with the semilunar valve .
- The semilunar valves do not open and close by any muscular effort of their own.
- **Simply semilunar valves**, prevent backflow of blood from the great arteries into the ventricles.

Heart Valves,,,



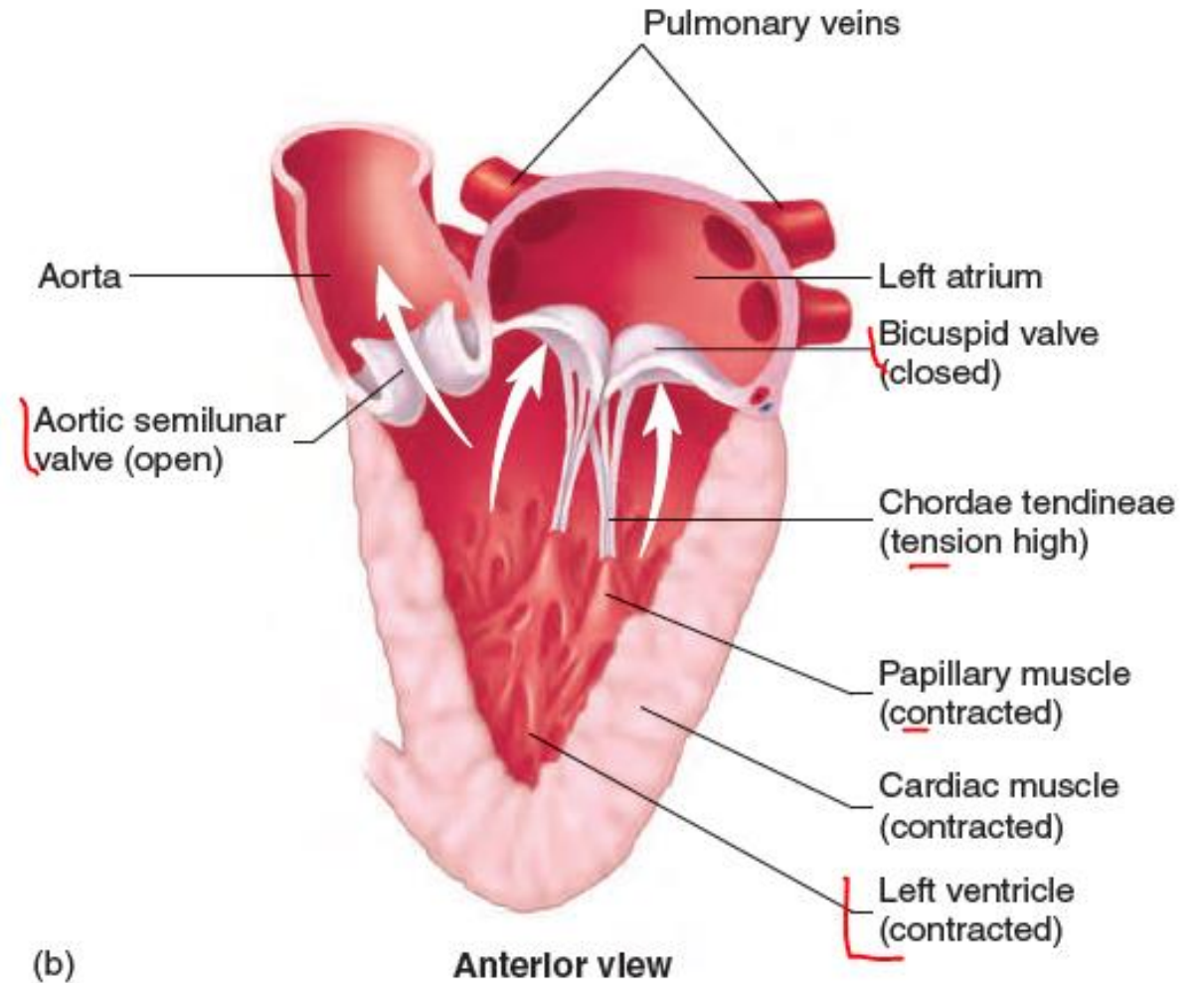
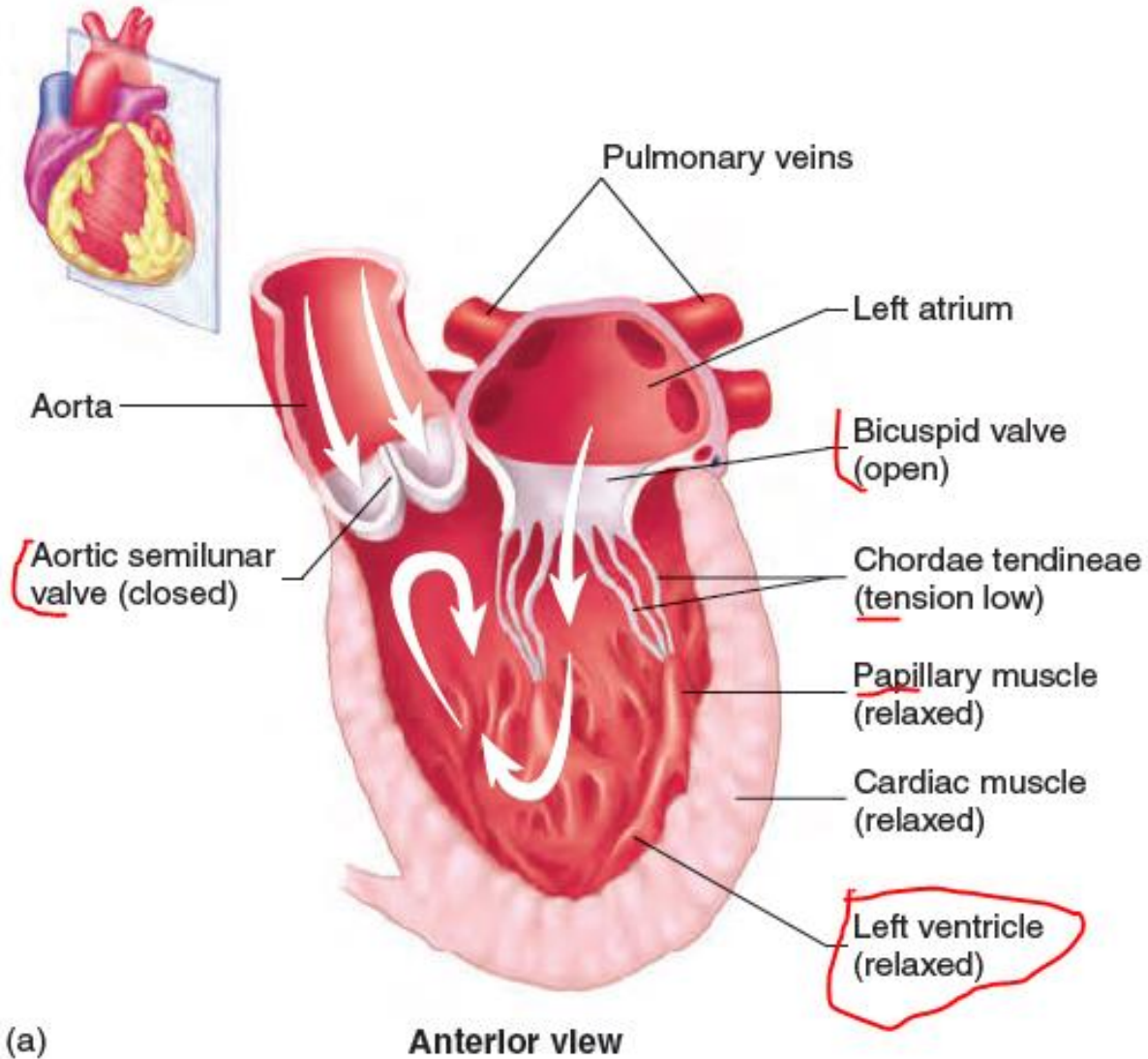
Heart Valves,,,



Atrioventricular Valves

- The opening between each atrium and its corresponding ventricle is guarded by an **atrioventricular valve** that is formed of dense irregular connective tissue.
- **These valves allow blood to flow from the atria into the ventricles but prevent it from flowing back into the atria.**
- When the ventricles relax, the higher pressure in the atria forces the AV valves to open, and blood flows from the atria into the ventricles.
- **In contrast**, when the ventricles contract, blood flows toward the atria and causes the AV valves to close.

Atrioventricular Valves,,,



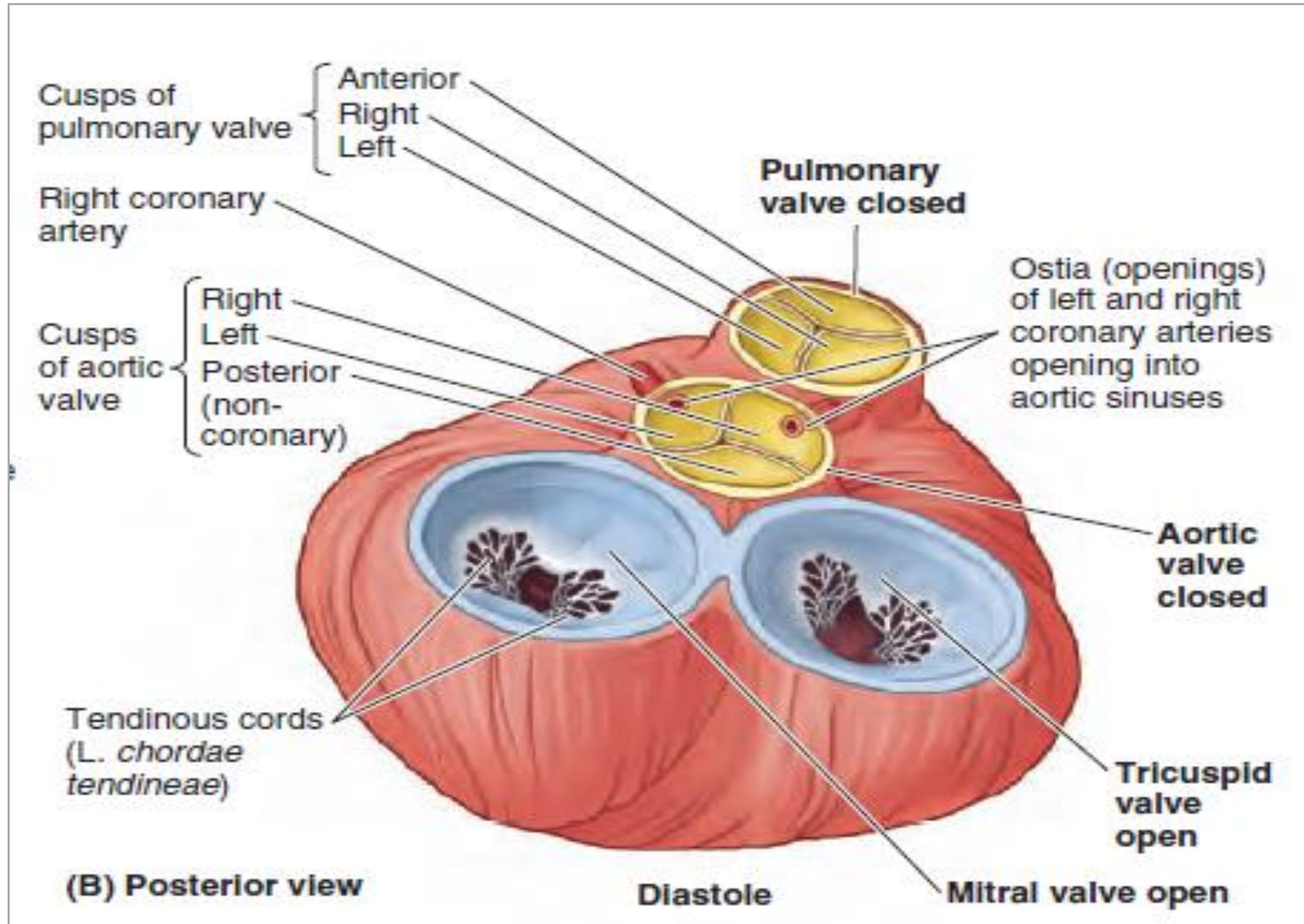
Atrioventricular Valves,,,

- ❖ **When the ventricles relax**, the pressures in the aorta and pulmonary trunk are higher than in the ventricles.
- Blood flows back from the aorta or pulmonary trunk toward the ventricles and enters the **pockets of the cusps**, causing them to bulge toward and meet in the center of the aorta or pulmonary trunk, this blocking blood flow back into the ventricles.
- When the ventricles contract, the increasing pressure within the ventricles forces the semilunar valves to open.

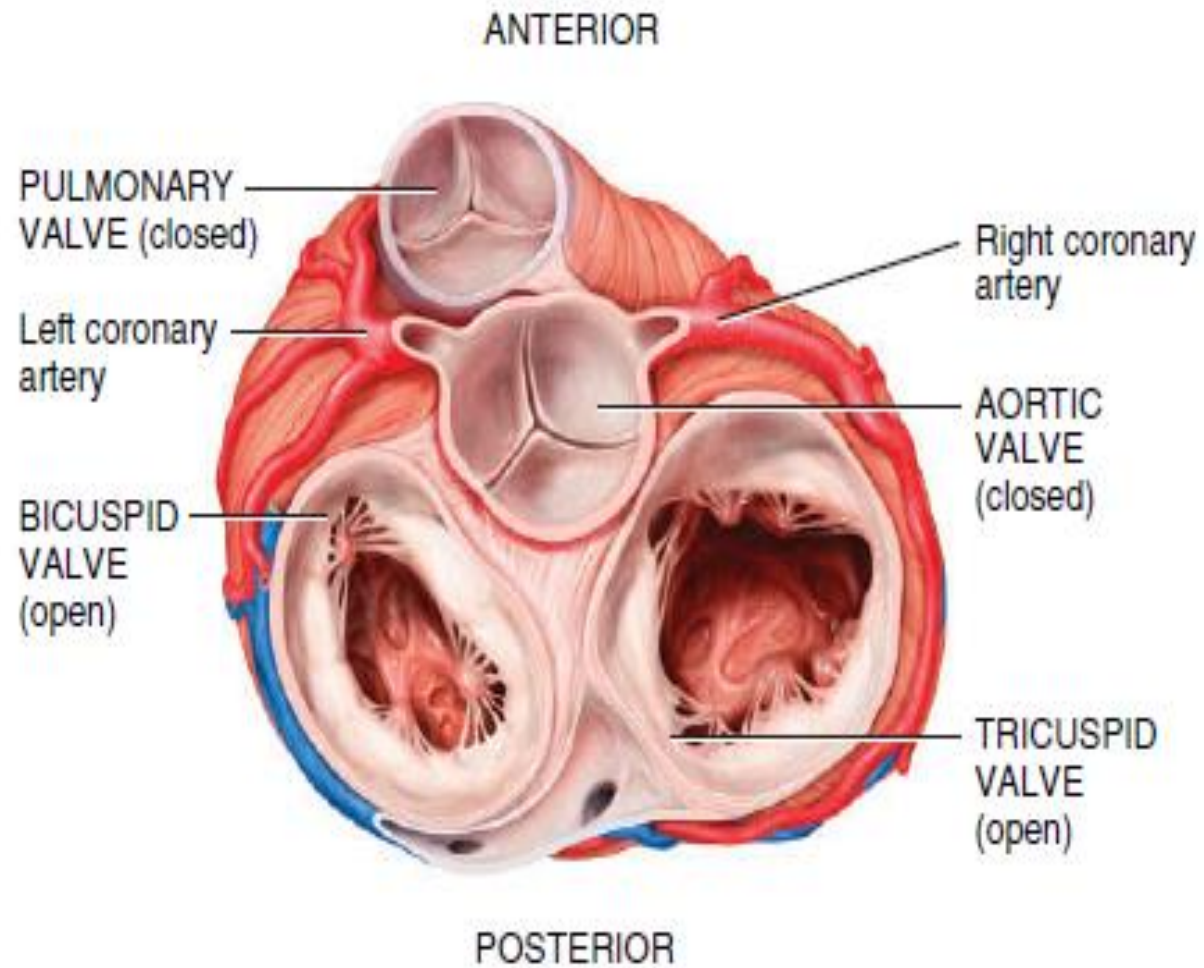
Semilunar valves

- The **semilunar valves**, have 3 pocketlike semilunar (half-moon-shaped) cusps.
- Each of 3 **semilunar cusps of the pulmonary valve** (*anterior, right & left*), is concave when viewed superiorly and the **aortic semilunar cusps**(*posterior, right & left*).
- **Semilunar cusps do not have tendinous cords to support them.**
- They are smaller in area than the cusps of the AV valves & the force exerted on them is less than half that exerted on the cusps of the **tricuspid & mitral** valves.
- The cusps project into the artery but are pressed toward (and not against) its walls as blood leaves the ventricle.

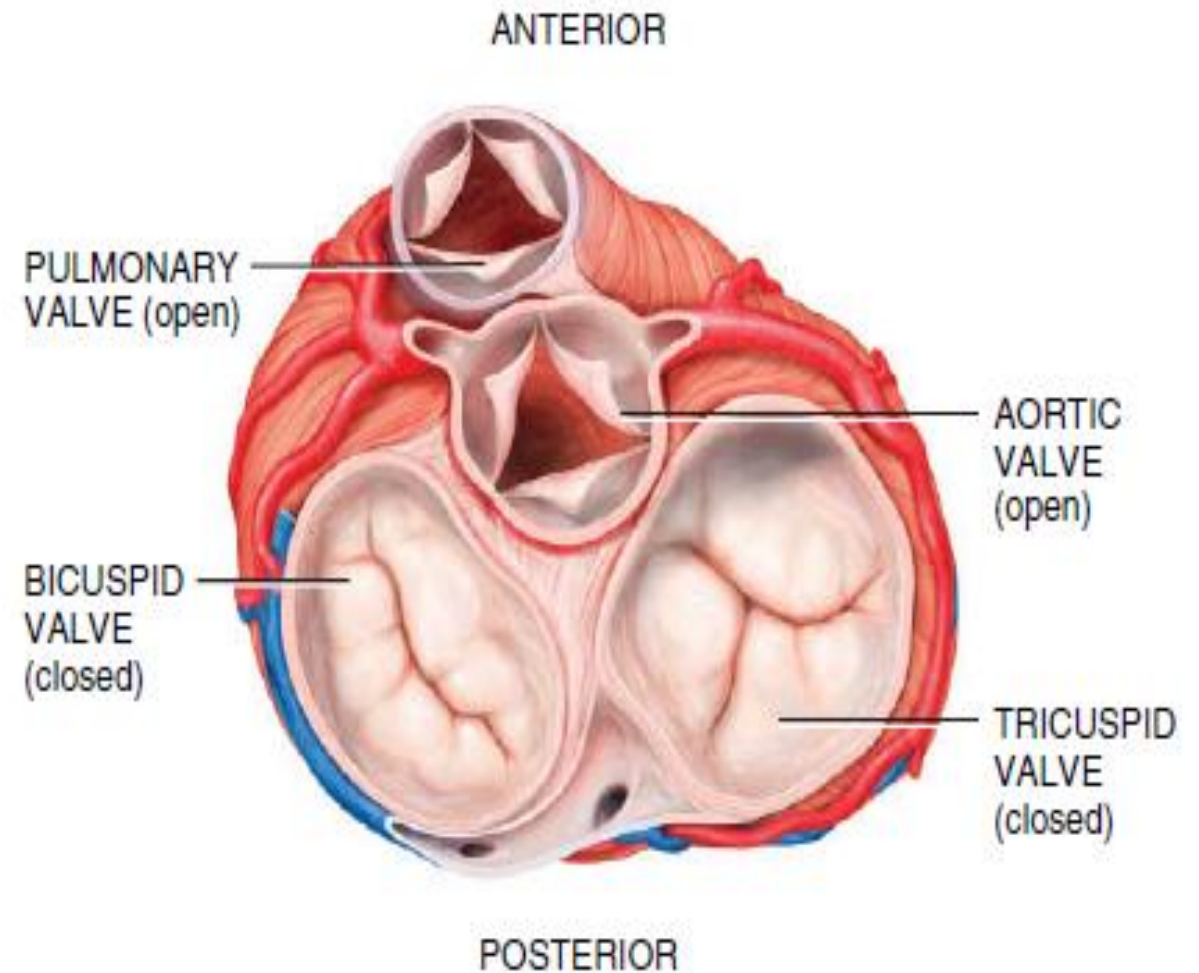
Semilunar valves,,,



Semilunar valves,,



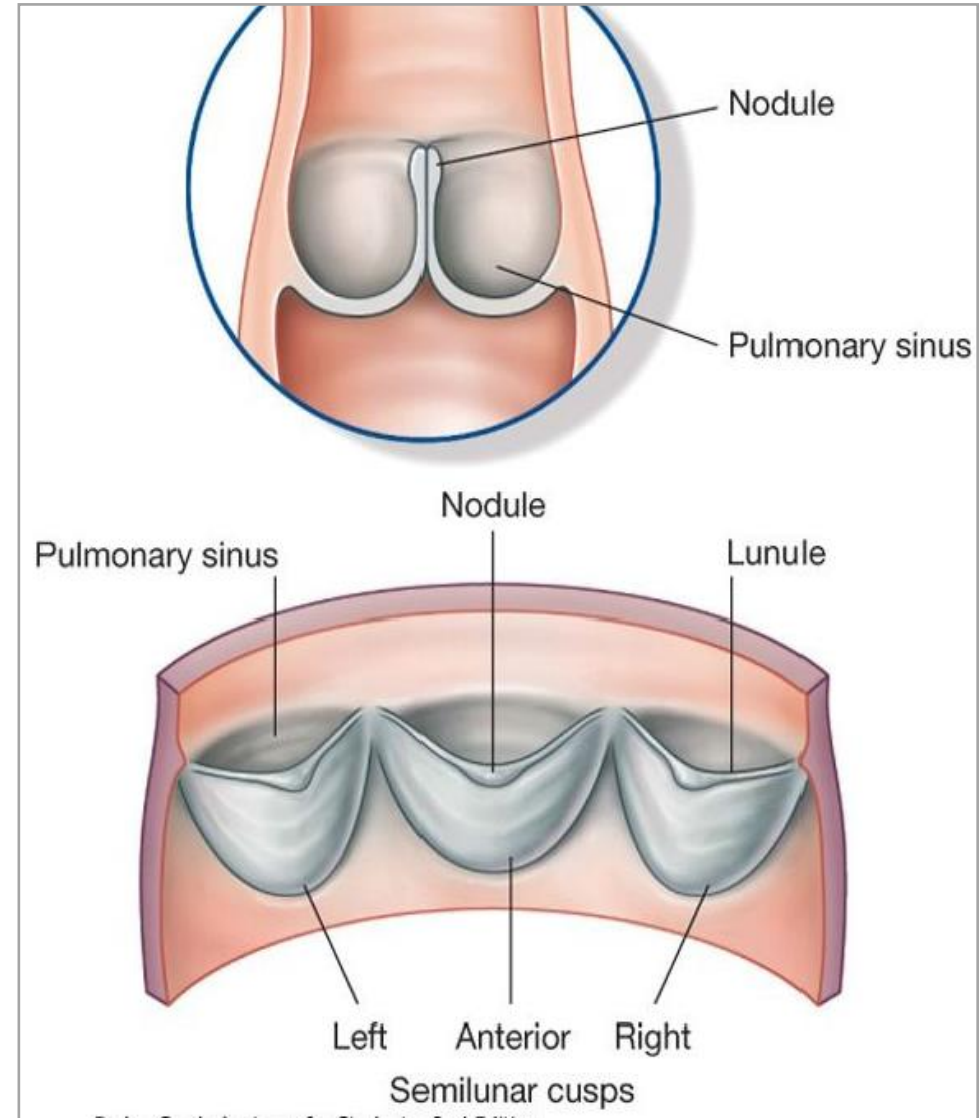
(d) Superior view with atria removed: pulmonary and aortic valves closed, bicuspid and tricuspid valves open

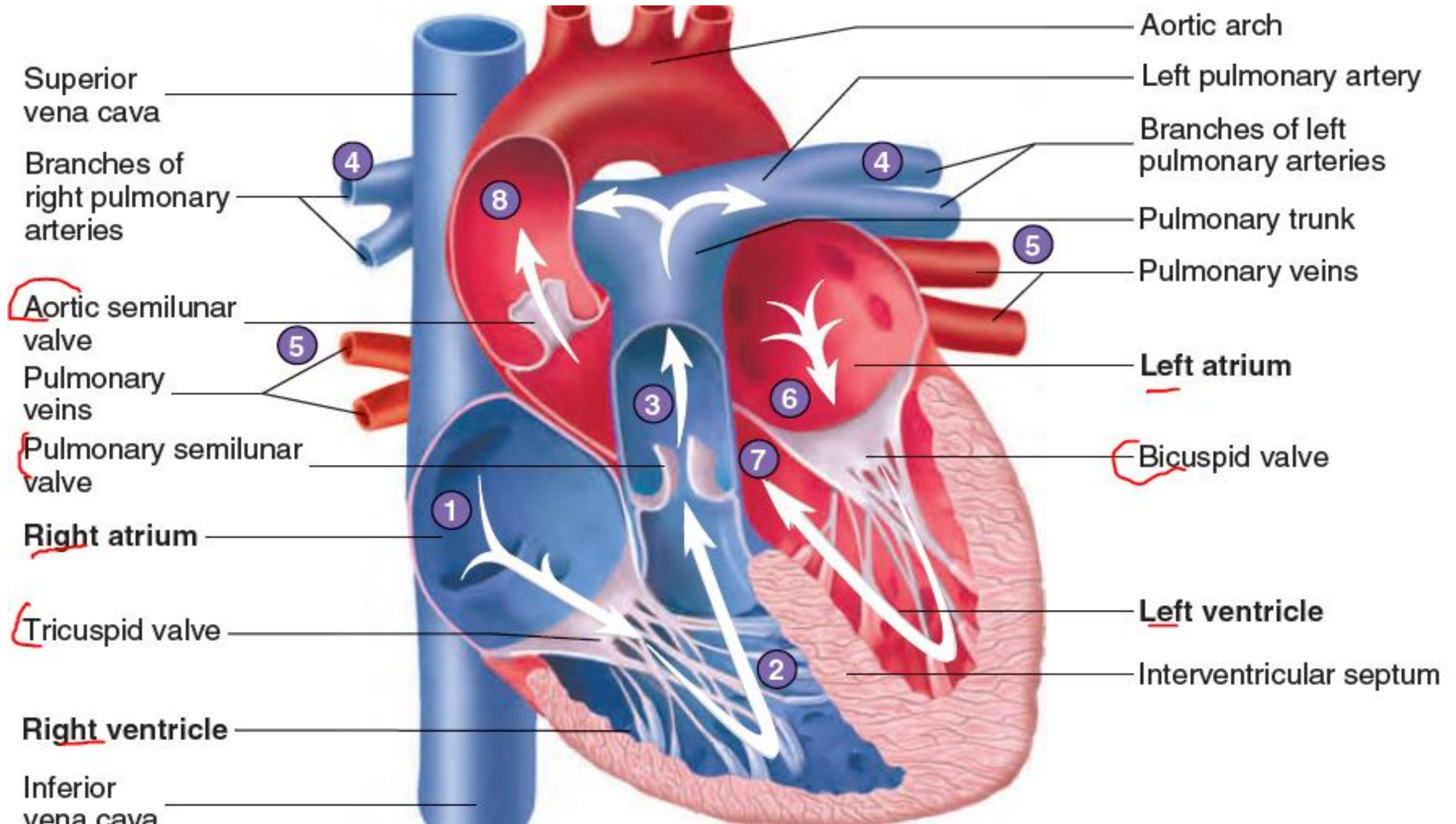


(e) Superior view with atria removed: pulmonary and aortic valves open, bicuspid and tricuspid valves closed

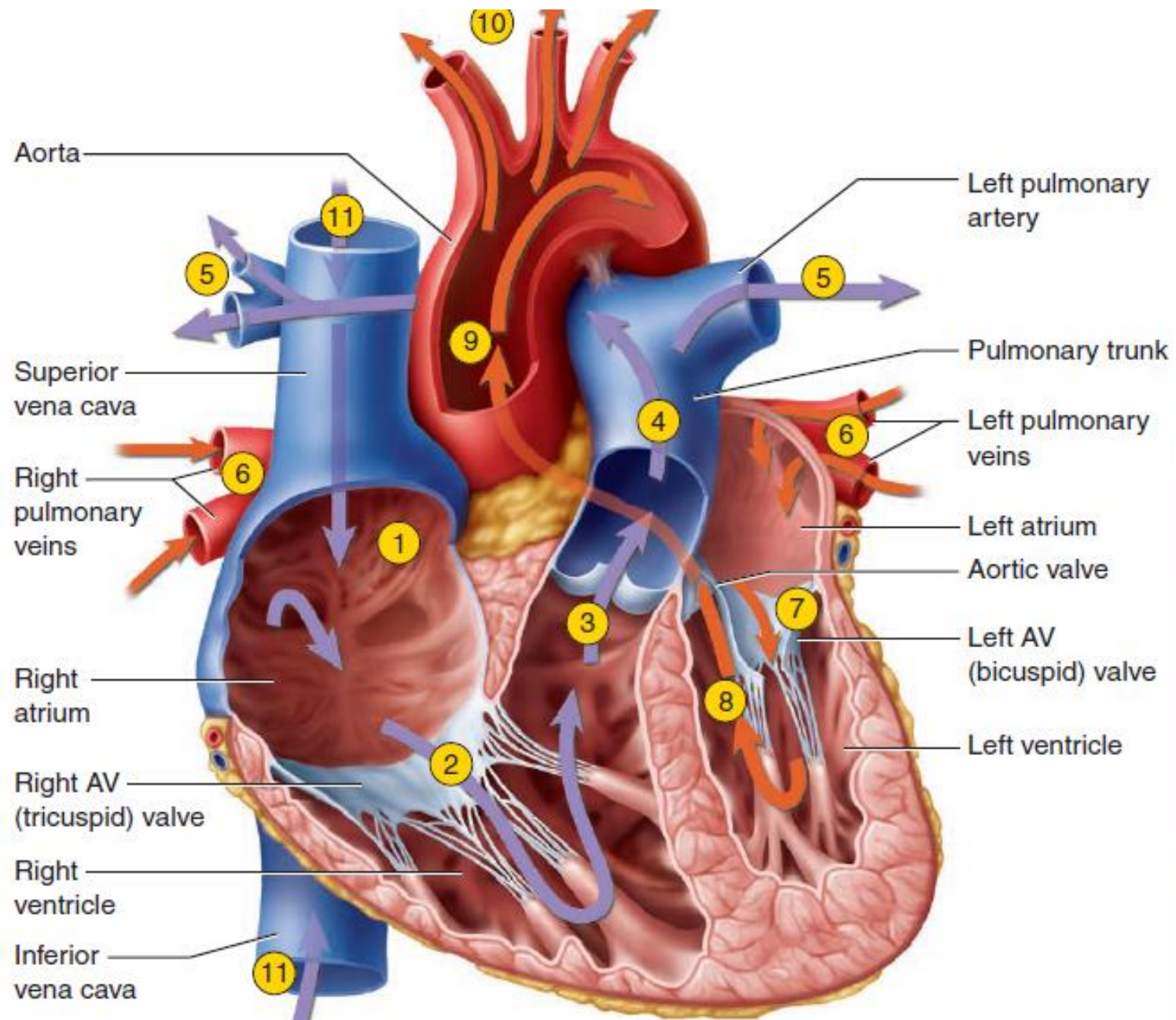
Semilunar valves,,,

- Immediately superior to each semilunar cusp, the walls of the **origins of the pulmonary trunk and aorta** are **slightly dilated, forming a sinus**.
- The **aortic sinuses** and **sinuses of the pulmonary trunk** (pulmonary sinuses) are the spaces at the origin of the pulmonary trunk & ascending aorta.





Route of Blood Flow Through the Heart



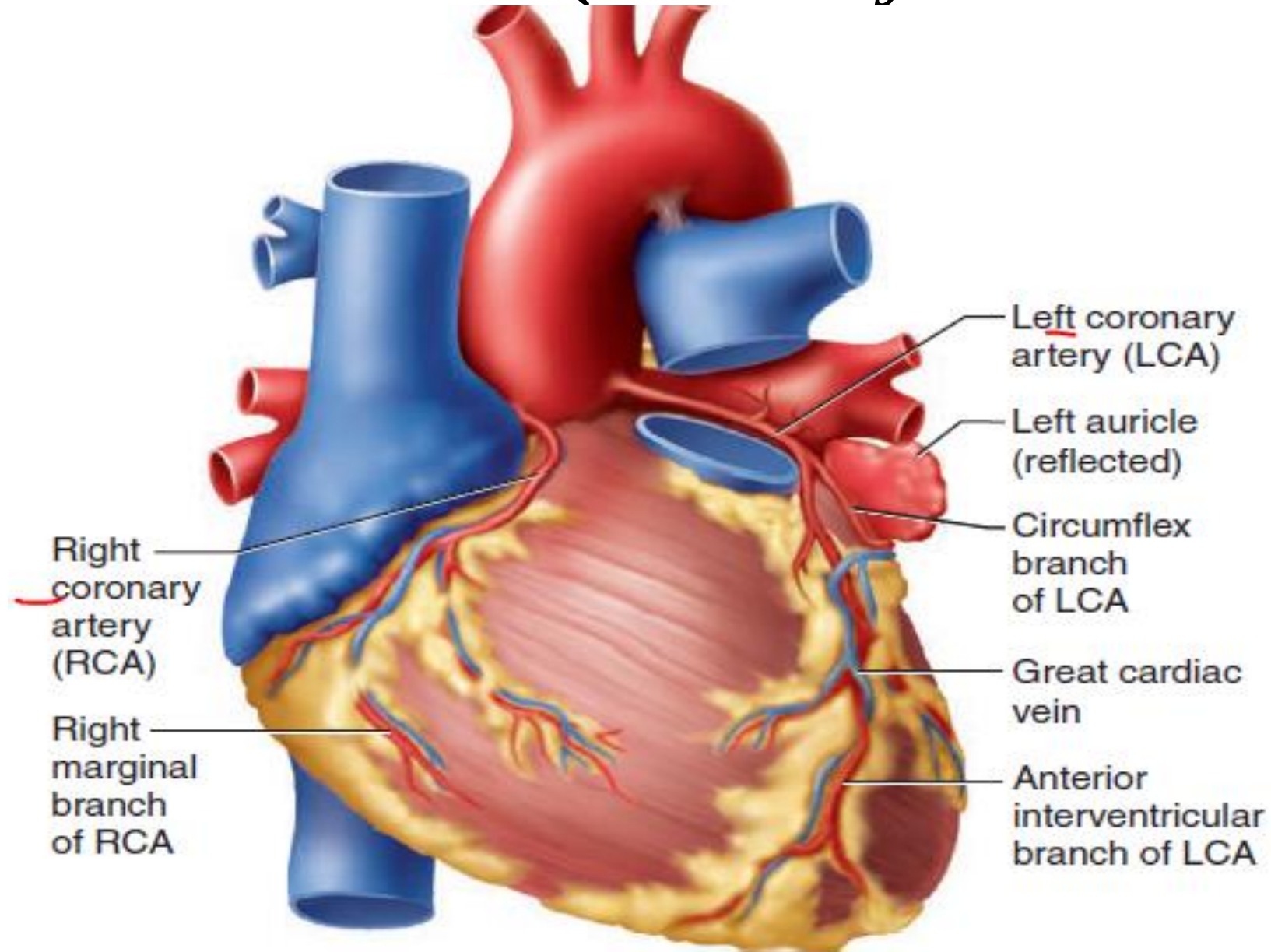
- ① Blood enters right atrium from superior and inferior venae cavae.
- ② Blood in right atrium flows through right AV valve into right ventricle.
- ③ Contraction of right ventricle forces pulmonary valve open.
- ④ Blood flows through pulmonary valve into pulmonary trunk.
- ⑤ Blood is distributed by right and left pulmonary arteries to the lungs, where it unloads CO_2 and loads O_2 .
- ⑥ Blood returns from lungs via pulmonary veins to left atrium.
- ⑦ Blood in left atrium flows through left AV valve into left ventricle.
- ⑧ Contraction of left ventricle (simultaneous with step ③) forces aortic valve open.
- ⑨ Blood flows through aortic valve into ascending aorta.
- ⑩ Blood in aorta is distributed to every organ in the body, where it unloads O_2 and loads CO_2 .
- ⑪ Blood returns to heart via venae cavae.

Route of Blood Flow Through the Heart

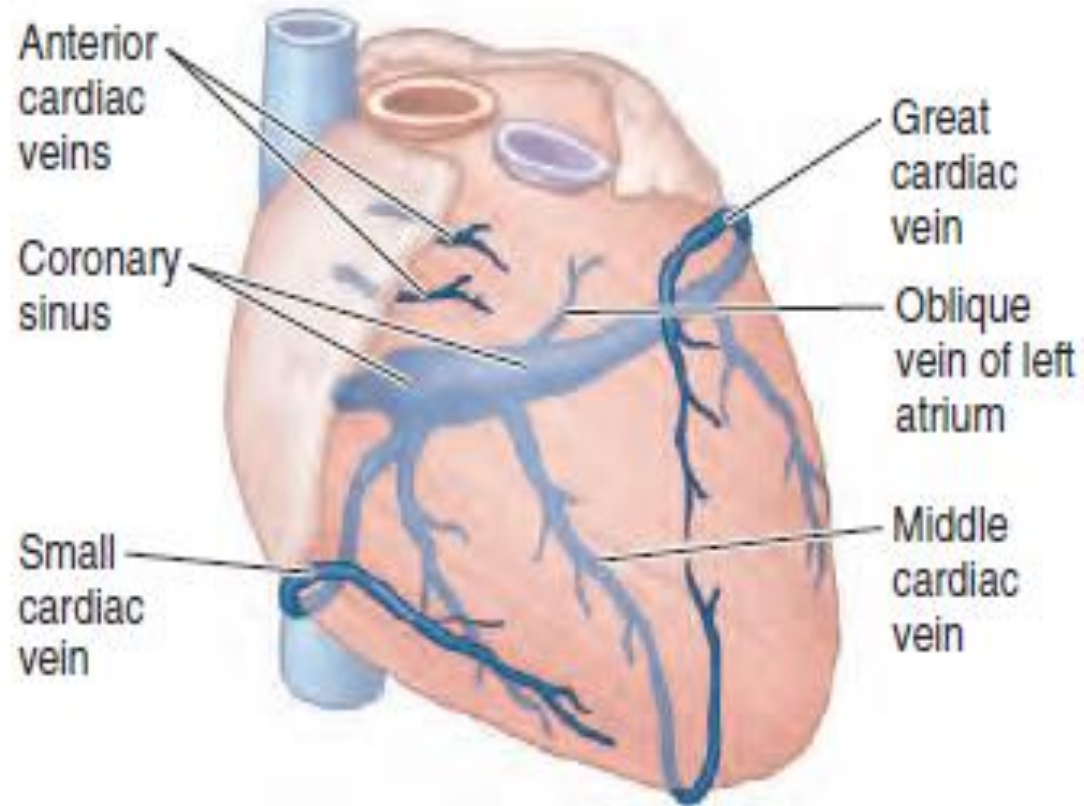
Vasculature of heart(*Coronary vasculature*)

- The cardiac muscle in the wall of the heart is thick and metabolically very active.
- The heart requires a constant supply of blood to nourish its own tissues
- The blood vessels of the heart comprise the **coronary arteries & cardiac veins**.
- **The right and left coronary** arteries are the first branches of the aorta, arising from the right and left aortic sinuses.
- **The two arteries branch into smaller arteries and arterioles, then to capillaries.**
- The coronary capillaries merge to form coronary veins, which empty blood into a large coronary sinus that returns blood to the right atrium.
- Blockage of a coronary artery may result in a heart attack.

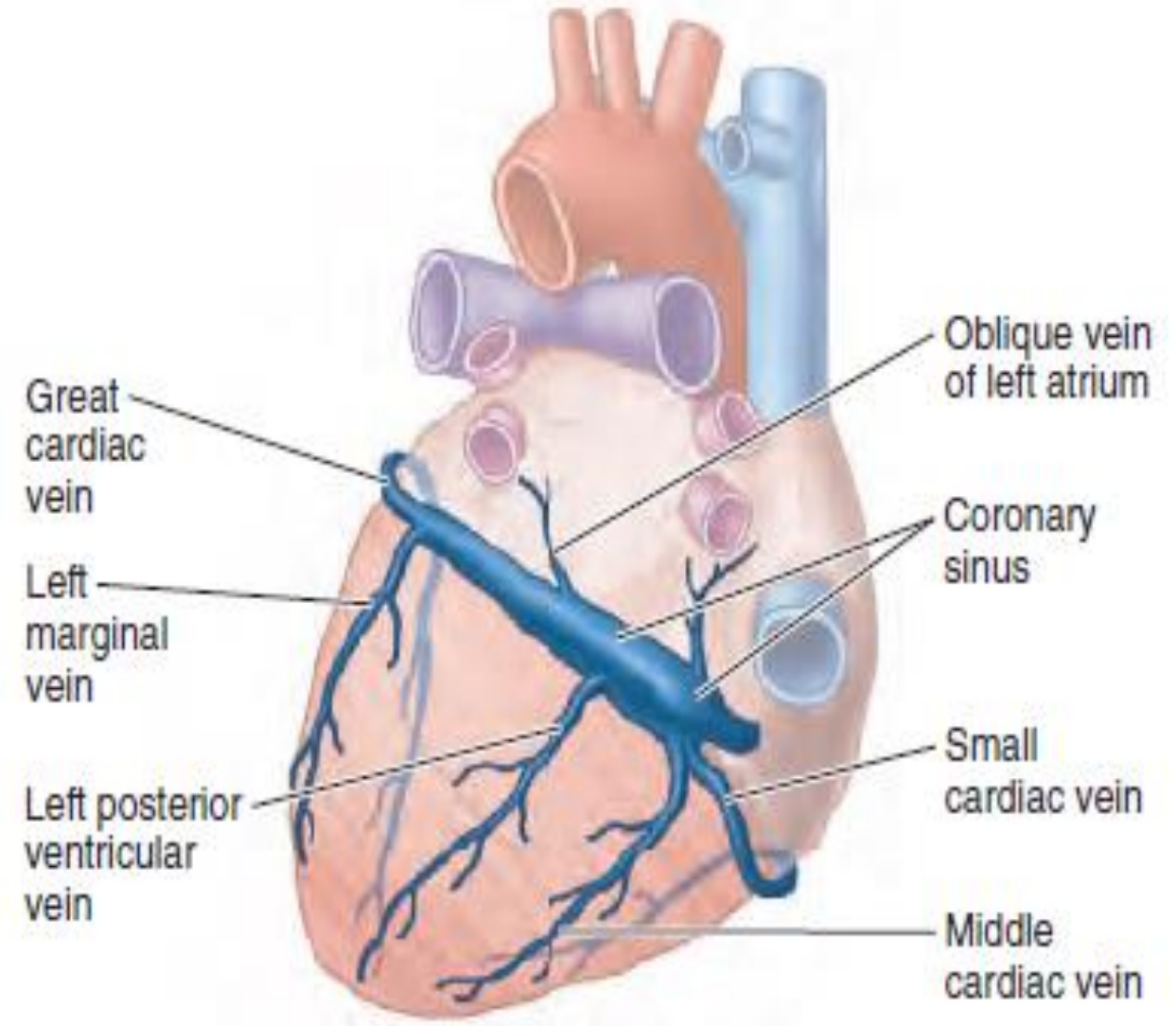
Vasculature of heart(*Coronary vasculature*),,,



Vasculature of heart(*Coronary vasculature*),,,



Anterior view

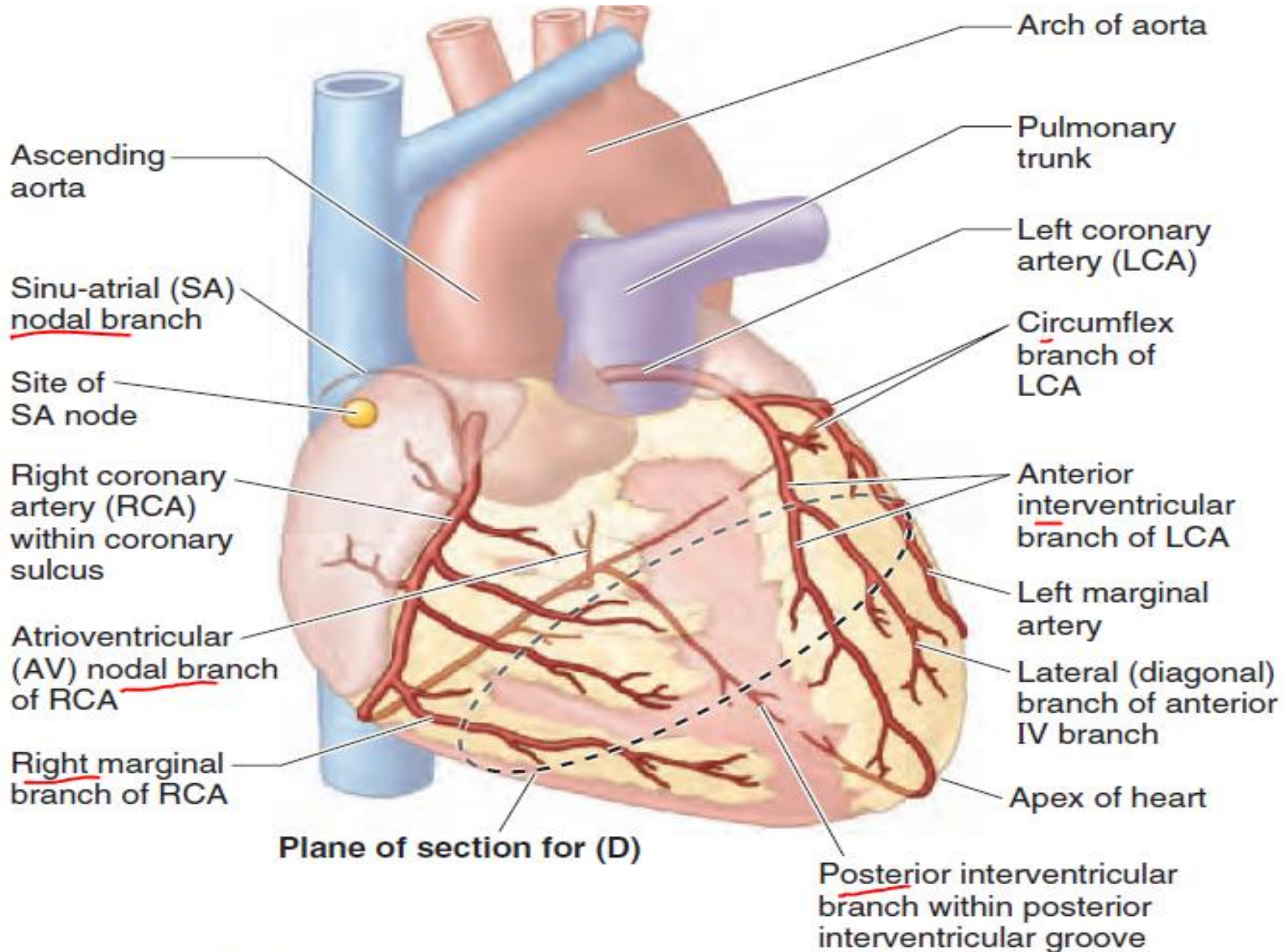


Postero-inferior view

The right coronary artery (RCA)

- The **right coronary artery (RCA)** arises from the right **aortic sinus** of the ascending aorta and passes to the right side of the pulmonary trunk, running in the coronary sulcus.
- Near its origin, the RCA usually gives off an ascending **sinu-atrial nodal branch** **then** RCA descends in the coronary sulcus & gives off the **Rt marginal branch**.
- After giving off this branch, the RCA turns to the Lt & continues in the coronary sulcus to the posterior aspect of the heart.

The right coronary artery (RCA),,,



(A) Anterior view

The right coronary artery (RCA),,,

- At the posterior aspect of the crux of the heart the junction of the inter atrial & IV septa b/n the 4 heart chambers the RCA gives rise to the **atrioventricular nodal branch**, which supplies the *AV node*.
- **Dominance of the coronary** arterial system is defined by which artery gives rise to the posterior IV branch (posterior descending artery).
- Dominance of the RCA is approximately 67%; the RCA gives rise to the large **posterior IV branch** w/c supplies adjacent areas of both ventricles & sends perforating **IV septal branches** into the IV septum.

The right coronary artery (RCA),,,

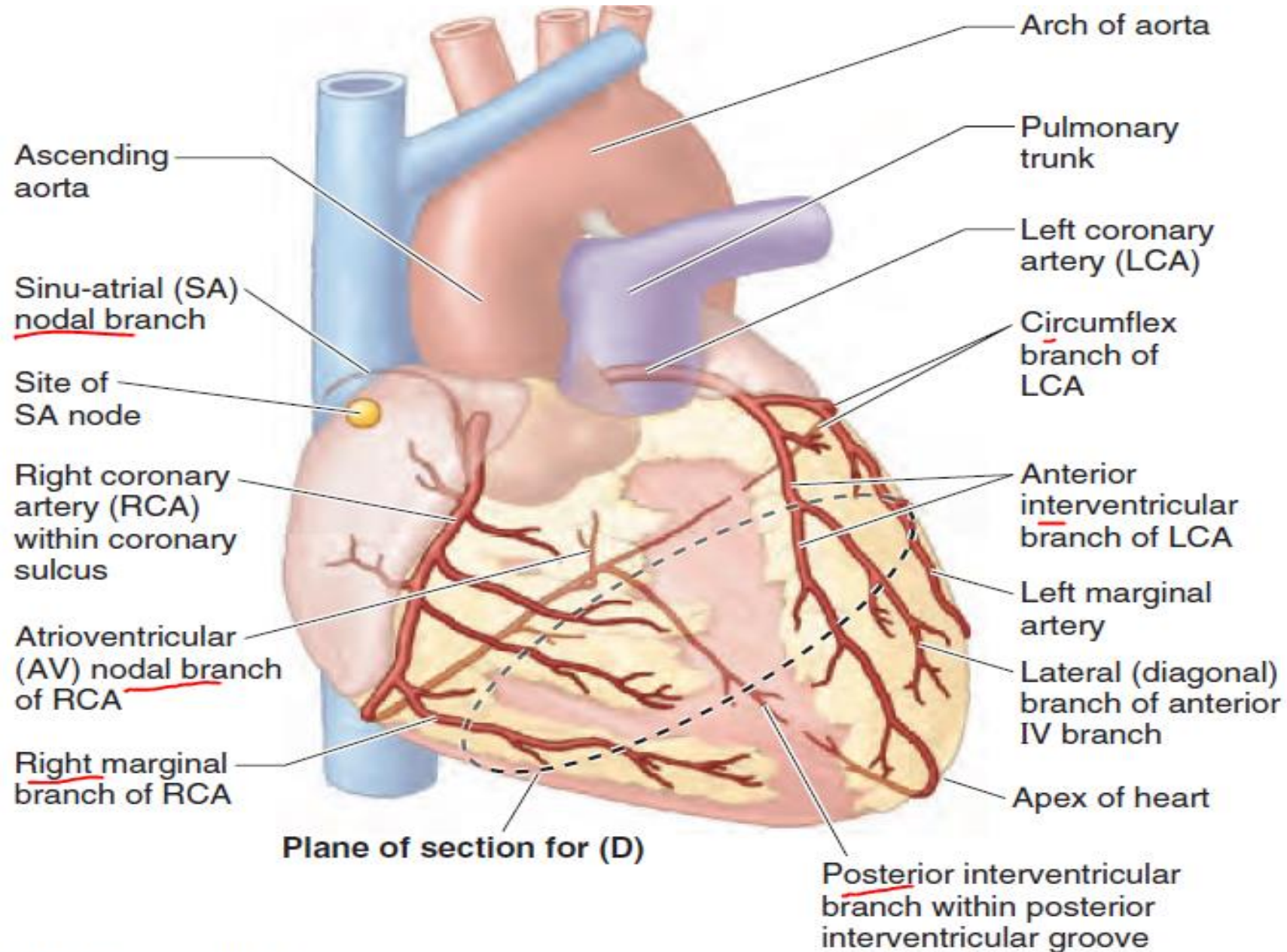
❖ Typically, *the RCA supplies* :

- ✓ The right atrium.
- ✓ Most of right ventricle.
- ✓ Part of the left ventricle (the diaphragmatic part).
- ✓ Part of the IV septum, usually the posterior 3rd .
- ✓ The SA node (in approximately 60% of people).
- ✓ The AV node (in approximately 80% of people)

The **left coronary artery (LCA)**

- The **left coronary artery(LCA)** arises from the *left aortic sinus* of the ascending aorta & passes b/n the Lt auricle & the Lt side of the pulmonary trunk before entering the coronary sulcus.
- ❖ **As it enters the coronary sulcus**, at the superior end of the anterior IV groove, the LCA divides into 2 branches, the *anterior IV branch* & the *circumflex branch*, **anterior IV branch** passes along the IV groove to the apex of the heart
- In approximately 40% of people, the **SA nodal branch** arises from the circumflex branch of the LCA.

The left coronary artery (LCA),,,



(A) Anterior view

The **left coronary artery (LCA)**,,,

- Around the inferior border of the heart it anastomoses with the posterior IV branch of the right coronary artery .
- **The anterior IV branch supplies adjacent** parts of both ventricles and, via IV septal branches, the anterior 2/3rd of the IVS.
- In many people, the anterior IV branch gives rise to a **lateral branch** (diagonal artery), which descends on the anterior surface of the heart
- The **left marginal** branch of the circumflex branch follows the left margin of the heart and supplies the left ventricle

The left coronary artery (LCA),,,

❖ Typically, the *LCA* supplies:

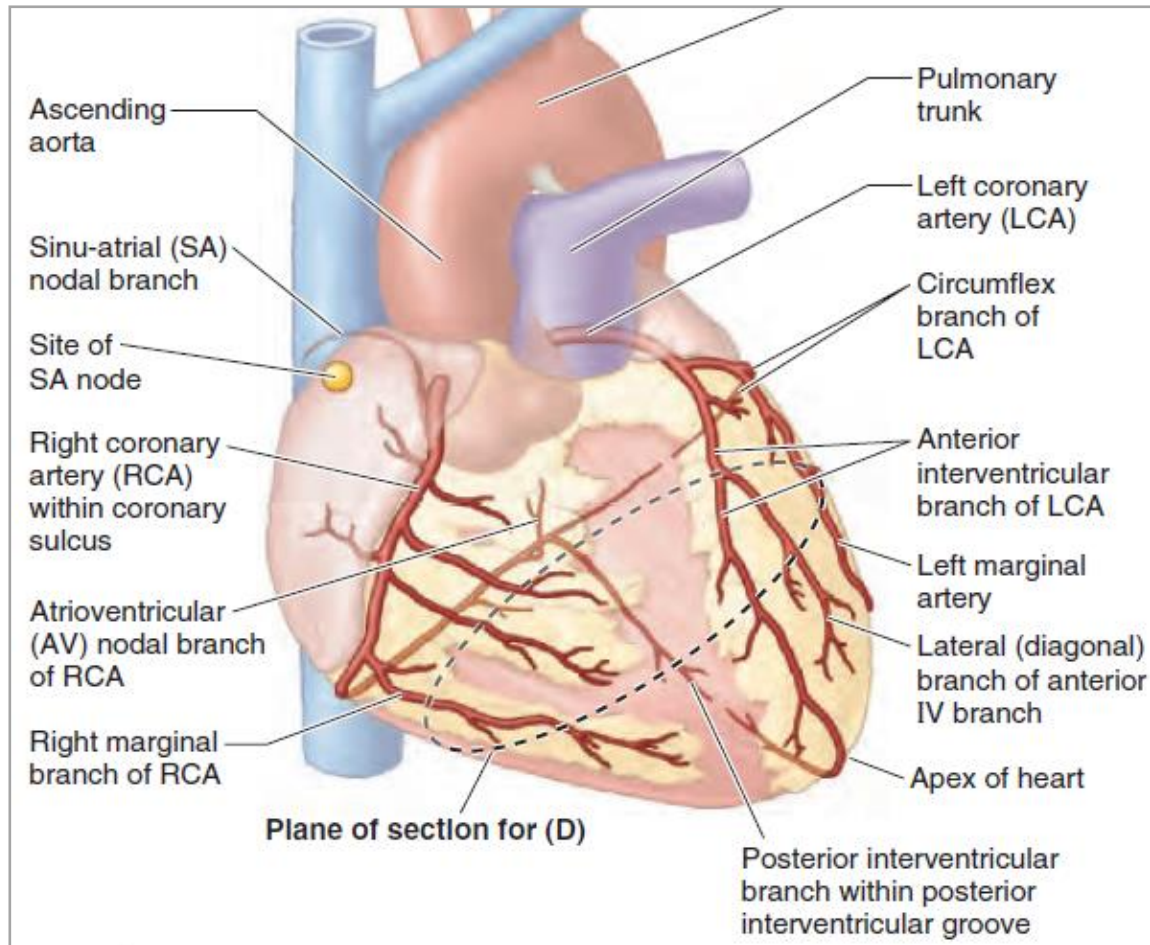
- ✓ The left atrium.
- ✓ Most of the left ventricle.
- ✓ Part of the right ventricle.
- ✓ Most of the IVS (usually its anterior two thirds), including the AV bundle of the conducting system of the heart, through its perforating IV septal branches.
- ✓ The SA node (in approximately 40% of people

Variations of the coronary arteries

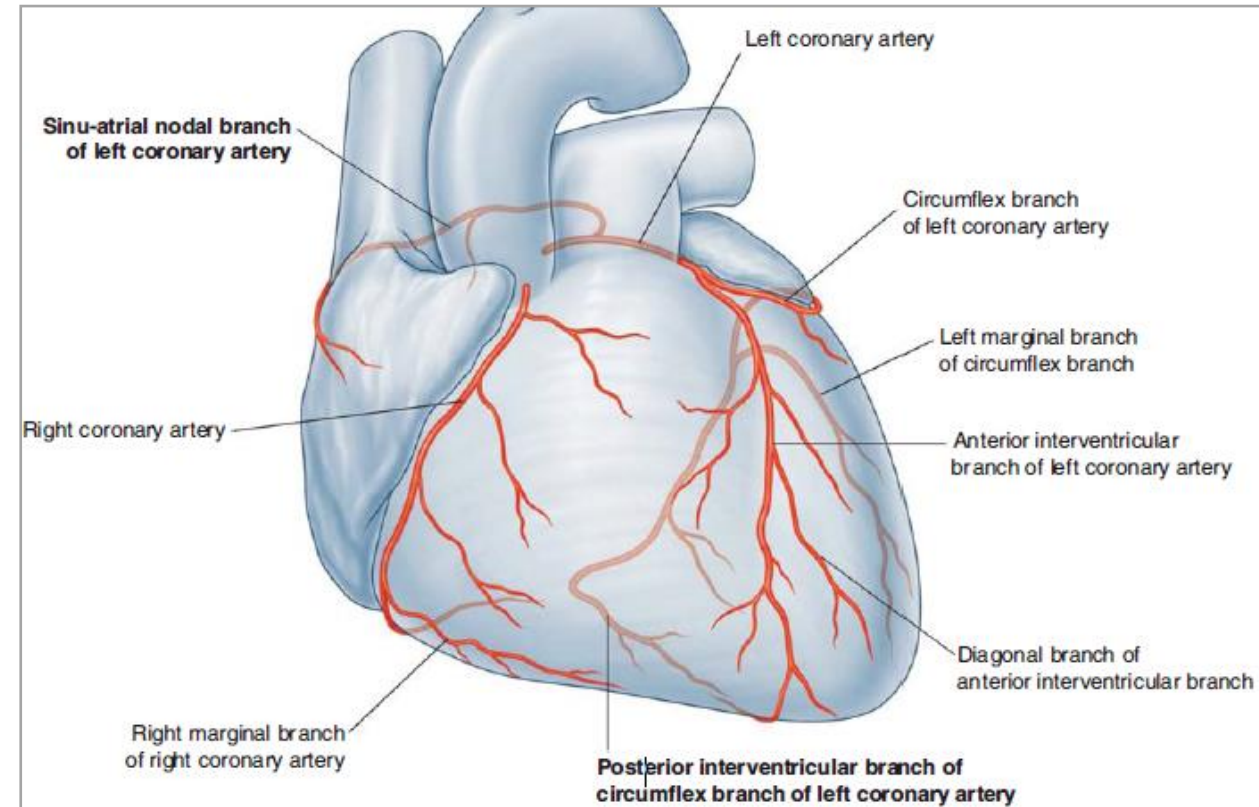
- Variations in the branching patterns and distribution of the coronary arteries are common.
- In the most common *Rt dominant pattern*, present in approximately 67% of people.
- In approximately 15% of hearts, the LCA is dominant.
- **There is codominance** in approximately 18% of people, in which branches of both the Rt & Lt CA reach the crux of the heart & give rise to branches that course in the posterior IV groove.
- In other people, the circumflex branch arises from the right aortic sinus .
- **Approximately 4% of people have an accessory coronary artery.**

Variations of the coronary arteries,,,

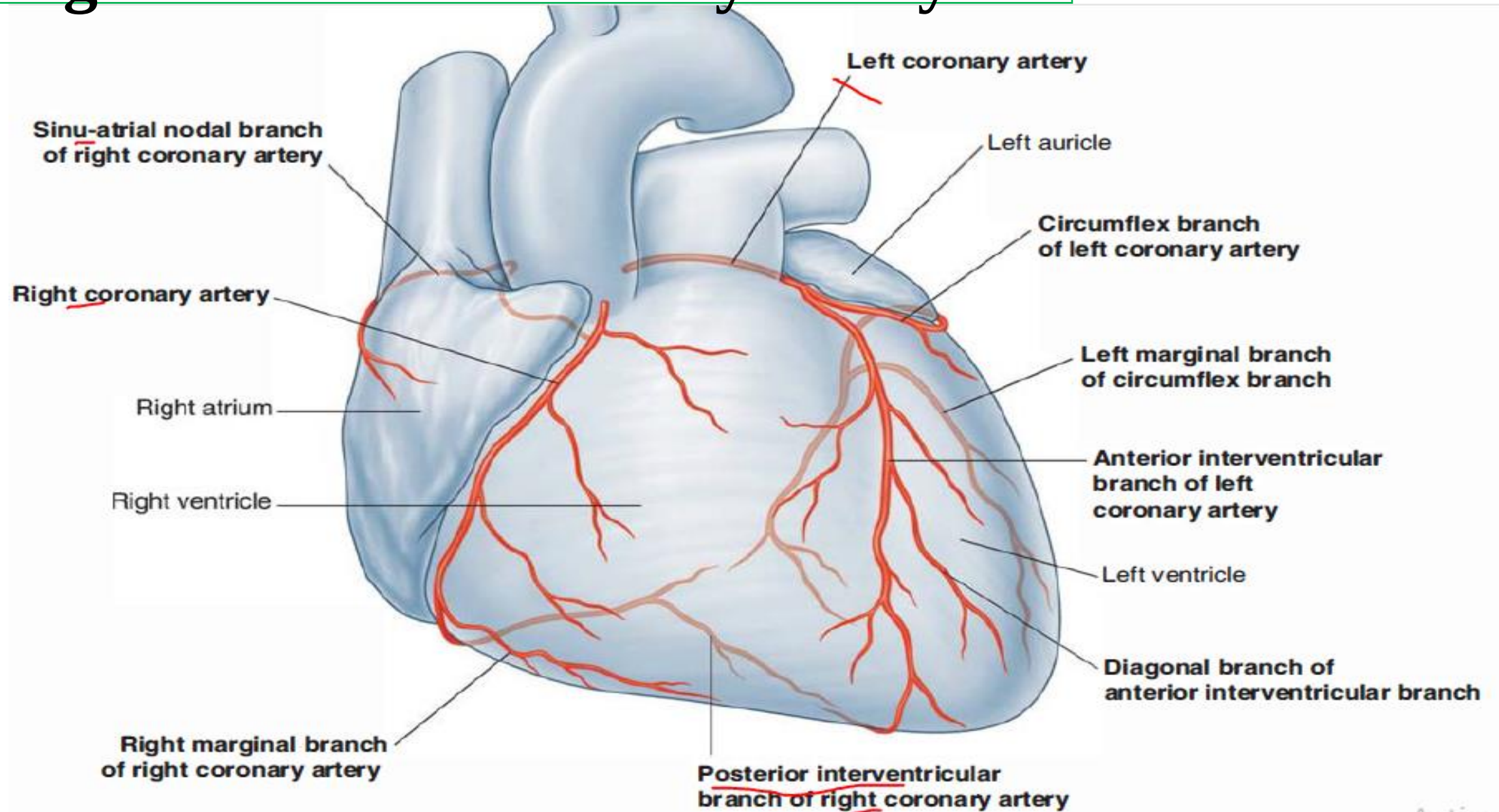
Right dominant coronary artery



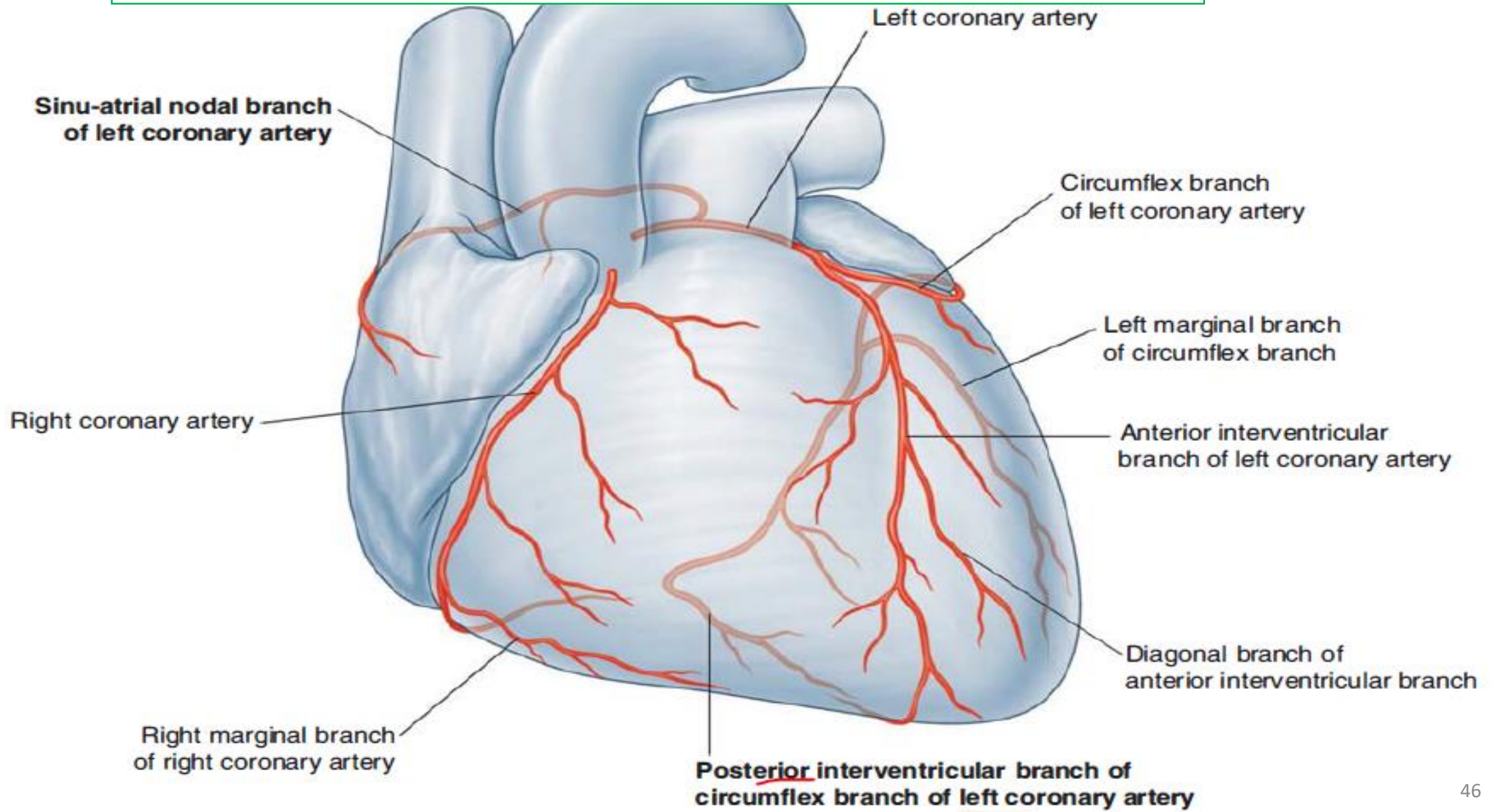
Left dominant coronary artery



□ Right dominant coronary artery



❑ Left dominant coronary artery

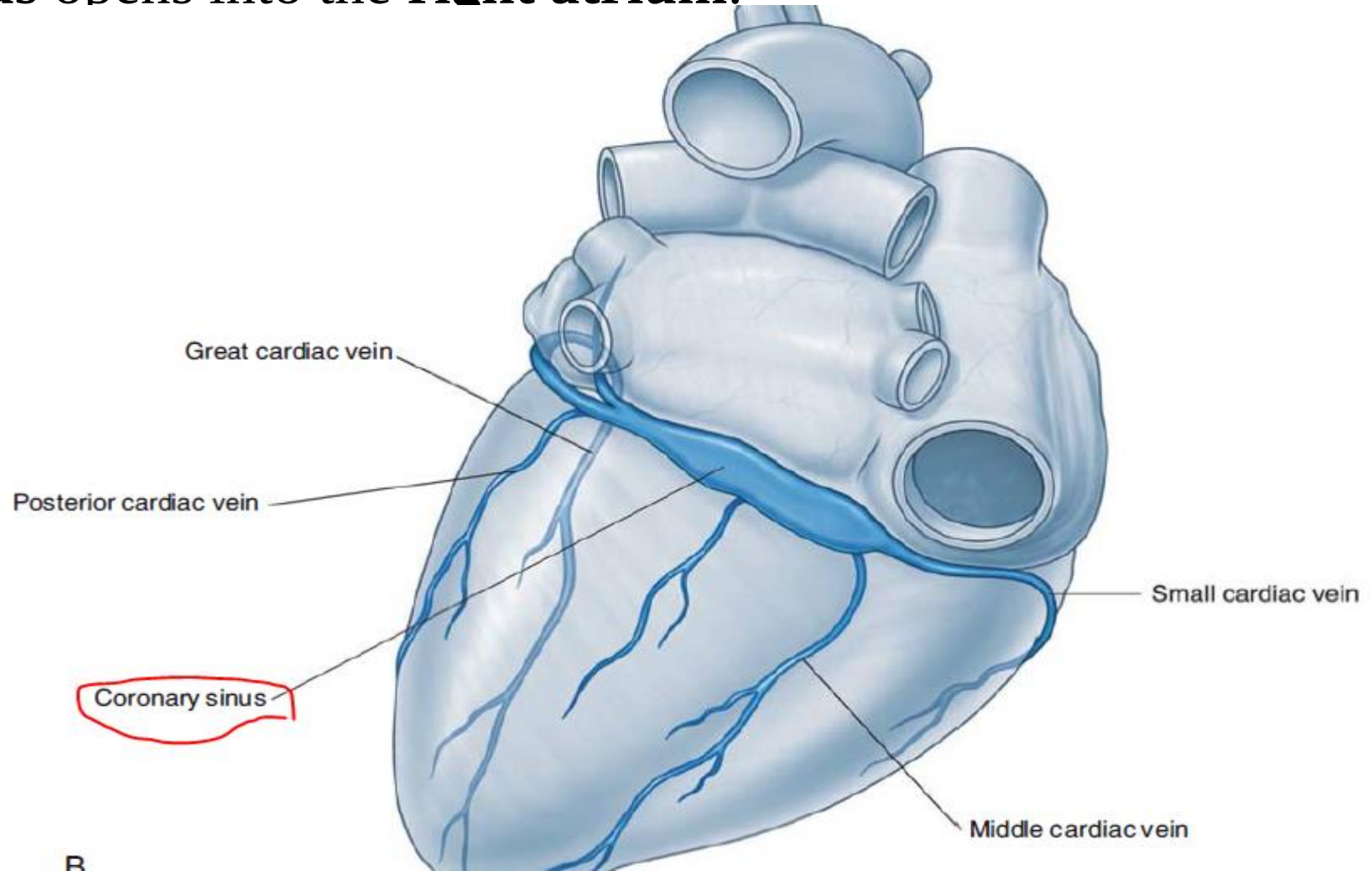


Coronary Collateral Circulation

- The branches of the coronary **arteries are generally considered** to be **functional end arteries** (arteries lacking sufficient anastomoses).
- **However, anastomoses do** exist between branches of the coronary arteries, subepicardial or myocardial, and **between** these arteries and **extracardiac vessels such as thoracic vessels.**
- **Anastomoses exist between** the terminations of the **right and the left** coronary arteries in the coronary sulcus and between the IV branches around the apex in approximately 10% of apparently normal hearts.
- **The potential for development of collateral circulation probably exists in most if not all hearts.**

Venous drainage of the heart

- About 2/3rd the heart is drained by the **coronary sinus** in the coronary sulcus, located posteriorly.
- ➔ The **coronary sinus** opens into the **right atrium**.

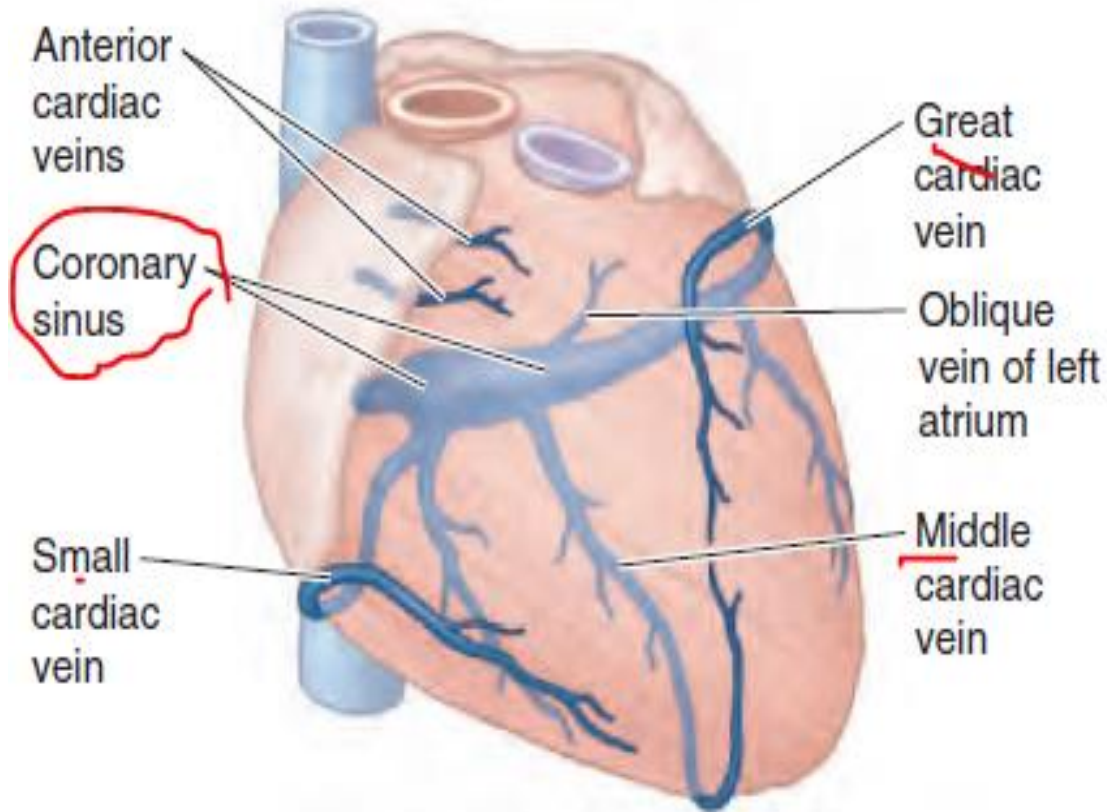


Venous drainage of the heart,,,

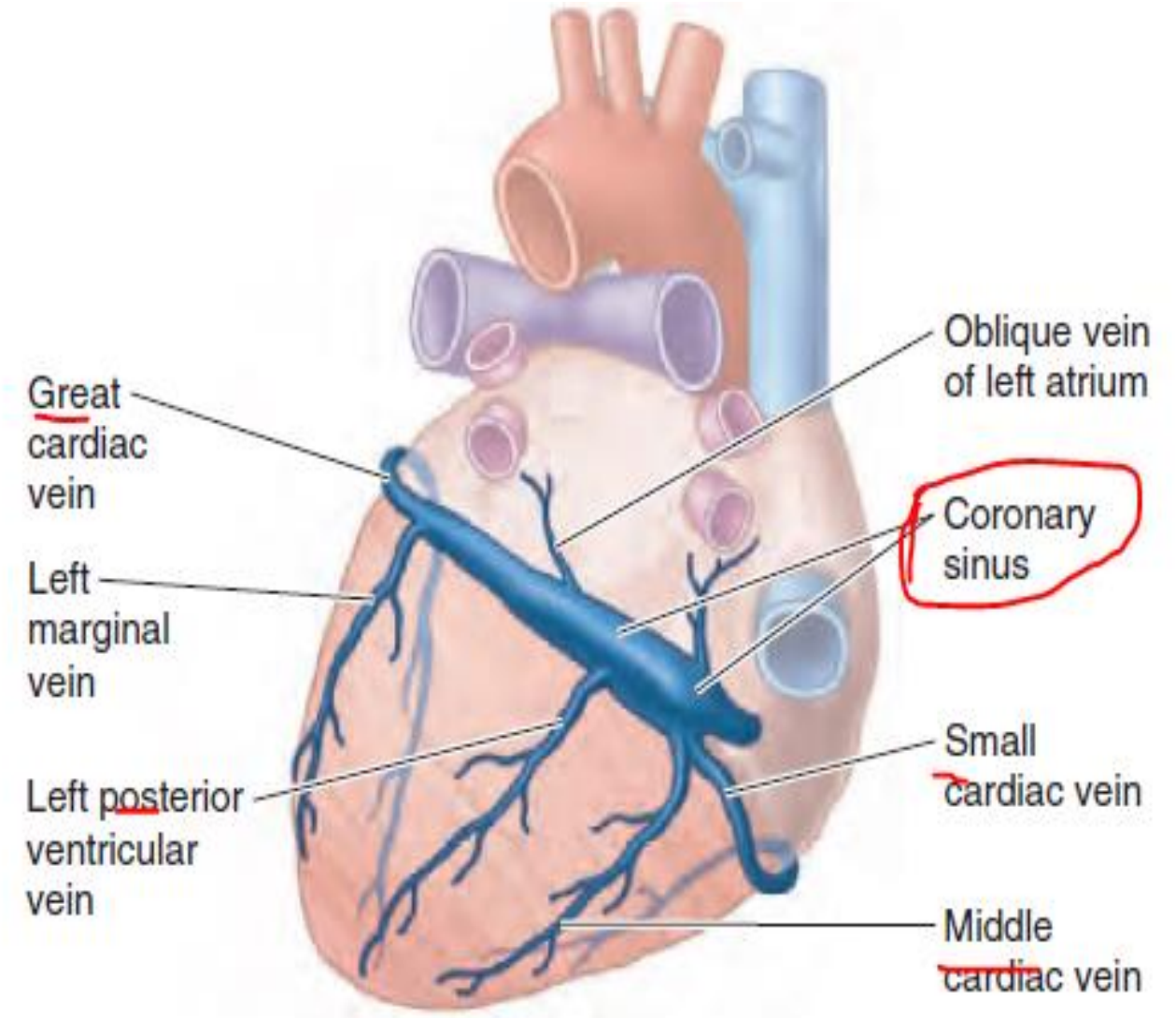
□ The **coronary sinus** receives four major tributaries:

- I. great cardiac vein* – accompanies anterior interventricular artery and drains the areas of the heart supplied by the LCA.
- II. middle cardiac vein* – accompanies posterior interventricular artery,
- III. small cardiac vein* – accompanies right marginal artery,
- IV. posterior left ventricular vein*, lies on the posterior surface of the left ventricle just to the left of the middle cardiac vein . It either enters the coronary sinus directly or joins the great cardiac vein

Venous drainage of the heart,,



Anterior view

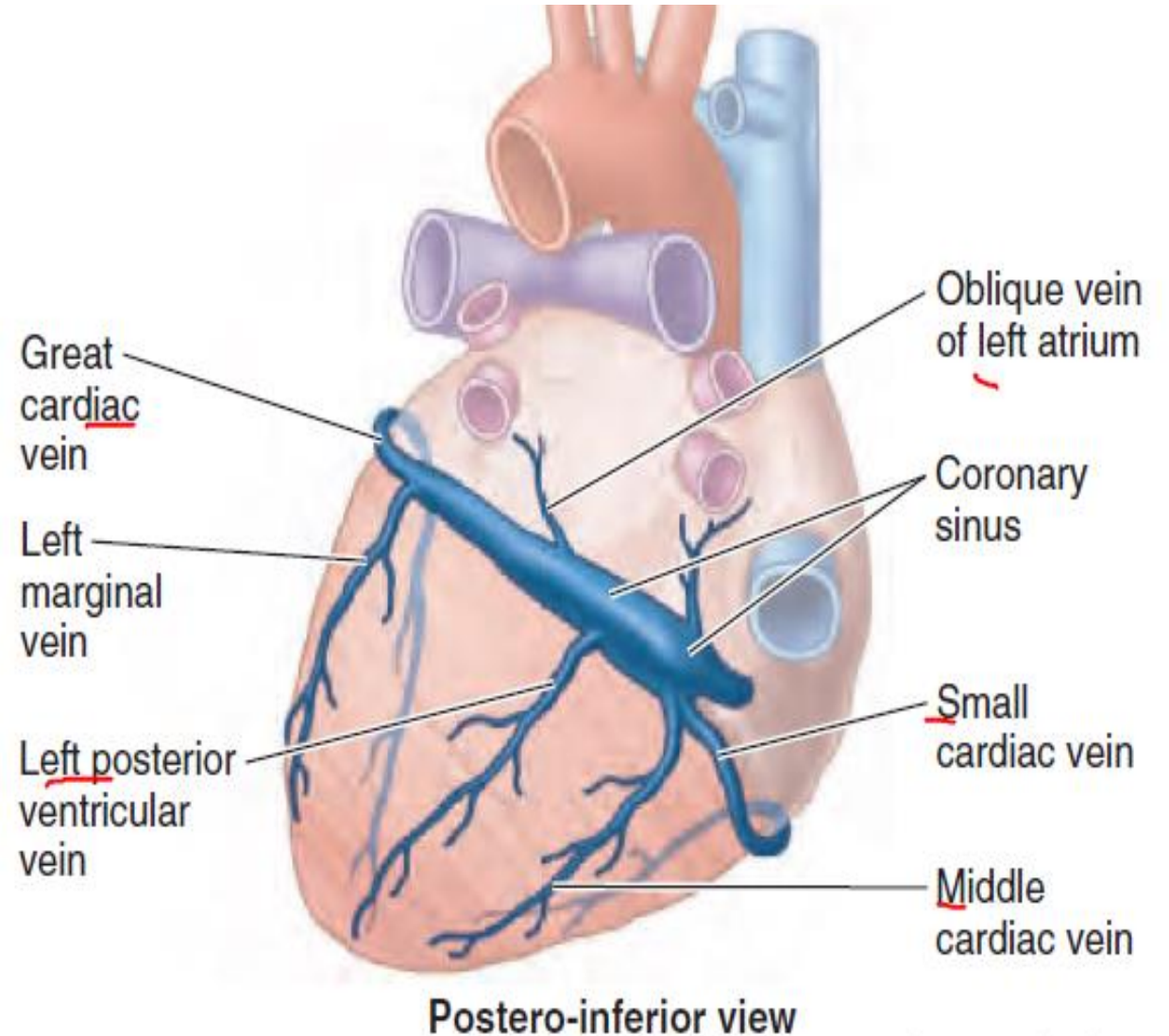
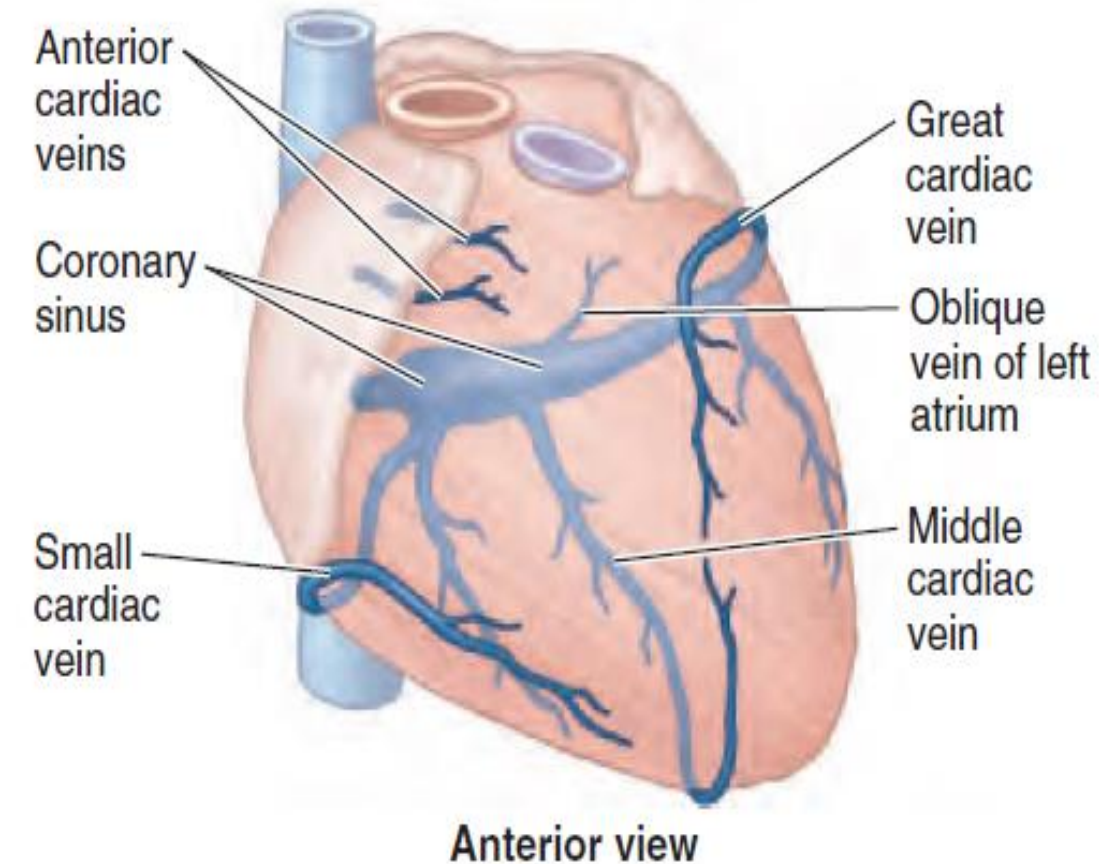


Postero-Inferior view

Venous drainage of the heart,,,

- ❑ The **oblique vein of the left atrium** (of Marshall) is a small vessel, relatively unimportant postnatally, that descends over the posterior wall of the left atrium and merges with the great cardiac vein to form the *coronary sinus* (defining the beginning of the sinus).
- The oblique vein is the remnant of the embryonic left SVC, which usually atrophies during the fetal period, but occasionally persists in adults, replacing or augmenting the right SVC.
- **About 1/2 of the blood drains** by small veins directly into the cardiac cavity, usually the atria:
 - ✓ *i) anterior cardiac vein --- drain into right atrium,*
 - ✓ *ii) venae cordis minimae (Thebesian veins) --- drain into all chambers, mainly atria.*

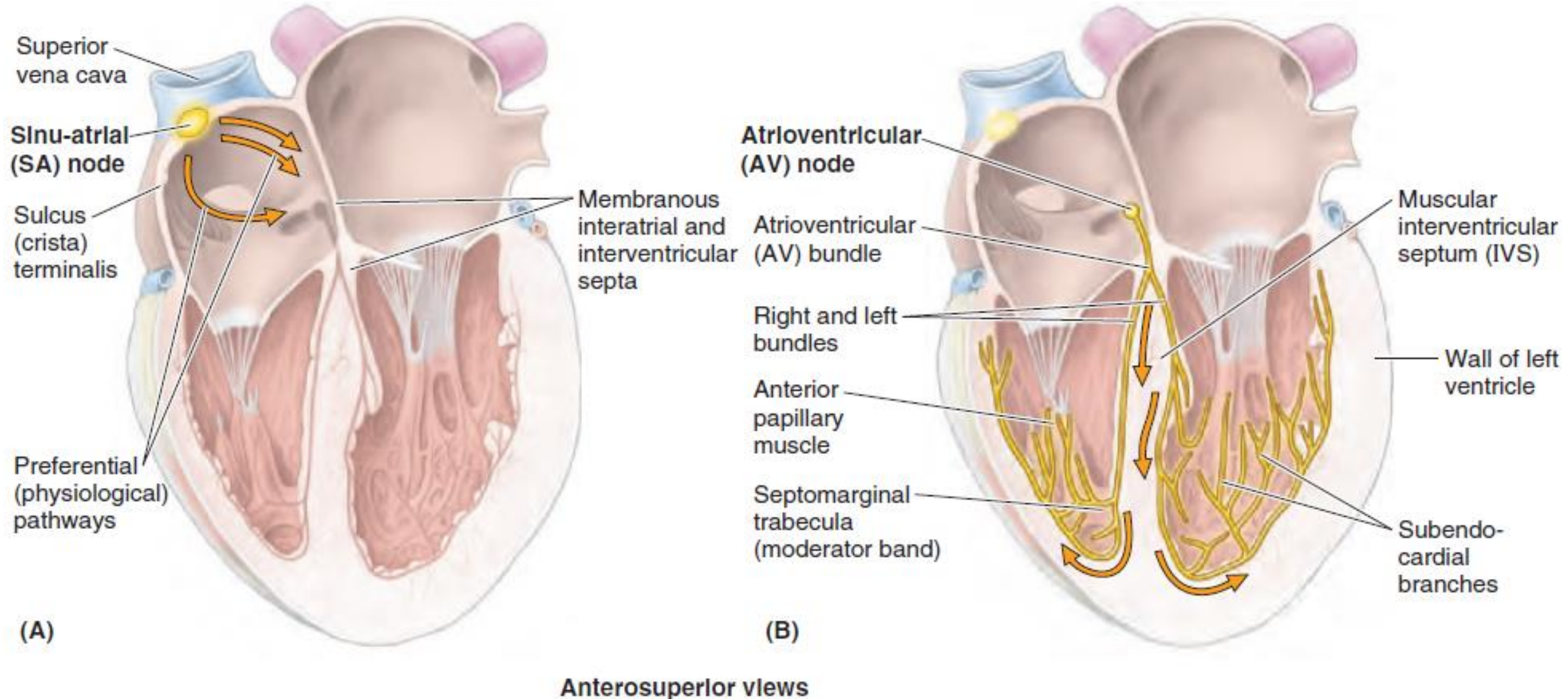
Venous drainage of the heart,,



Conducting system of the heart

- ✓ In the cardiac cycle, the atrium and ventricle work together as one pump.
- ✓ The cardiac conduction system initiates and coordinates contraction
- ✓ The **conducting system of the heart** generates and transmits the impulses.
- ❖ **The conduction system consists of nodes and networks of specialized cardiac muscle cells organized into four basic components:**
 - ✓ the sinu-atrial node
 - ✓ the atrioventricular node
 - ✓ the atrioventricular bundle with its right and left bundle branches;
 - ✓ the subendocardial plexus of conduction cells (the Purkinje fibers).

Conducting system of heart,,



Conducting system of heart,,

- The conducting system initiates by *nodal tissue* that propagated by the cardiac striated muscle cells.
- The **sinu-atrial (SA) node** is located anterolaterally at the junction of the SVC& Rt atrium near the superior end of the *sulcus terminalis*.
- ❖ **The SA node**, specialized cardiac muscle fibers & associated fibroelastic connective tissue, is the *pacemaker of the heart*.
- *The SA node initiates and regulates the impulses for the contractions of the heart*, giving off an impulse approximately 70 times per minute in most people most of the time.

Conducting system of heart,,,

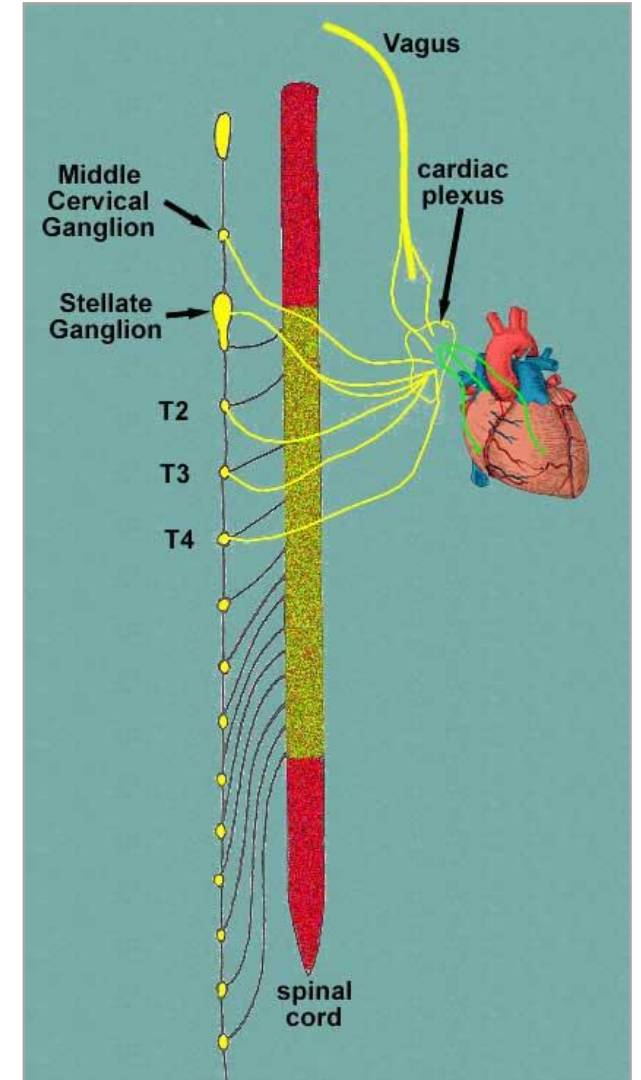
- ❖ The SA node **is stimulated** by the **sympathetic division** to accelerate the heart rate & is **inhibited by the parasympathetic** autonomic nervous system division.
- The AV node is located in the posteroinferior region of the interatrial septum.
- The **signal** generated by the SA node passes through the walls of the Rt atrium, propagated by the cardiac muscle (**myogenic conduction**) w/c transmits the signal rapidly from the SA node to the AV node.
- The AV node then distributes the signal to the ventricles through the **AV bundle**.

Conducting system of heart,,

- The AV bundle, the bridge b/n the atrial& **ventricular myocardium**, passes from AV node thru the *fibrous skeleton of the heart*& along the membranous part of the IVS.
- At the junction of the membranous & muscular parts of the IVS, the AV bundle divides into **right** and **left bundles**.
- These branches then ramify into **subendocardial branches** (Purkinje fibers), w/c extend into the walls of the respective ventricles and the subendocardial branches of the *right bundle* stimulate the muscle of the IVS.

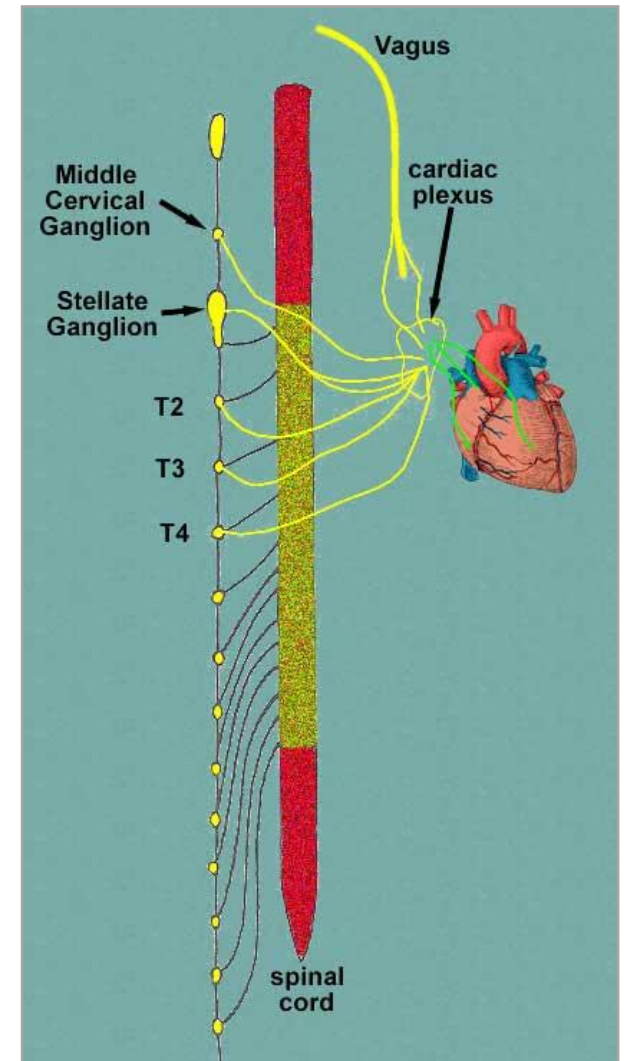
Innervation of the heart

- **Autonomic nervous system (ANS)** is directly responsible for regulating: *heart rate; force of each contraction; cardiac output:*
 - ➔ Strength & frequency of heart beat.
- Both **parasympathetic & sympathetic** parts are involved.
- **Sympathetic fibers** arise from segments **T2-T4** of the spinal cord & are distributed thru **middle cervical & cervico-thoracic (stellate) ganglia & first four ganglia of the thoracic sympathetic chain.**



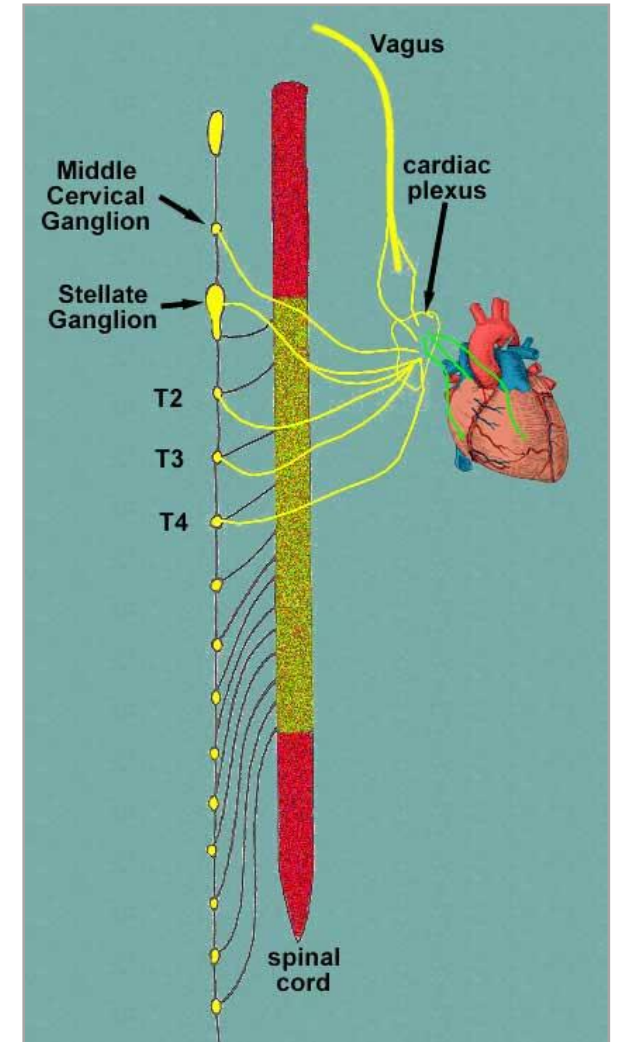
Innervation of the heart,,

- Sympathetic fibers pass into **cardiac plexus** & from there to **SA node** and **cardiac muscle**.
- Effect of sympathetic nerves at **SA node** is an **increase in heart rate**.
- Effect on the **muscle** is an increase in rise of pressure within the ventricle, thus **increasing stroke volume**, dilatation of coronary aa.

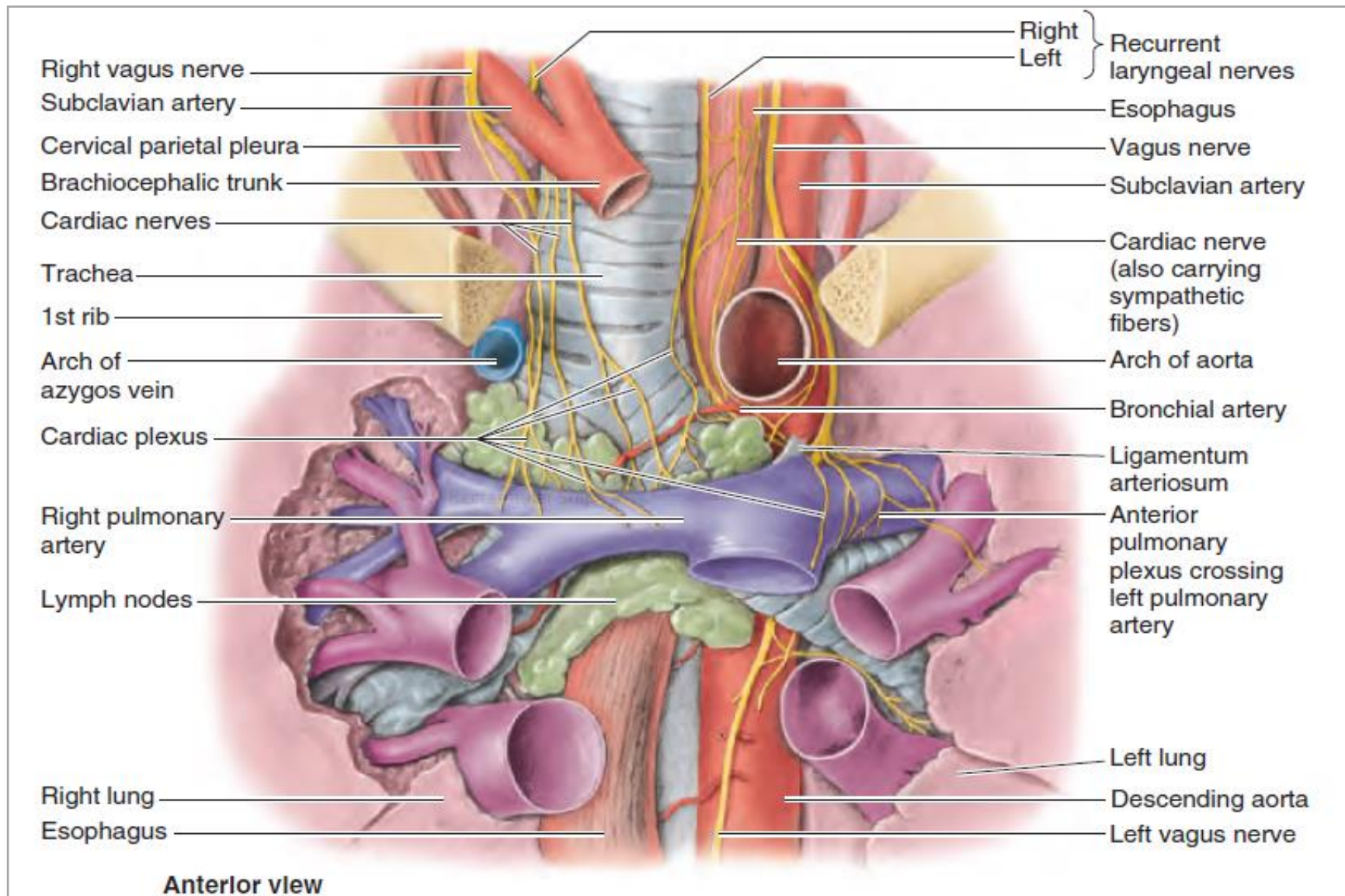


Innervation of the heart,,,

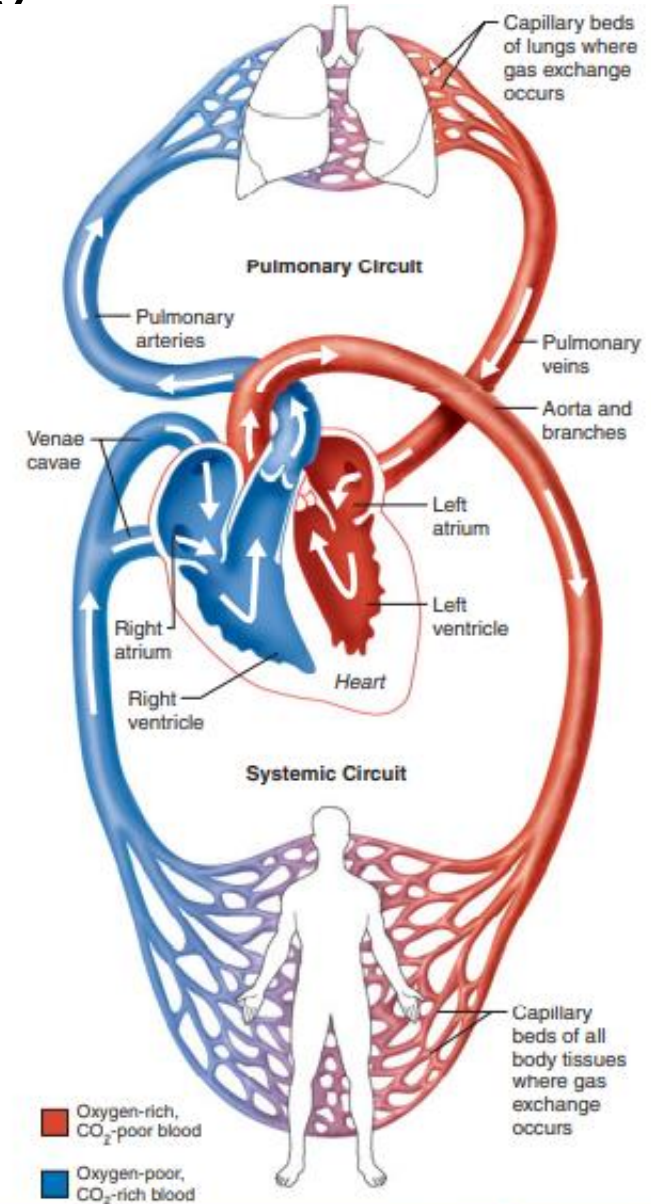
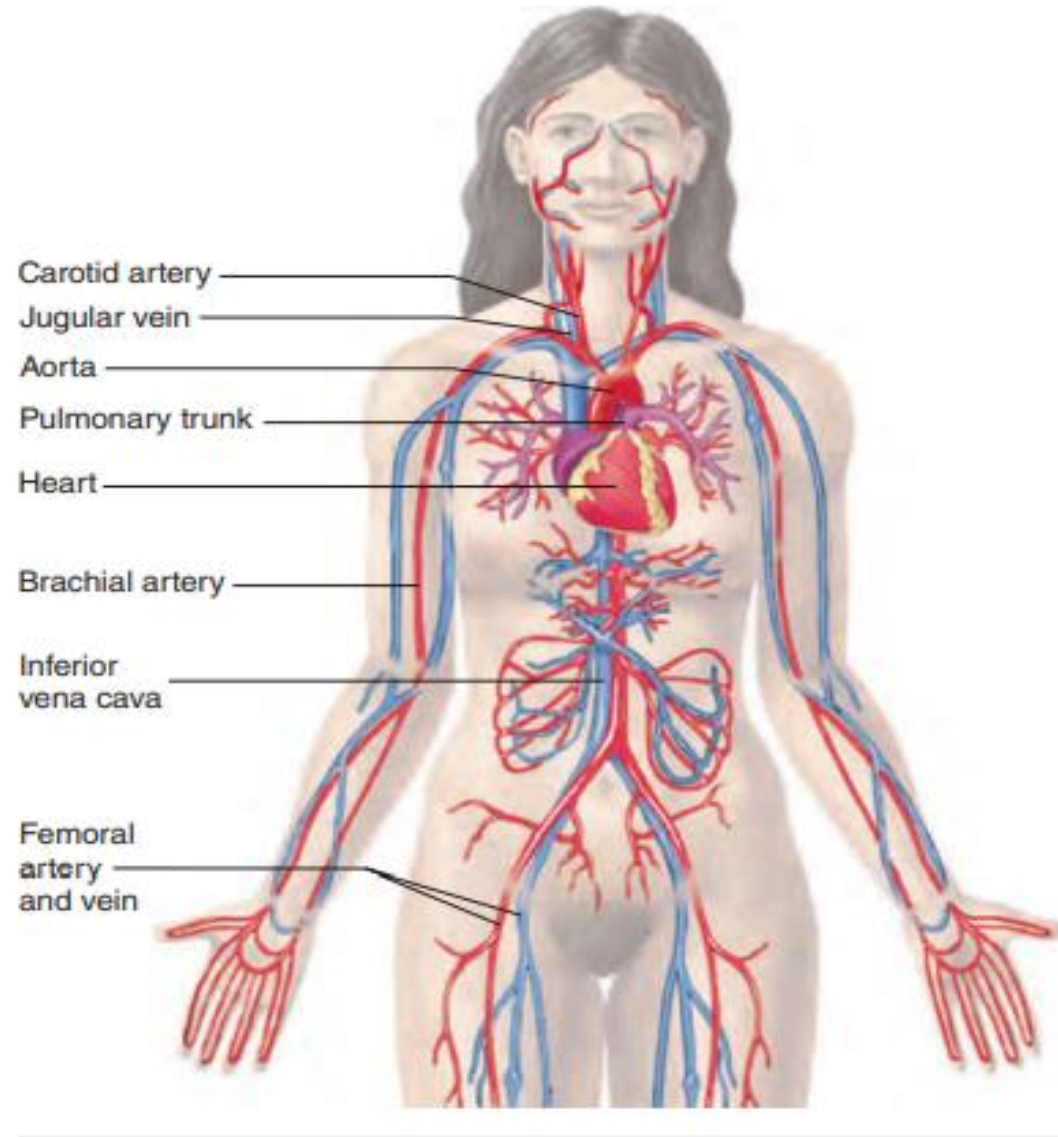
- **Parasympathetic stimulation:** → *decreases heart rate; reduces force of contraction; constricts the coronary arteries:*
 1. Effect of **vagus n.** at the **SA node** is the opposite of **sympathetic nerves** → **decreases the heart rate.**
 2. It also decreases excitability around the **AV node:** → **slows down transmission.**
 3. Effect on the **muscle** is decrease in force of contraction.
- **Strong vagal** stimulation at **AV node** may produce **AV block.**



Innervation of the heart,,



Major blood vessels of the body



Major blood vessels of the body

- The blood vessels of the body form a **closed delivery system** powered by the pumping heart.
- Blood vessels are not rigid tubes that simply direct the flow of blood but rather are dynamic structures.
- Blood vessels are part of the cardiovascular system, that transport blood throughout the body and provide the physical site where gases, nutrients, and other substances are exchanged with body cells.
- They form a closed system of tubes that carry blood from the heart to the tissue cells and back to the heart.

Types of Blood Vessels

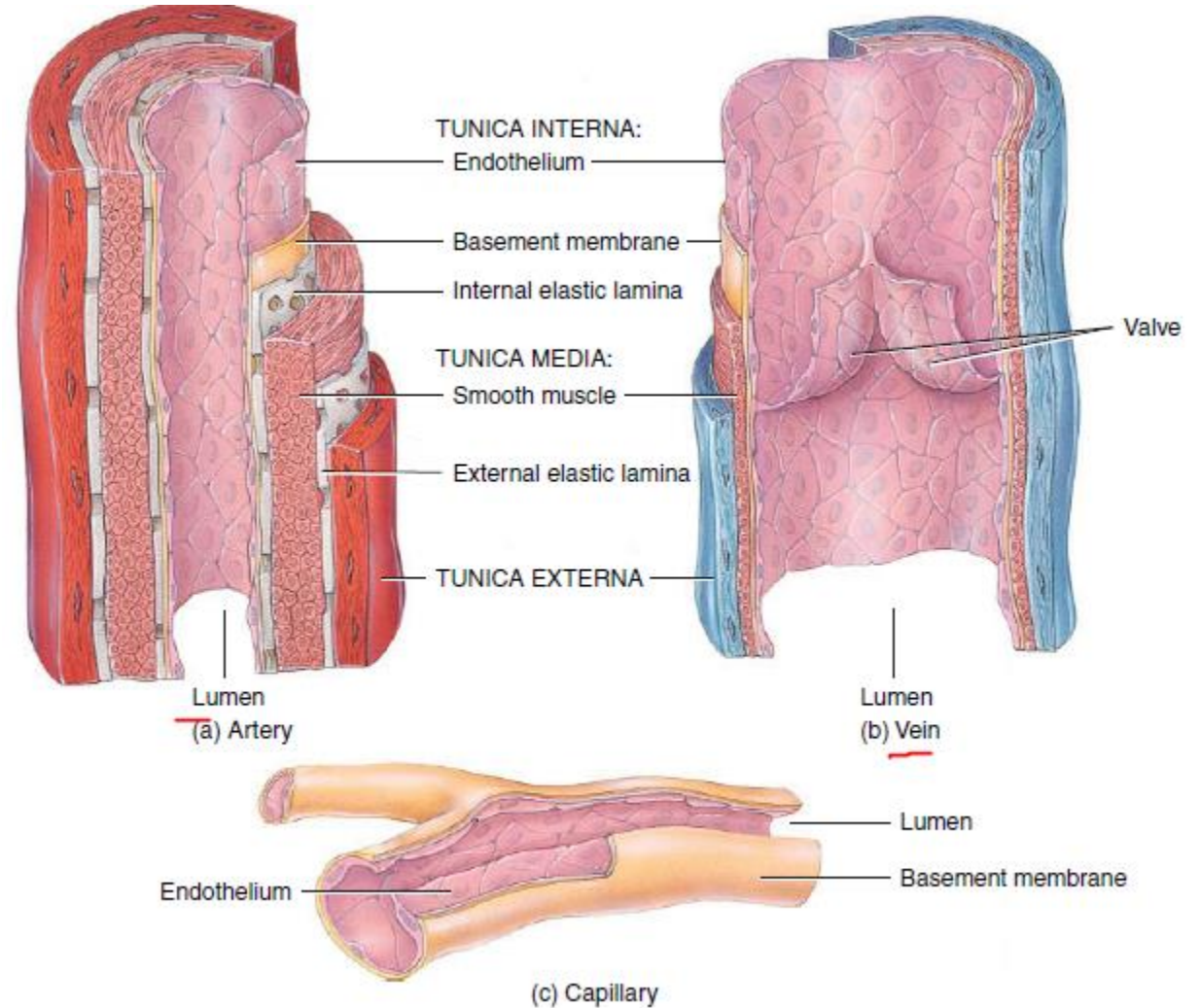
- Based on their function there are three basic types of blood vessels: arteries, capillaries, and veins.
- **Arteries** are the **efferent** vessels of the cardiovascular system—that is, vessels that carry blood away from the heart.
- **Veins** are the **afferent** vessels that carry blood back to the heart.
- **Capillaries** are microscopic, thin-walled vessels that connect the smallest arteries to the smallest veins.
- They are thin walled that allow the exchange of substances between the blood and body tissues.

Basic Structure of a Blood Vessel

❑ Except in **capillaries and venules**, blood vessel walls consist of three layers, or **tunics**.

❖ From the inner to the outer wall, the tunics are:

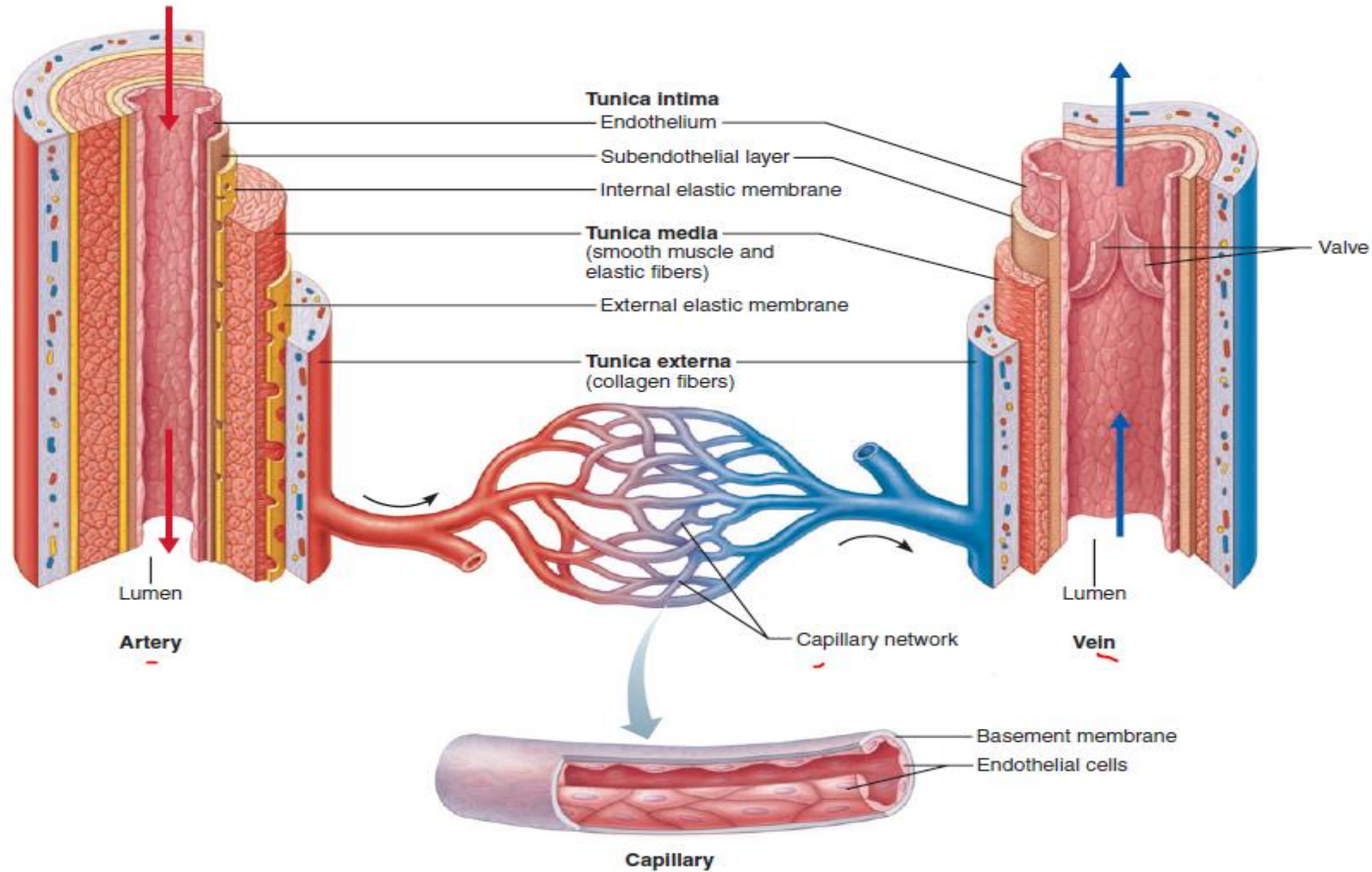
- I. the tunica intima(intima),
- II. the tunica media, and
- III. the tunica adventitia, or tunica externa



Types of Blood Vessels

- ❖ **There are three basic types of blood vessels: arteries, capillaries, and veins.**
- They form a closed system of tubes that carry blood from the heart to the tissue cells and back to the heart
- **Modifications of this basic design account for the five types of** blood vessels and the structural and functional differences among the various vessel types.
- **The 5 main types of blood vessels are arteries, arterioles, capillaries, venules& veins**
- ❖ Always remember that structural variations correlate to the differences in function that occur throughout the cardiovascular system.

Types of Blood Vessels,,,



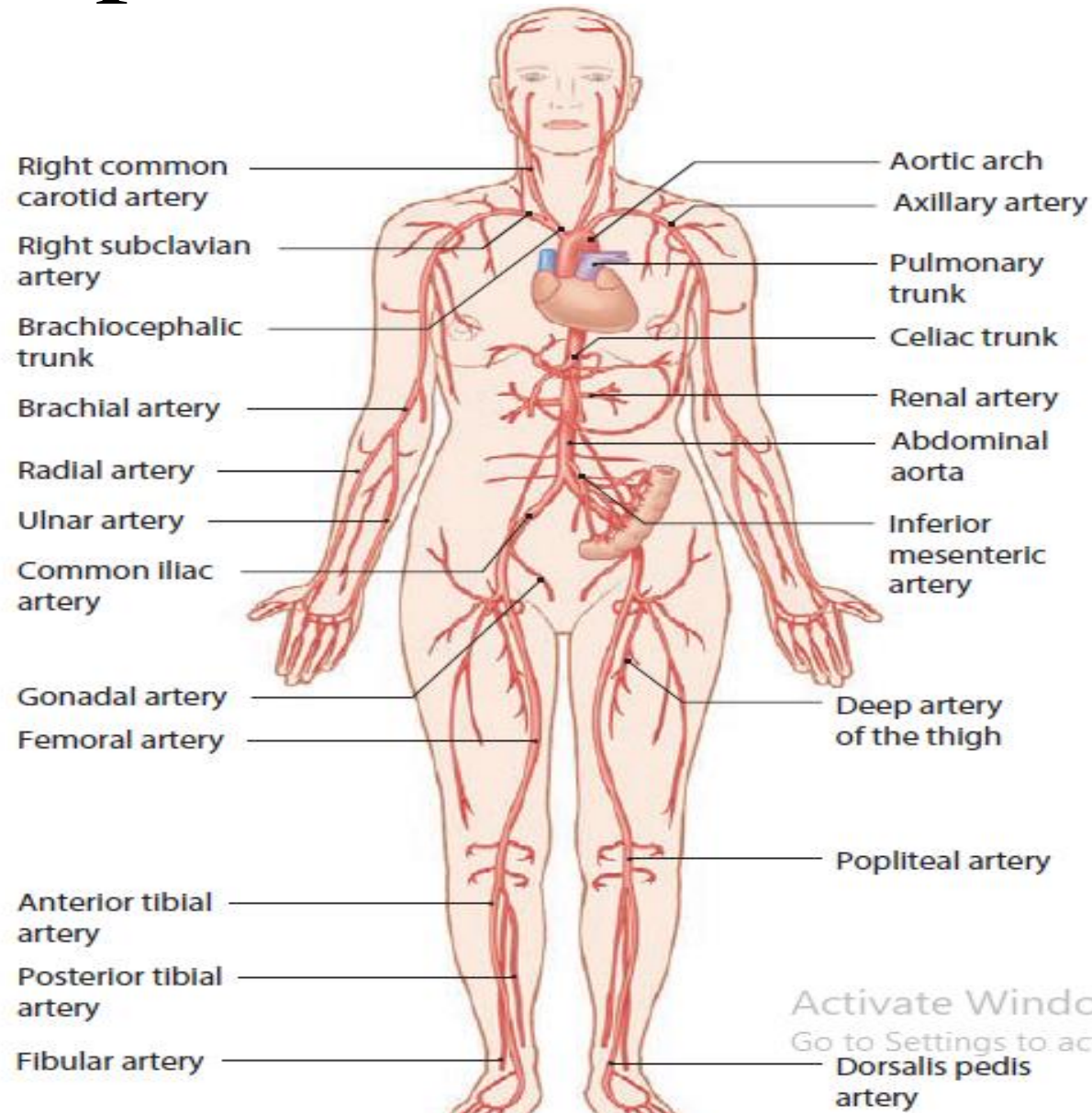
Types of Blood Vessels,,,

- **Large, elastic arteries** leave the heart and divide into medium-sized, *muscular arteries* that branch out into the various regions of the body.
- **Medium-sized** arteries then divide into small arteries, which in turn divide into still smaller arteries called **arterioles**
- As the arterioles enter a tissue, they branch into numerous tiny vessels called **capillaries**.
- The thin walls of capillaries allow the exchange of substances between the blood and body tissues. Groups of capillaries within a tissue reunite to form small veins called **venules** .
- These in turn merge to form progressively larger blood **vessels called veins**.
- **Veins** are the blood vessels that convey blood from the tissues *back to the heart*.⁶⁸

Types of Blood Vessels,,

- Groups of capillaries within a tissue reunite to form small veins called **venules**.
- These in turn merge to form progressively larger blood vessels called veins.
- **Veins** are the blood vessels that convey blood from the tissues *back to the heart*.
- **Arteries** are vessels that carry blood away from the heart.
- **It is a common misconception that *all* arteries carry oxygen-rich blood, whereas *all* veins carry oxygen-poor blood.**
- This notion is correct for the systemic circuit, but it is not correct for the pulmonary circuit, whose arteries carry oxygen-poor blood to the lungs for oxygenation

Principal Arteries of the body



Elastic Arteries

- **Elastic arteries** are the largest arteries in the body.
- These vessels are characterized by well-defined internal and external elastic laminae, along with a thick tunica media that is dominated by elastic fibers, called the **elastic lamellae**.
- Elastic arteries include the **two major** arteries that exit the heart **the aorta and the pulmonary trunk**, along with the aorta's major initial branches, such as the **brachiocephalic, subclavian, common carotid, and common iliac arteries**.
- As blood is ejected from the heart into elastic arteries, their walls stretch, easily accommodating the surge of blood.
- As they stretch, the elastic fibers momentarily store mechanical energy

Elastic Arteries,,,

Arteries of the head and trunk

Internal carotid artery

External carotid artery

Common carotid arteries

Vertebral artery

Subclavian artery

Brachiocephalic trunk

Aortic arch

Ascending aorta

Coronary artery

Thoracic aorta (above diaphragm)

Celiac trunk

Abdominal aorta

Superior mesenteric artery

Renal artery

Gonadal artery

Inferior mesenteric artery

Common iliac artery

Internal iliac artery

Arteries that supply the upper limb

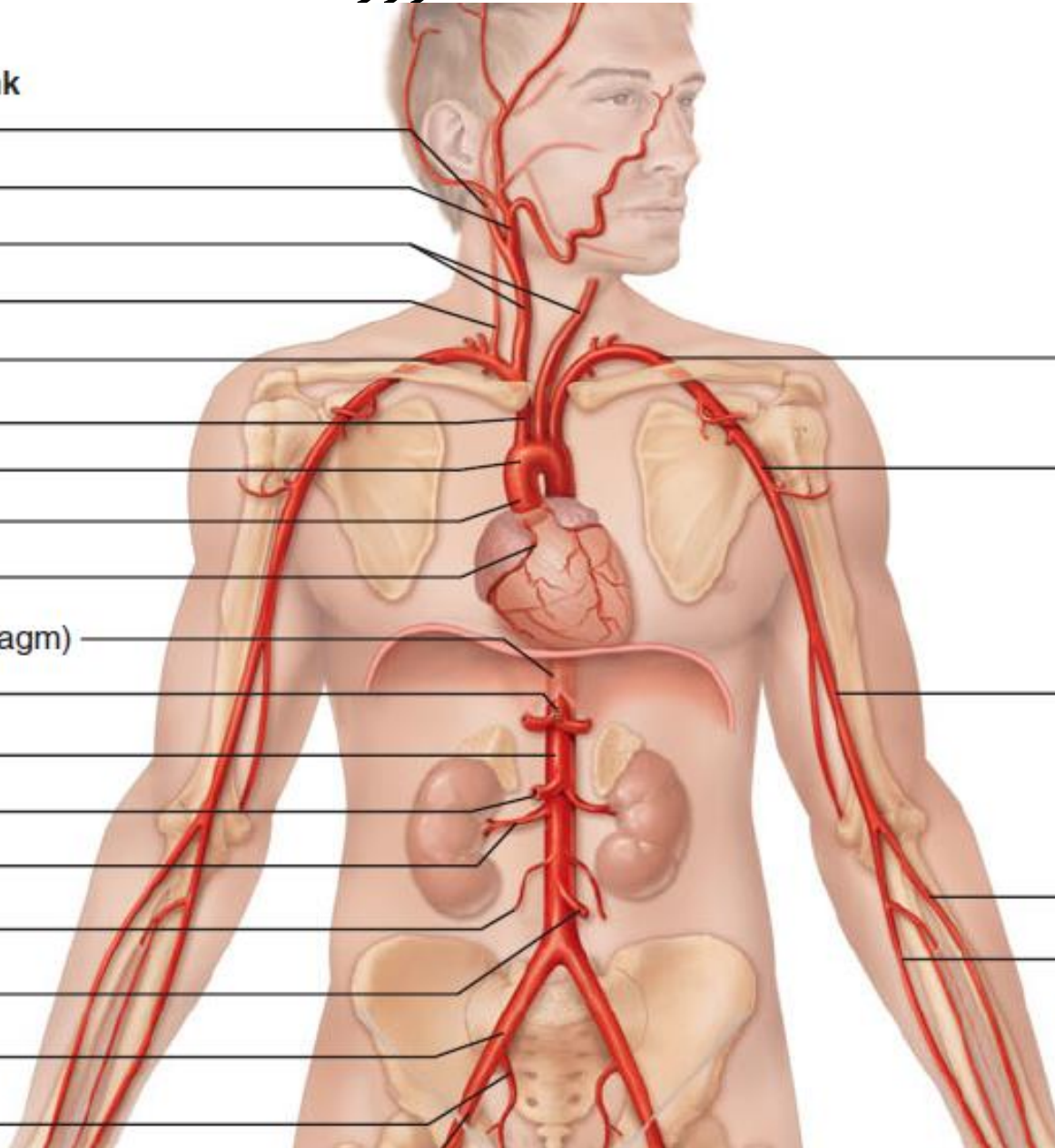
Subclavian artery

Axillary artery

Brachial artery

Radial artery

Ulnar artery



Arteries of the head and trunk

- Internal carotid
- External carotid
- Left common carotid
- Brachiocephalic
- Aortic arch
- Pulmonary trunk
- Left coronary
- Right coronary
- Thoracic aorta
- Celiac trunk
- Splenic
- Renal (kidney not shown)
- Superior mesenteric
- Abdominal aorta
- Inferior mesenteric
- Common iliac
- Internal iliac
- External iliac

Arteries of the upper limb

- Subclavian
- Axillary
- Brachial
- Radial
- Ulnar

Arteries of the lower limb

Muscular Arteries

- **Muscular (distributing) arteries** lie distal to the elastic arteries and supply groups of organs, **individual organs, and parts of organs.**
- These “middle-sized” arteries constitute **most of the named arteries.**
- They range in diameter from about 1 cm to 0.3 mm.
- These “middle-sized” arteries are called **muscular arteries** because their tunica media contains more smooth muscle and fewer elastic fibers than elastic arteries.
- By actively changing the diameter of the artery, this muscular layer regulates the amount of blood flowing to the organ supplied according to the specific needs of that organ.

Anastomoses

- Most tissues of the body receive blood from more than one artery.
- The union of the branches of two or more arteries supplying the same body region is called an **anastomosis**.
- ❖ **Anastomoses** between arteries provide alternative routes for blood to reach a tissue or organ.
- If blood flow stops for a short time when normal movements compress a vessel, or if a vessel is blocked by disease, injury, or surgery, then circulation to a part of the body is not necessarily stopped.
- The alternative route of blood flow to a body part through an anastomosis is known as **collateral circulation**.
- **Anastomoses may also occur between veins and between arterioles and venules.**
- Arteries that do not anastomose are known as **end arteries**.

Vasa vasorum

- The walls of blood vessels contain living cells and therefore require a blood supply of their own
- For this purpose, the larger arteries and veins have tiny arteries, capillaries, and veins in their tunica externa.
- These little vessels, the **vasa vasorum** “vessels of the vessels”, arise either as tiny branches from the same vessel or as small branches from other, nearby vessels and nourish the outer half of the wall of the larger vessel.
- The inner half, by contrast, gets its nutrients by diffusion from the blood in the vessel’s own lumen.
- Small blood vessels need no vasa vasorum because their walls are entirely supplied by luminal blood.

The aorta

- The **aorta** is the largest artery of the body, with a diameter of 2–3 cm .

- **Its three principal divisions are:**

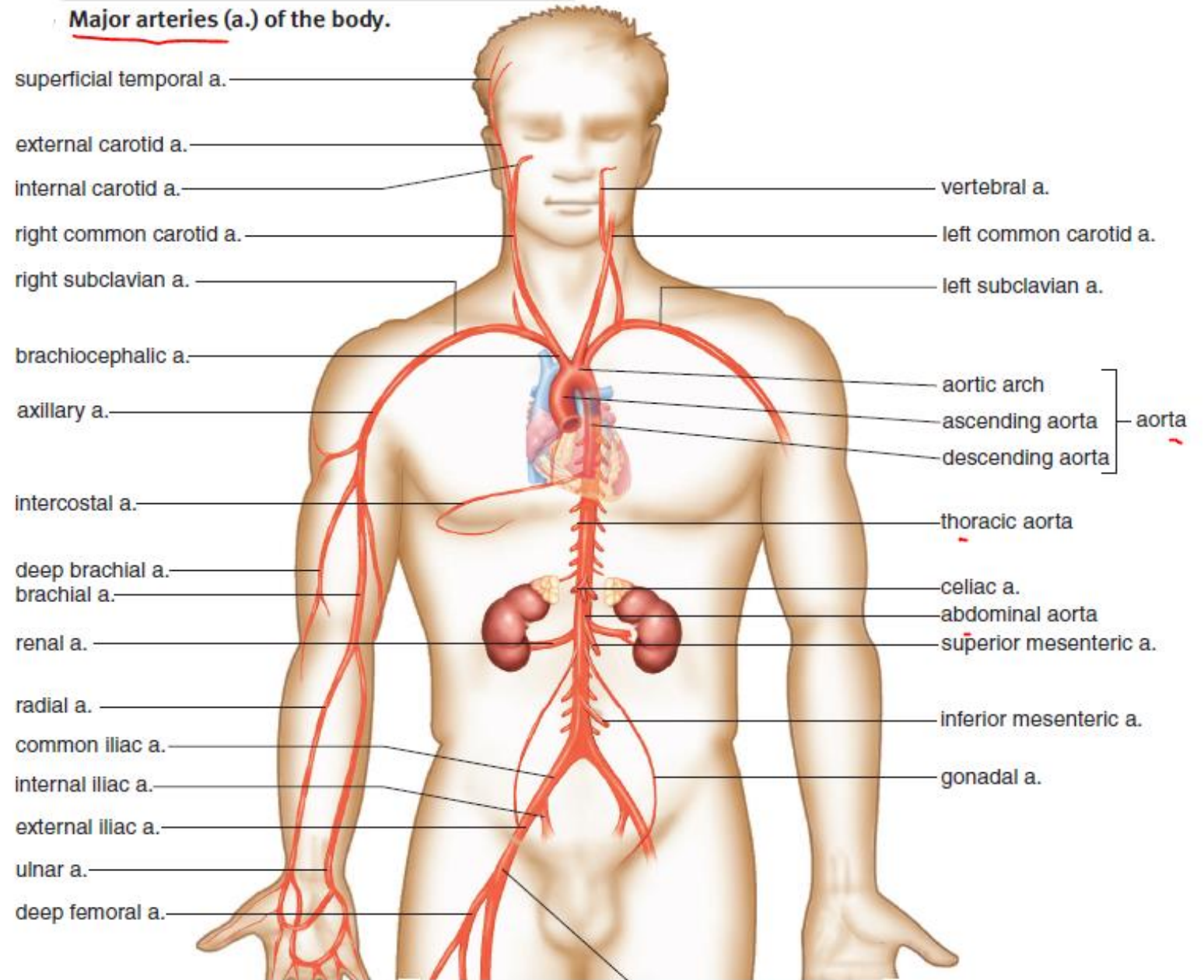
I. Ascending aorta,

II. Arch of the aorta,

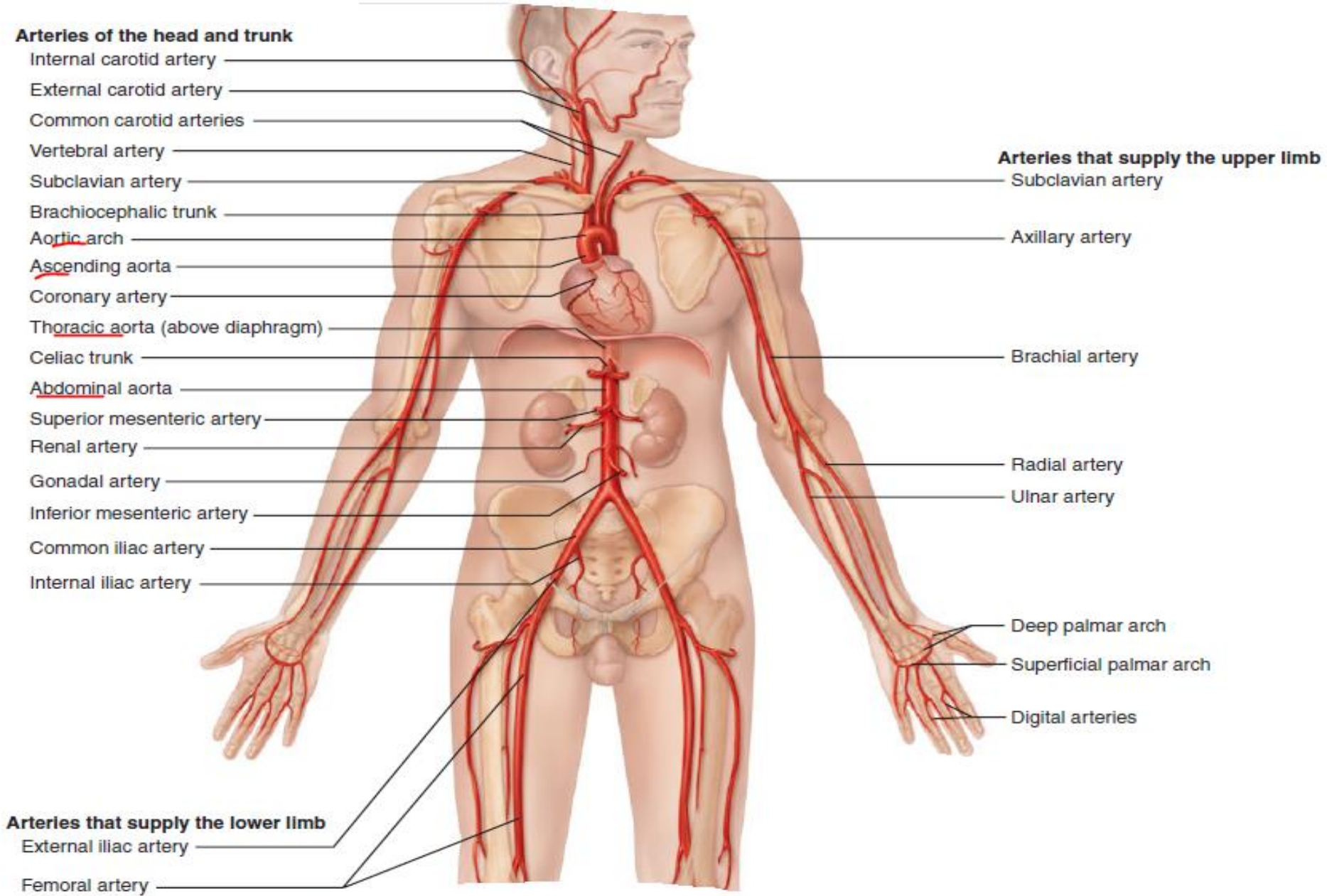
III. The descending aorta; the last segment is further divided into the **thoracic aorta and the abdominal aorta**

- The beginning of the aorta contains the aortic valve.
- All arteries of the systemic circulation branch directly or indirectly from the **aorta**.

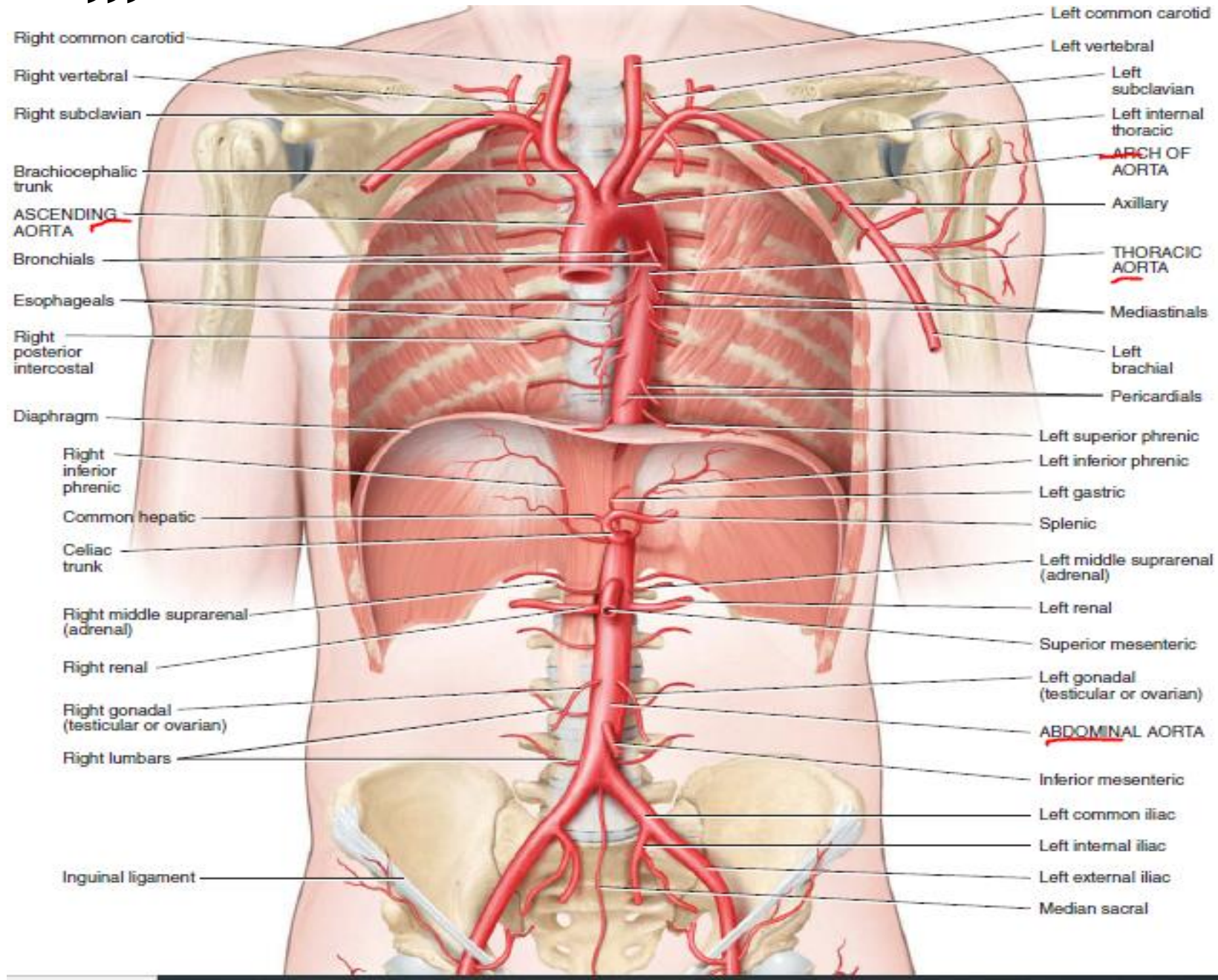
The aorta,,,



The aorta,,



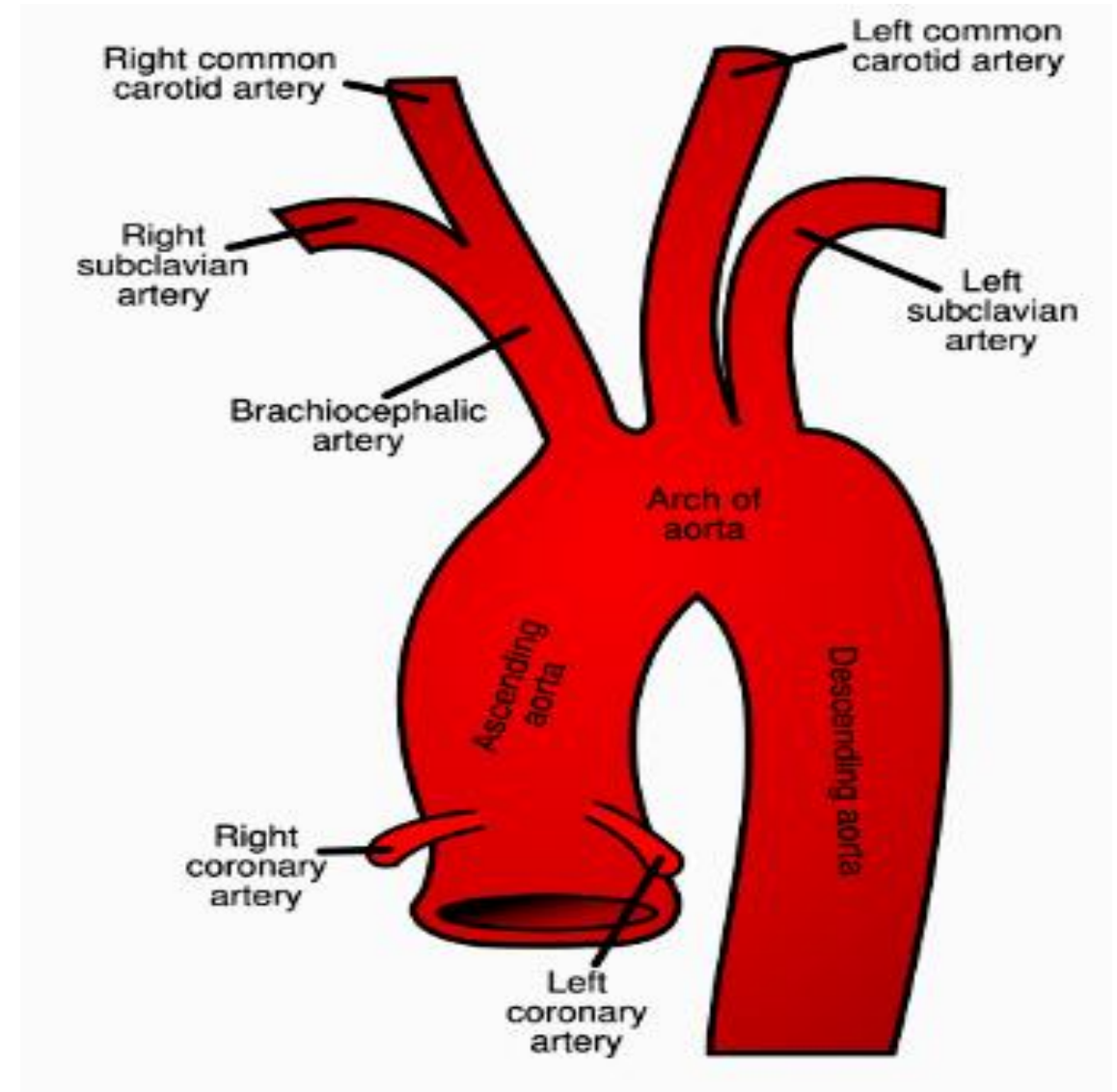
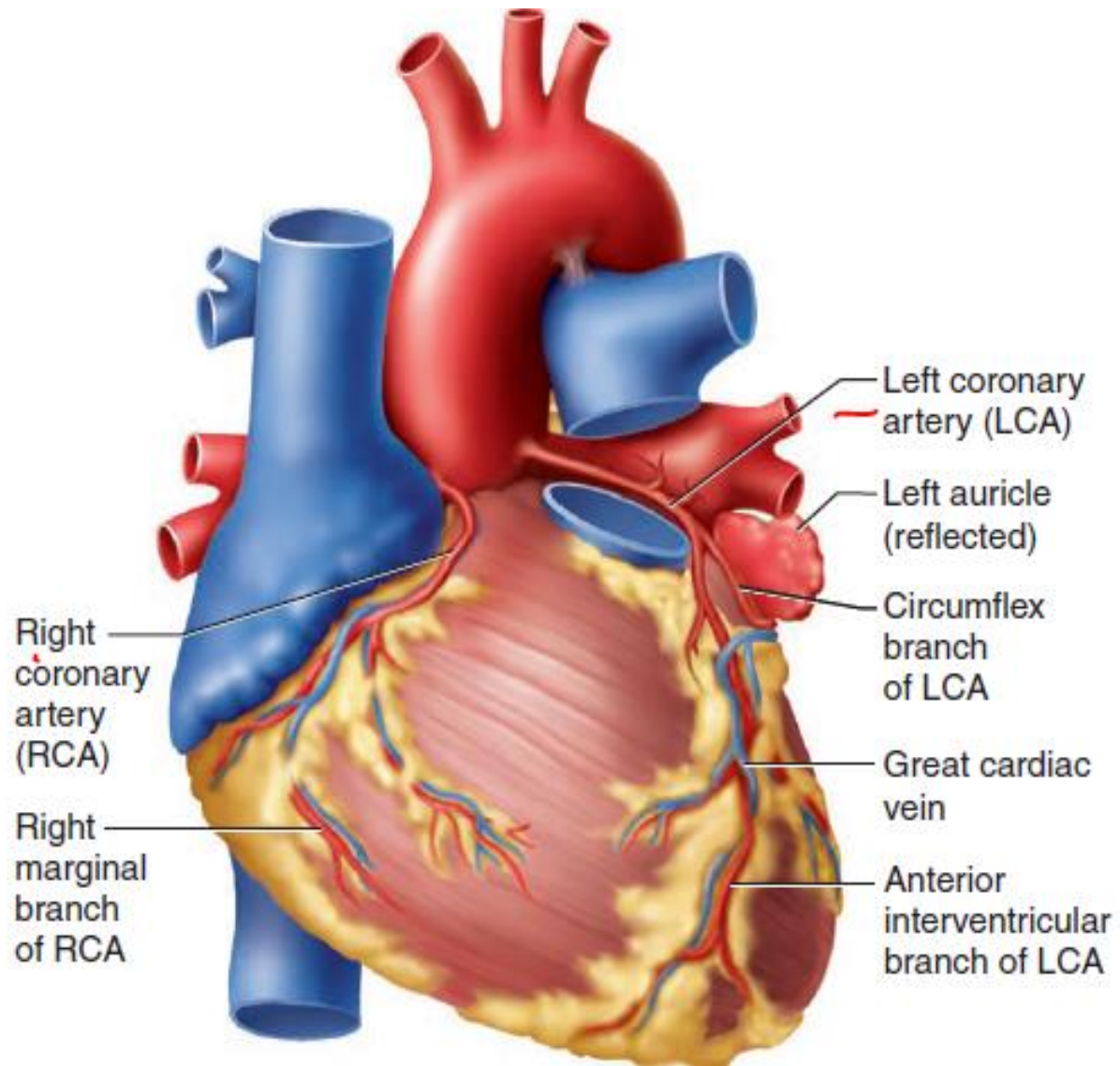
The aorta,,



Ascending aorta

- The portion of the aorta that emerges from the left ventricle posterior to the pulmonary trunk is the **ascending aorta**.
- The right and left **coronary arteries** are the only branches of the ascending aorta.
- The right and left **coronary arteries** are the first branches of the ascending aorta, just beyond the aortic semilunar valve
- Then the ascending aorta arches to the left, forming the **arch of the aorta**, which descends and ends at the level of the intervertebral disc between the fourth and fifth thoracic vertebrae

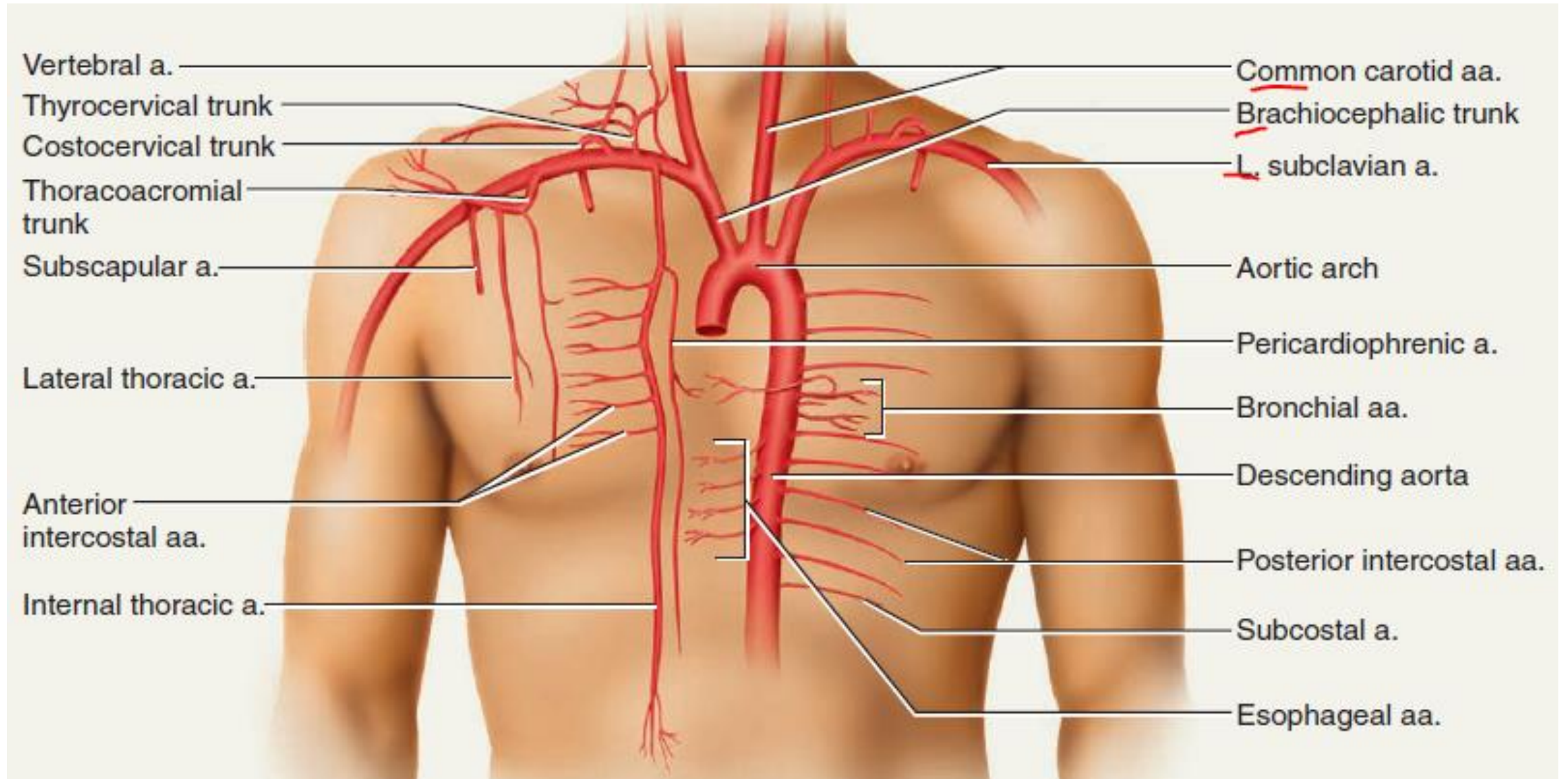
Ascending aorta,,



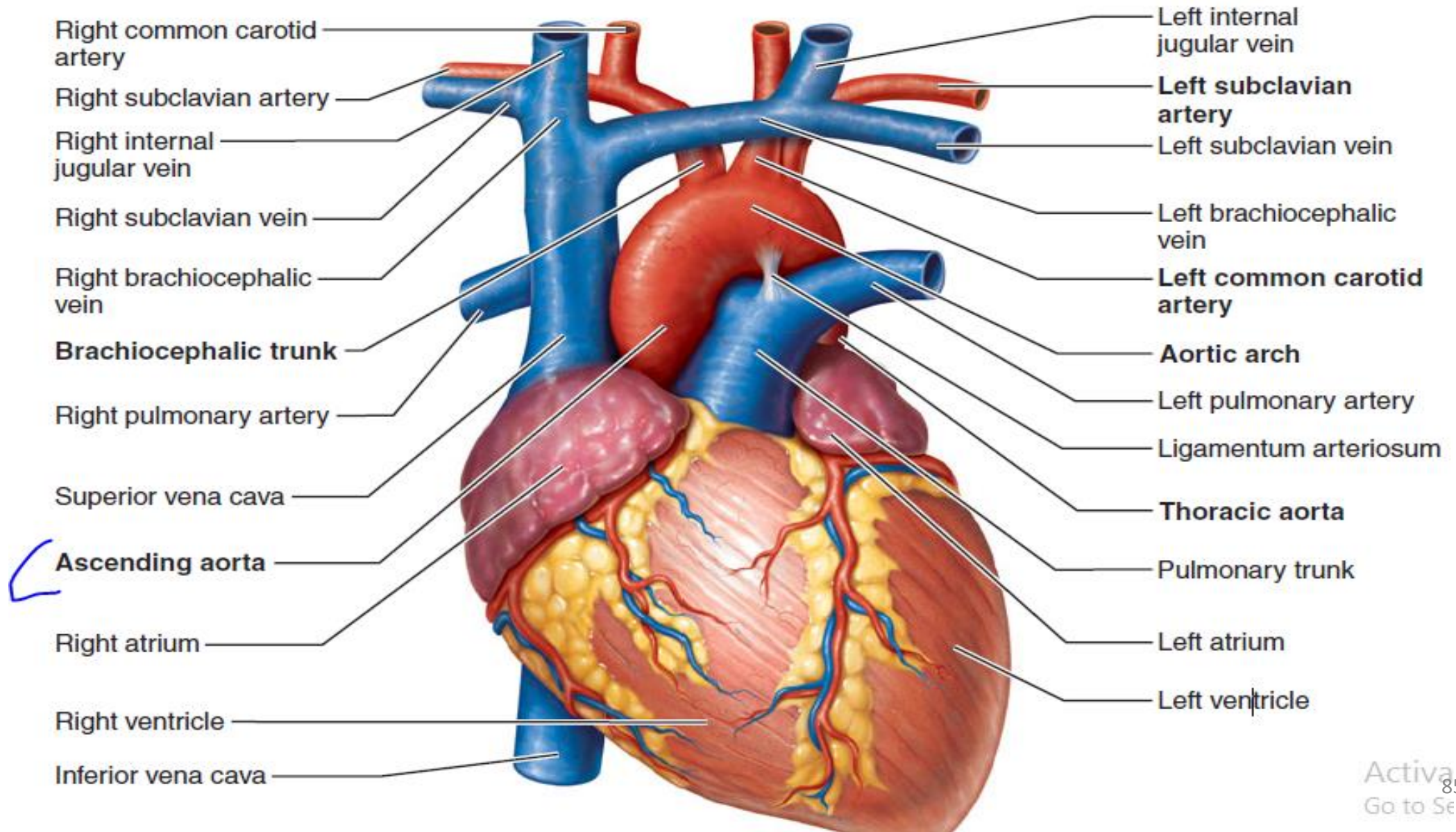
Arch of the aorta

- The arch of the aorta is a continuation of the ascending aorta
- The aorta arches posteriorly and to the left as the **aortic arch**.
- Three major arteries arise from the aortic arch are:
 - Brachiocephalic artery—supplies the upper limbs and head regions
 - Left common carotid artery
 - Left subclavian artery
- These three branches of the aortic arch supply the head and neck, upper limbs, and the superior part of the thoracic wall.
- Note that the brachiocephalic trunk on the right has no corresponding artery on the left because the left common carotid and subclavian arteries arise directly from the aorta.

Arch of the aorta, ,,



Arch of the aorta, ,,



Arch of the aorta, ,,

- This is the typical branching pattern of these vessels off the aortic arch; however, as with all vessels, there is some variability from this pattern.
- The most frequent variation is the branching of the left common carotid artery from the brachiocephalic trunk.
- ❖ **Less commonly, four large arteries** (the right and left common carotids, and the right and left subclavian arteries) arise separately from the aortic arch, or the left common carotid artery and the left subclavian artery arise from a left brachiocephalic trunk.

Descending aorta

- **The descending aorta** continuing from the aortic arch, runs posterior to the heart and at first to the left of the vertebral column and then anterior to it.
- It has two parts, the *thoracic aorta* and the *abdominal aorta*
- ❖ It is the longest part of the aorta.
- It is called the *thoracic aorta* above the diaphragm and the *abdominal aorta* below it.
- It ends in the lower abdominal cavity by forking into the ***right and left common iliac arteries***.

Descending aorta,,,

Arteries of the head and trunk

Internal carotid artery —
External carotid artery —
Common carotid arteries —
Vertebral artery —
Subclavian artery —
Brachiocephalic trunk —
Aortic arch —
Ascending aorta —
Coronary artery —
Thoracic aorta (above diaphragm) —
Celiac trunk —
Abdominal aorta —
Superior mesenteric artery —
Renal artery —
Gonadal artery —
Inferior mesenteric artery —
Common iliac artery —
Internal iliac artery —

Arteries that supply the upper limb

Subclavian artery

Axillary artery

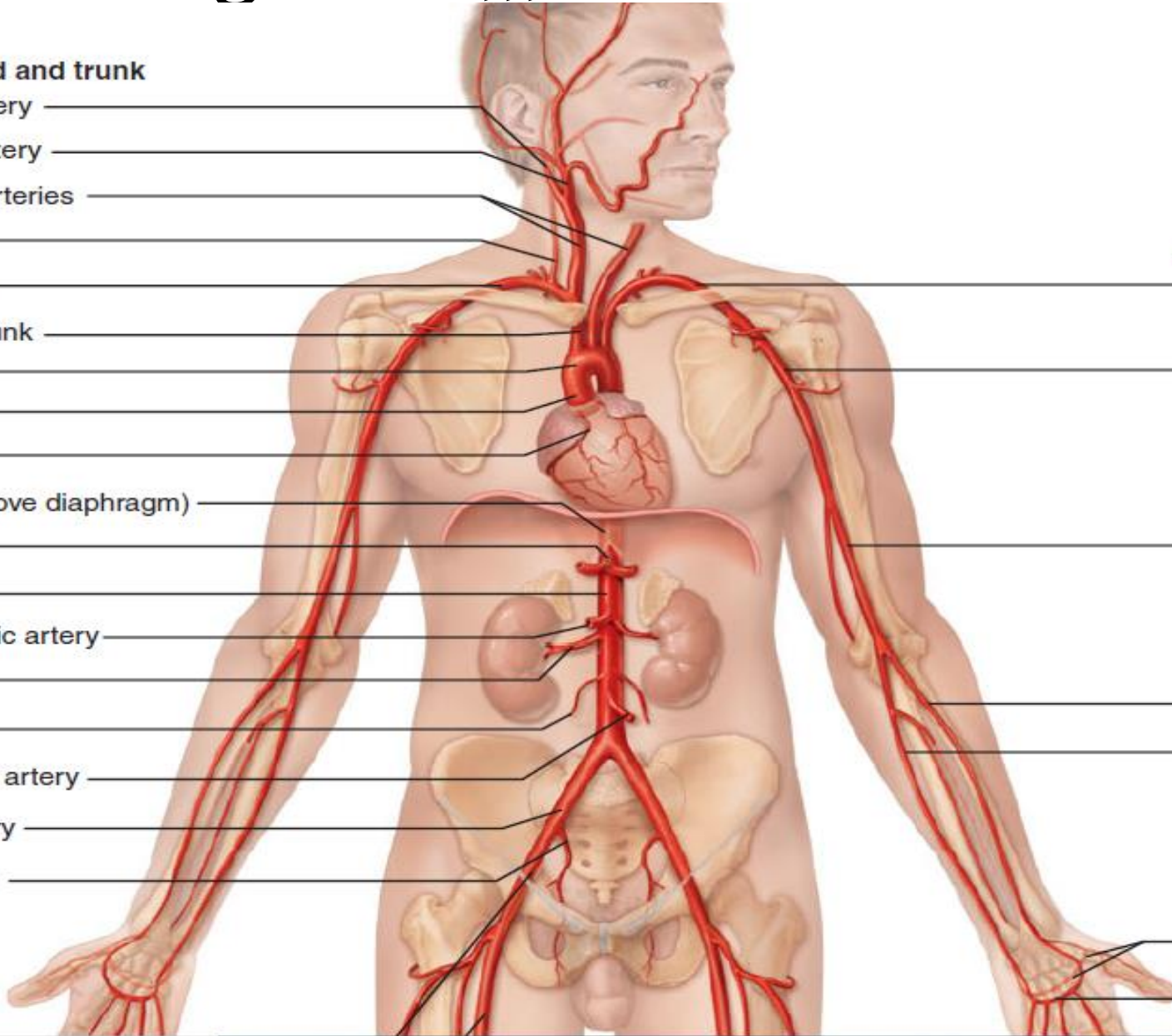
Brachial artery

Radial artery

Ulnar artery

Deep palmar arch

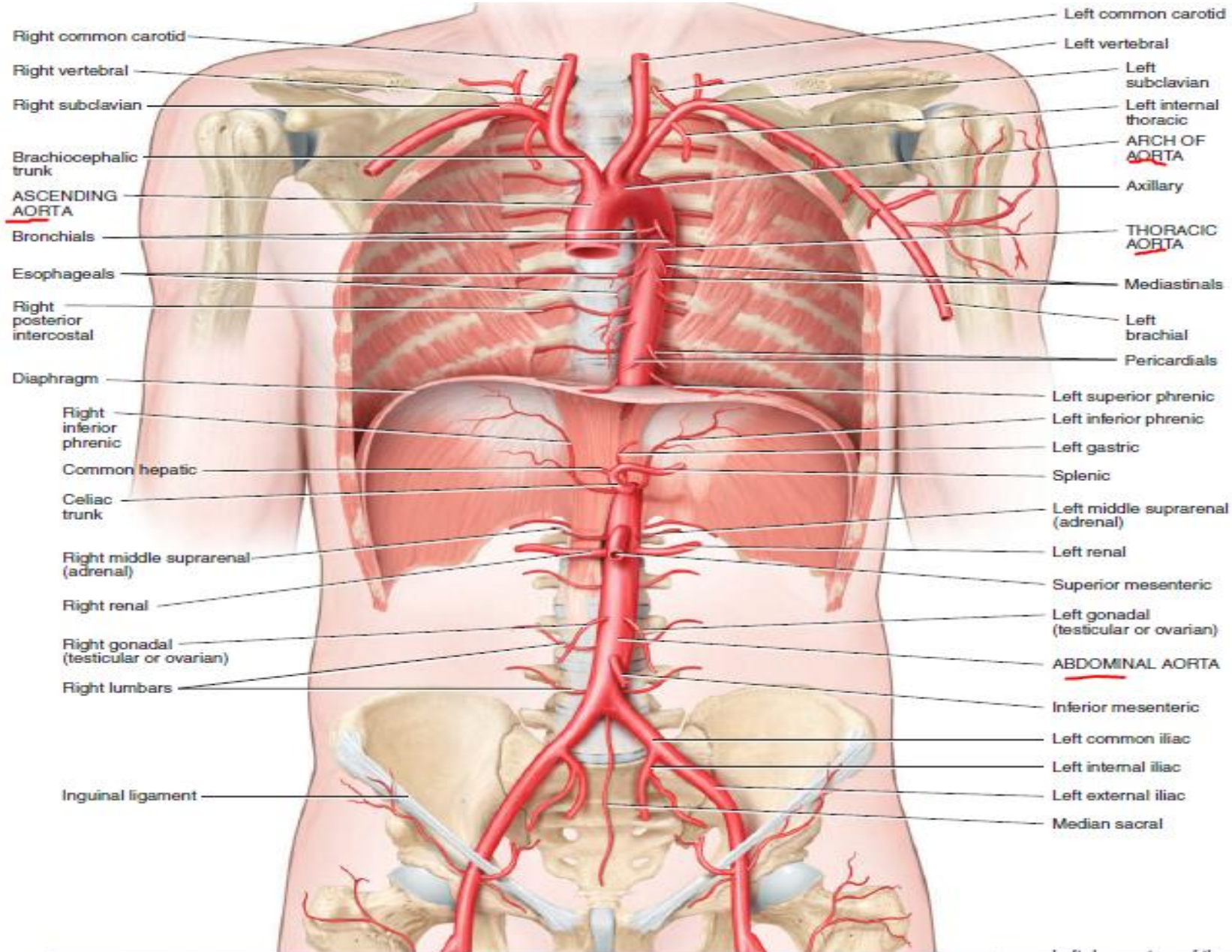
Superficial palmar arch



Descending aorta,,

- The **thoracic aorta** descends on the bodies of the thoracic vertebrae (T5–T12) just to the left of the midline.
- Along the way, it sends many small branches to the thoracic organs and body wall.
- The thoracic aorta passes through the diaphragm at the level of vertebra T12 and enters the abdominal cavity as the **abdominal aorta**, which lies on the lumbar vertebral bodies in the midline.
- **The abdominal aorta ends at the level of vertebra L4**, where it divides into the right and left *common iliac arteries*, which supply the pelvis and lower limbs.
- Branches of the thoracic aorta include **pericardial, esophageal, bronchial, mediastinal, posterior intercostal arteries, subcostal and superior phrenic arteries,**

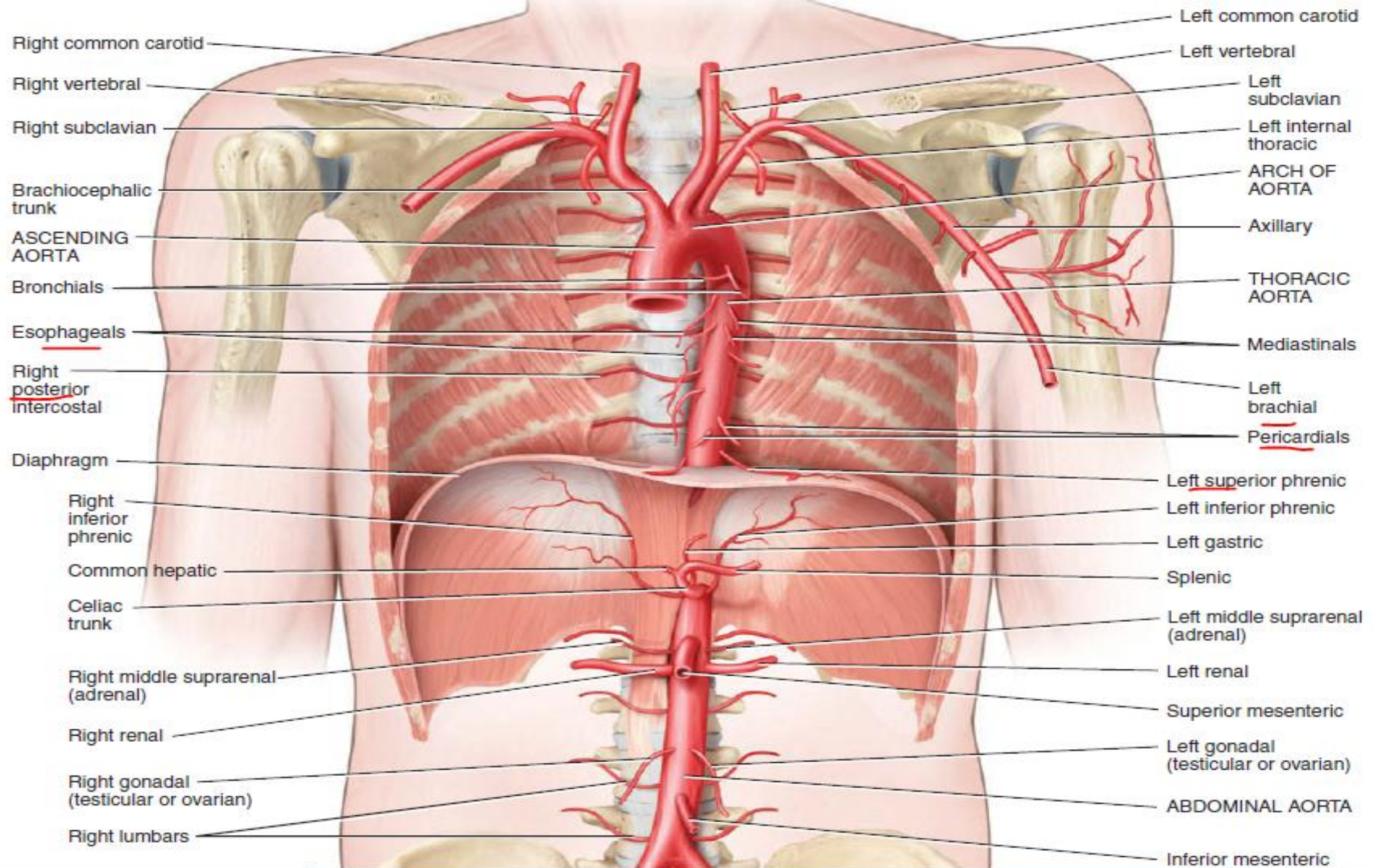
Descending aorta,,

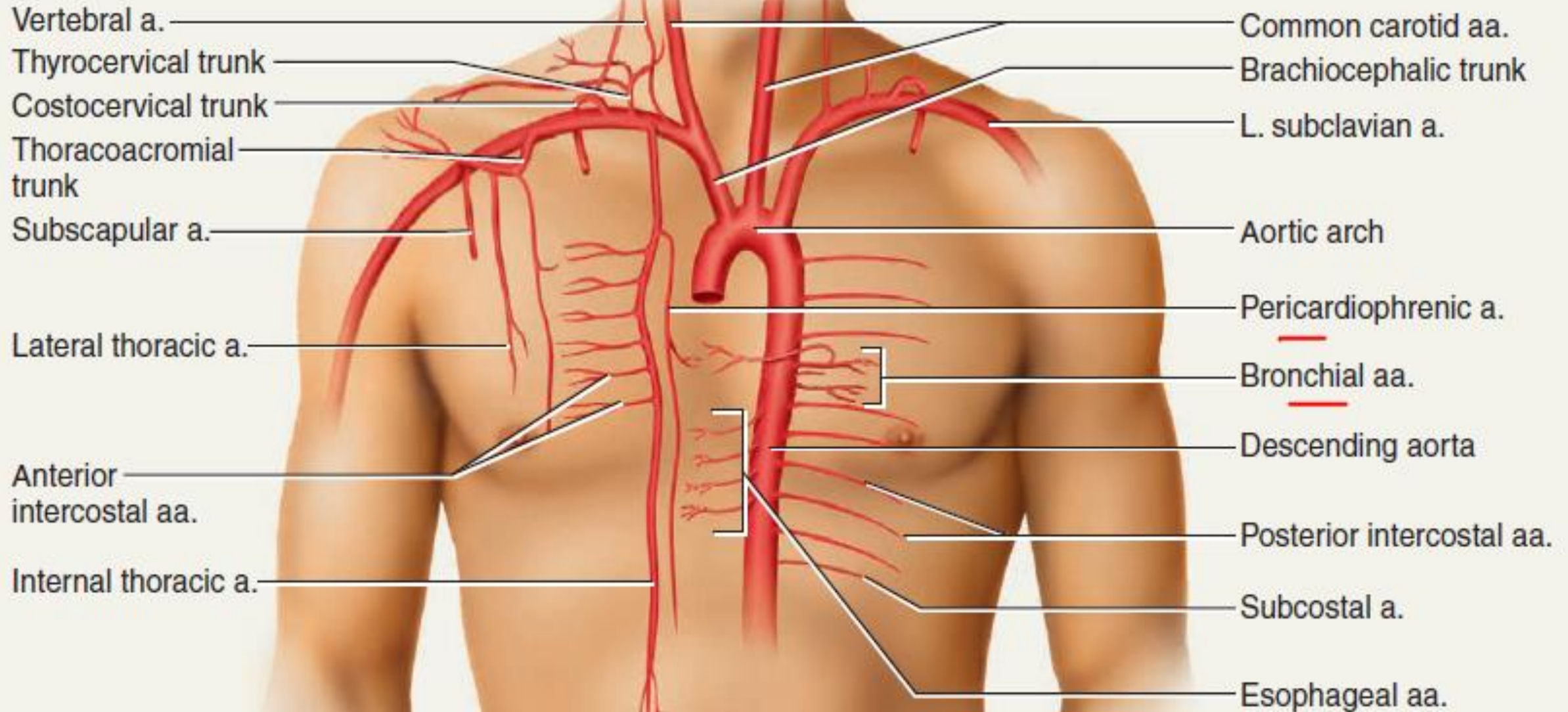


Thoracic Aorta and its Branches

❖ **The branches of the thoracic aorta can be divided into two groups:**

- **The visceral arteries** supply the thoracic organs, and the **parietal arteries** supply the thoracic wall.
- ❑ **The visceral branches** of the thoracic aorta supply the esophagus, the trachea, the parietal pericardium, and part of the lung.
- ❑ **The major parietal arteries** are the **posterior intercostal arteries**, which arise from the thoracic aorta and extend between the ribs.
- They supply the intercostal muscles, the vertebrae, the spinal cord, and the deep muscles of the back.
- The **superior phrenic arteries** supply the diaphragm.

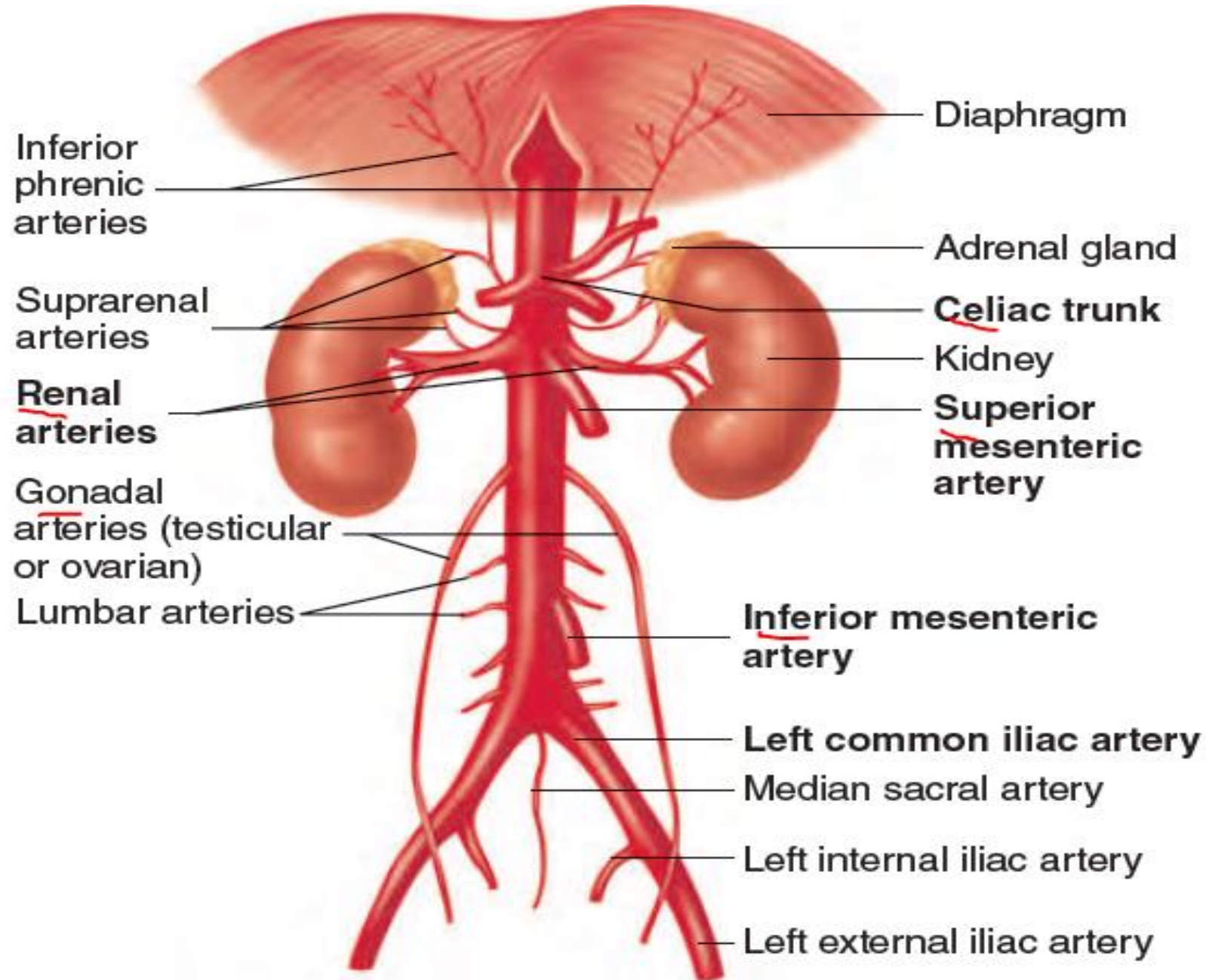




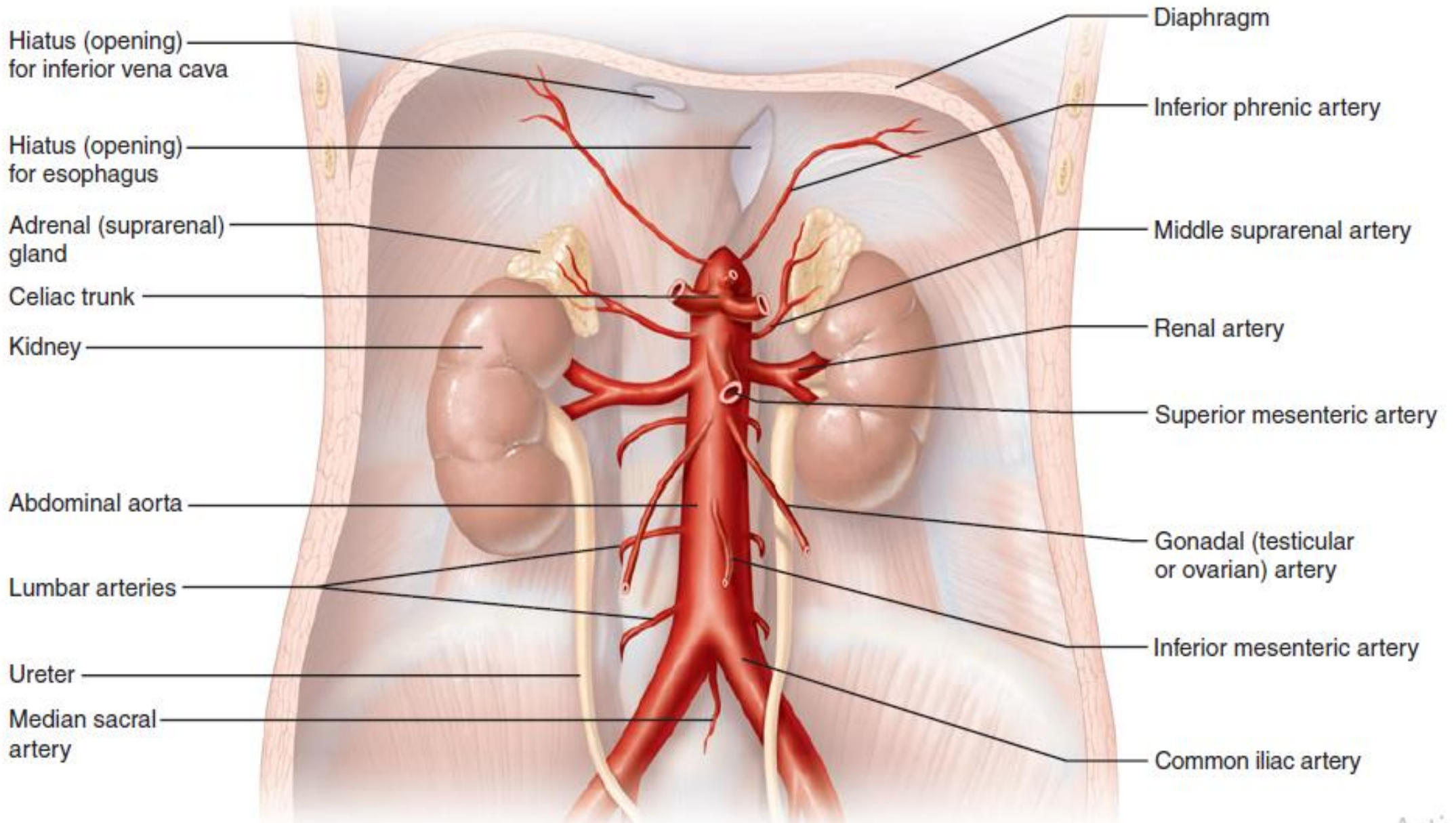
Abdominal Aorta and its Branches

- The branches of the abdominal aorta, **divided into visceral and parietal groups.**
- ❖ **The visceral arteries are divided into paired and unpaired branches.**
- **There are 3 major unpaired branches:** the celiac trunk, the superior mesenteric artery, and the inferior mesenteric artery.
- ❖ **There are 3 paired visceral branches of the abdominal aorta. The renal, suprarenal & gonadal arteries(testicular or ovarian) arteries.**
- **The parietal branches of the abdominal aorta are: The inferior phrenic arteries, the lumbar arteries and the median sacral artery.**

Abdominal Aorta and its Branches,,



Abdominal Aorta and its Branches,,



Table

The Aorta and Its Principal Branches

Portion of Aorta	Major Branch	Regions Supplied
<u>A</u> scending aorta	Left and right coronary arteries	Heart
<u>A</u> ortic arch	Brachiocephalic artery Right common carotid Right subclavian Left common carotid artery Left subclavian artery	Right side of head Right arm Left side of head Left arm
<u>D</u> escending aorta		
<u>T</u> horacic aorta	Intercostal artery	Thoracic wall
<u>A</u> bdominal aorta	Celiac artery Superior mesenteric artery Renal artery Gonadal artery Inferior mesenteric artery Common iliac artery	Stomach, spleen, and liver Small and large intestines (ascending and transverse colons) Kidney Ovary or testis Lower digestive system (transverse and descending colons, and rectum) Pelvic organs and legs

